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SUPERSEDING
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MILITARY SPECIFICATION

MULTIPLEXER-COMBINER, TD-1234(P)/TTC

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the construction and testing necessary for the manufacture of Multiplexer- Combiner, TD-1234(P)/TTC. This unit is also known as Remote Multiplexer Combiner (RMC) and is one element of the Digital Group Multiplex (DGM) family of cable system equipments developed by the Joint Tactical Command, Control and Communications Agency (JTC^3A, formerly TRI-TAC). See 6.1 for intended use of the TD-1234 (P)/TTC.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications - Electronics Command, ATTN: AMSEL-ED-LC, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

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SPECIFICATION

Military

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronics Equipment
MIL-C-55583	Cable Assembly, Special Purpose CX-11230()/G, Cable Assembly Adapter CS-10734()/G
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A (Notice 4)	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 (Notice 3)	Electromagnetic Interference Characteristics Measurement of
MIL-STD-781C	Reliability Tests, Expontential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration System Requirements

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-E-876632	Combiner, Remote Multiplexer, TD-1234(P)/TTC
SM-D-884567	Circuit Card Assembly, Multi Vdc
SM-D-884575	CCA, DC Input
SM-D-884579	CCA, AC Input
SM-D-884585	CCA, 52/180 Vdc
SM-D-959323	CCA, Analog Applique Unit
SM-E-986232	CCA, Group Modem
SM-E-986235	CCA, Loop Modem
SM-E-986239	CCA, Timing Generator
SM-E-986241	CCA, Bite 2
SM-E-986247	CCA, MUX/DEMUX
SM-E-986250	CCA, Group Framing

(Copies of specifications, standards, handbooks, drawings and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with the requirements this specification and of DL-SM-D-876632 including the requirements for parts, materials and processes therein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirement 30 and 64.

3.3 Parts, materials, and processes - general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and drawing DL-SM-D-876632.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. (See 4.16) A unique serial number is required on each nomenclatured item and CCA. The numbers shall be consecutively assigned.

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3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled unit shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no Built-In Test Equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.13 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Diphase group output characteristics. (See 4.7.1) The equipment shall provide two diphase group output, a low-speed output and a high-speed output. The high-speed outputs bit rates shall be 256, 288, 512 or 576 kb/s when the channel bit rate is 32 kb/s. The high-speed output bit rates shall be 128, 144, 256, or 288 kb/s when the channel bit rate is 16 kb/s. The low-speed output bit rates shall be 144, 256, or 288 kb/s when the loop bit rate is 32 kb/s. The low-speed output bit rates shall be 72, 128, or 144 kb/s when the loop bit rate is 16 kb/s.

3.7.1.1 Output voltage. (See 4.7.1.1) The high and low speed output signal shall be 3 volts $\pm$30 percent peak-to-peak into a 58-ohm resistive load.

3.7.1.2 Output rise and fall time. (See 4.7.1.2) The high and low speed output rise and fall time shall be 130 nanoseconds maximum 10 to 90 percent into a 58-ohm resistive load.

3.7.1.3 Output impedance. (See 4.7.1.3) The high and low speed output impedance shall be 58 ohm $\pm$ 15 percent from 1000 Hz to 300 kHz and 50 to 100 ohms from 300 to 1000 Hz.

3.7.2 Diphase group input characteristics. (See 4.7.2) The equipment shall provide two diphase group inputs, a low-speed input and a high-speed input. The high-speed input bit rates shall be 256, 288, 512 or 576 kb/s when the channel bit rate is 32 kb/s. The high-speed input bit rates shall be 128, 144, 256, or 288 kb/s when the channel bit rate is 16 kb/s. The low-speed input bit rates shall be 144, 256, or 288 when the loop bit rate is 32 kb/s. The low-speed input bit rates shall be 72, 128, or 144 kb/s when the loop bit rate is 16 kb/s.

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3.7.2.1 Input impedance. (See 4.7.2.1) The high and low speed input impedance shall be 58 ohms ±15 percent from 1000 Hz to 300 kHz and 50 to 100 ohms from 300 to 1000 Hz.

3.7.2.2 Received signal level. (See 4.7.2.2) The unit shall operate properly on the signal specified in 3.7.1 at the end of 3.2 km of CX-11230 cable (see MIL-C-55583) with an additional 4 dB of attenuation. Proper operation shall be defined as a channel bit error rate (BER) of zero errors in 30 seconds.

3.7.3 Station clock input characteristics. (See 4.7.3) The input shall accept a balanced input at the nominal high-speed group bit rate.

3.7.3.1 Receiver dynamic range. (See 4.7.3.1) The receiver shall deliver a logic 1 output for a line-to-line voltage between +50 millivolts and +4.0 volts and a logic 0 for a line-to-line voltage between -50 millivolts and -4.0 volts.

3.7.3.2 Input load impedance. (See 4.7.3.2) The input load impedance shall be 100 ohms ±10 percent line-to-line. The resistive termination shall float; i.e., the center tap is not grounded.

3.7.3.3 Timing modes. (See 4.7.3.3)

3.7.3.3.1 Loop back mode. Transmit timing shall be derived from the received group signal. The multiplexer shall accept external timing frequency errors up to ±50 PPM from the nominal.

3.7.3.3.2 Station clock mode. (See 4.7.3.1) Transmit timing shall be derived from an external timing source. The received group signal shall be slaved to this timing source by external means. When the external timing signal is absent, the transmit timing shall be derived from an internal timing source. The frequency of this timing source shall be within ±40 ppm of the nominal frequency.

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3.7.4 Orderwire interface. (See 4.7.4) There shall be a maintenance orderwire in the equipment on the CX-11230 cable interface. Two modes shall be provided, an active through mode, and a passive termination mode. With the unit power on and no TS-3647/G connected, the active through mode shall be automatically selected. In this mode, the orderwire signals coming in on the high group shall be passed through to the low group and vice versa. Isolation between the high and low group signals and amplification for the orderwire signals shall be provided. If the unit power fails or the TS-3647/G is connected to the equipment, the passive termination mode shall be automatically selected. In this mode, the TS-3647/G shall be connected to either the high group or the low group, selectable by the operator, and the through connection shall be broken. The passive termination mode shall operate properly on either group independent of equipment unit power.

3.7.4.1 Input Impedance. (See 4.7.4.1) The orderwire interface input impedance shall be 600 ohms ±10% at a frequency of 1 kHz.

3.7.4.2 Output impedance. (See 4.7.4.2) The orderwire interface output impedance shall be 600 ohms ±10% at a frequency of 1 kHz.

3.7.4.3 Receive level. (See 4.7.4.3) The orderwire receiver level shall be between -10 and -13 dBm with a +21.3 dBm input level at 1.6 kHz. The diphase group output shall be looped through less than 5 feet of cable.

3.7.4.4 Bandwidth. (See 4.7.4.4) The frequency response of an orderwire link, as described in 3.7.4.3, shall be as follows:

Frequency	Transmission Loss (From reference)
300	-5 to +8 dB
400	±5 dB
1000	Reference
3000	±5 dB
3400	-5 to +8 dB

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3.7.4.5. Signal-to-noise ratio. (See 4.7.4.5) With random group traffic, the signal to noise ratio C-message weighted shall be greater than 11 dB at 72 kb/s and 20 dB at 144 kb/s.

3.7.4.6 Transmit level. (See 4.7.4.6) For an input at either group of +1½ dBm, the output at the other group shall be +5 dBm ±½ dB. For an input at the passive interface of +21.3 dBm, the output at the appropriate group shall be +5 dBm ±1 dB.

3.7.5 Digital channel characteristics. (See 4.7.5)

3.7.5.1 Output voltage. (See 4.7.5.1) The output voltage shall be 3 volts ±15% peak-to-peak measured into a 125 ohm load.

3.7.5.2 Output waveshape. (See 4.7.5.2) Measured into a 125 ohm resistive load, the rise and fall times (10% to 90%) shall be between 0.65 and 3.0 microseconds. The droop shall be less than 5% of the peak-to-peak amplitude.

3.7.5.3 Input and output impedance. (See 4.7.5.3) The input and output impedance magnitude shall be 125 ohms ±15% over the band from 8 to 56 kHz.

3.7.5.4 Input voltage. (See 4.7.5.4) Proper operation shall occur over the input voltage range of 0.15 to 4.0 volts peak-to-peak. Proper operation shall be defined as a Bit Error Rate (BER) of $1X10^{6}$ or less.

3.7.5.5 Idle loop. (See 4.7.5.5) When transmission from the DSVT has ceased on a loop, the loop modem shall send continuous ones on its high group output.

3.7.6 Loop power feed. (See 4.7.6) Maximum output voltage shall be 56 Vdc into a 11,200 ohm resistive load. Minimum output currents shall be 68.8 mA into a 680 ohm resistive load. The loop outputs shall be short-circuit proof, both line-to-line and line-to-ground. The powerfeed circuitry shall tolerate an initial current surge of 100 mA for up to 45 ms without power feed shutoff.

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3.7.7 Analog channel characteristics. (See 4.7.7)

3.7.7.1 Analog channel input/output impedance. (See 4.7.7.1) The impedance at the 4-wire analog channel input and output terminals shall be 600 ohms resistive with a minimum return loss of 26 dB over the frequency range 300 to 3400 Hz.

3.7.7.2 Longitudinal balance. (See 4.7.7.2) The longitudinal balance at the analog channel input and output terminals shall be not less than 40 dB when measured at a tone level of 0 dBm over the frequency range of 300 to 3400 Hz.

3.7.7.3 Frequency response. (See 4.7.7.3) The characteristics of the input/output filter should be as follows:

Frequency, Hz	Input Atten (dB0)	Output Atten (dB)
76	6 min	6 min
200	3 ± 1	3 ± 1
1000	ref	ref
3500	3 ± 1	0 to 4
8800	12 min	45 min

3.7.7.4 Analog channel input and output test tone level and frequency. (See 4.7.7.4) The nominal test tone input and output level shall be -17 dBm at 820 Hz.

3.7.7.5 Envelope delay distortion. (See 4.7.7.5) The envelope delay relative to 800 Hz of the input and output filters added together shall be less than 100 μs from 800 to 3000 Hz.

3.7.7.6 Input-output linearity. (See 4.7.7.6) The output level of each channel (see 3.7.7.11.3 and 3.7.7.11.4) shall be within ±1.0 dB of the input signal level from full load tone (see 3.7.7.10) to 35 dB below full load tone when the input and output level selector switches are set to a sum of 0 dB.

3.7.7.7 Loopback switch. (See 4.7.7.7) A loopback switch shall be provided for each channel. It shall permit the digital output of the encoder to be looped to the digital input of the decoder. The loopback switches shall be operable when the modules are inserted into the equipment.

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3.7.7.8 Noise. (See 4.7.8.7)

3.7.7.8.1 Idle channel noise. (See 4.7.7.8.1) With one or more analog modules inserted in the equipment and the remaining module positions filled with the digital modules and the analog channels operating in loopback mode, the signal measured on the analog inputs and outputs shall not exceed 33 dBrnC when all analog ports are terminated in 600 ohms and input and output level selector switches are set to 0 dB.

3.7.7.8.2 Crosstalk level.(See 4.7.7.8.2) The inter-channel and intra-channel crosstalk plus noise measured at either the near end or the far end of two analog channels shall not exceed 34 dBrnC when sending the 820 Hz test tone at full load.

3.7.7.9 Peak clipping.(See 4.7.7.9) For a +4 dBm input signal at 200 Hz, the output signal shall have a total distortion of less than 10% at a channel rate of 32 kb/s and with the input level selector switch set to -10 dB and the output level selector switch set to +10 dB.

3.7.7.10 Full load tone.(See 4.7.7.10) The nominal full tone input and output level shall be 0 dBm at a frequency of 820 Hz with the input level selector switch set to -10 dB, and the output level selector switch set to +10 dB. With a full load tone, the Continuous Variable Slope Delta (CVSD) converter approaches the slope overload condition.

3.7.7.11 Analog channel signal processing.(See 4.7.7.11)

3.7.7.11.1 Clocking rate.(See 4.7.7.11.1) The channel clocking rate of the CVSD converter shall be 16 or 32 kb/s, selectable.

3.7.7.11.2 Dynamic range.(See 3.7.7.11.2) The dynamic range of the CVSD shall be such that the repetitive test patterns shown below shall produce 666 2/3 Hz tone outputs having a ratio of 23 ± 1 dB.

Channel rate kb/s	Digital Pattern (Hexadecimal Equivalent) *											
32 low output	D	D	5	2	4	9	2	2	A	D	B	6
high output	A	E	F	F	F	B	5	1	0	0	0	4
16 low output	6	E	A	9	2	5						
high output	B	F	D	4	0	2						

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3.7.7.11.3 Output level control. (See 4.7.7.11.3) The output level of the CVSD decoder shall be selectable in four steps independent of the level setting at the encoder. The output level selector shall be labeled -3, 0, +10, and +20. With either of the two repetitive digital patterns (shown below) applied to the decoder, the analog output into a 600 ohm load shall be -20, -17, -7, or +3 dBm ±0.5 dB, when the output level selector is set at -3, 0, +10 or +20 respectively.

Channel rate kb/s	Digital Pattern (Hexadecimal Equivalent)*
32	F D A A 1 0 2 5 5 E
16	F B 4 1 2

*The serial bit patterns are read from left to right and are expressed as hexadecimal characters, each character is transmitted in time sequence as read, MSB first and LSB last.

3.7.7.11.4 Input level control. (See 4.7.7.11.4) The operating level of the CVSD encoder shall be adjustable in four steps. The input level selector shall be labeled +3, 0, -10 and -20. When the channel is in the loopback mode, the input levels and level settings shall produce the output levels shown below with a tolerance of ±1.0 dB.

Input Level (dBm)	Encoder Level Setting	Decoder Level Setting	Output Level (dBm)
-20	+3	+20	+3
-17	0	+20	+3
- 7	-10	+20	+3
+ 3	-20	+20	+3
-20	+ 3	+10	-7
-17	0	+10	-7
- 7	-10	+10	-7
+ 3	-20	+10	-7
-20	+ 3	0	-17
-17	0	0	-17
- 7	-10	0	-17
+ 3	-20	0	-17
-20	+ 3	-3	-20
-17	0	-3	-20
- 7	-10	-3	-20
+ 3	-20	-3	-20

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3.7.8 Framing. (See 4.7.8)

3.7.8.1 Format. (See 4.7.8.1) Channel zero of the equipment groups shall be designated as the overhead channel. The high-speed groups overhead channel shall be composed of eight equal rate sub-channels with the format shown in Figure 1. When the voice digitization rate is 32 kb/s, the overhead subchannel rate shall be 4 kb/s. When the voice digitization rate is 16 kb/s, the overhead subchannel rate shall be 2 kb/s. Subchannel 0 shall be the frame subchannel. Subchannel 7 shall be used for the equipment system control telemetry. Subchannel 6 shall be used for the system control telemetry of an equipment or Multiplexer TD-1233/TTC directly connected via the input group. Subchannel 5 shall be used for the system control telemetry of a Multiplexer TD-1233/TTC which is connected to the low-speed group terminal of the directly connected TD-1234 i.e., TD-1234 "B" (see Figure 1).

3.7.8.2 Frame acquisition time. (See 4.7.8.2) The frame acquisition time is defined as the period from the start of frame search until the multiplexer at each end of a link is capable of restoring traffic in both directions on the link between them. The equipment shall acquire frame synchronization within 50/100 ms at frame rates of 4/2 kb/s with a minimum probability of 0.9, in an error environment of 0.1% BER random errors. The equipment shall also acquire frame synchronization within 100/150 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9, in an error environment of 20% BER random bursts interspersed with 0.1% BER random error intervals. The time between the start of successive bursts shall be random between 50 ms and 1 second, and the burst length shall be 5% of the time between the start of that burst and the start of the next burst. The above times do not include cryptographic resync delays, or any preset delay intended to assist in flywheeling through fades.

3.7.8.3 Frame maintenance. (See 4.7.8.3) While bit count integrity is maintained, "out of sync" shall not be declared more often than once per 240 hours (on the average) in an error environment of 20% BER bursts interspersed with 0.1% BER random error intervals. The time between the start of successive bursts shall be random between 50 ms and 1 second, and the burst length shall be 5% of the time between the start of that burst and the start of the next burst.

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3.7.8.4 Detection of out of sync. (See 4.7.8.4) When bit count integrity is lost and with random data on all channels, the equipment shall detect out of sync within 150 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9. When out of sync occurs on either group, the equipment shall continue to transmit frame pattern on the remaining group, as appropriate, to maintain frame synchronization of this link.

3.7.8.5 Idle filter pattern. (See 4.7.8.5) When the equipment is out of frame it shall set its channel outputs to all ones.

3.7.8.6 False frame lock. (See 4.7.8.6) When the high group signal is looped back, the equipment shall be capable of clearing any false frame lock condition in the multiplexer connected to the low group.

3.7.9 Prime power interface. (See 4.7.9) The dc primary power shall have a steady state voltage of 28 (+4, -6) volts. The ac primary power source shall be 115 volts ±10%, single phase. The frequency of the ac voltage shall be 50 Hz -5%, to 400 Hz +5%. The ac primary power shall conform to MIL-STD-1332 for Type I, Class 1, and 2, Mode I, II and III generator sets. The initial power to the equipment shall not exceed 210 watts.

3.7.10 Current protection. (See 4.7.10) The low and high cable interface circuits shall withstand without damage to the equipment an injected current of 90 mA on either center conductor of the CX-11230 cable.

3.7.11 Power feed (low-speed group). (See 4.7.11.1) The equipment shall be capable of supplying the power requirements of a connected TD-1233()/TTC via the low speed group cable.

3.7.11.1 Power feed output over-voltage. (See 4.7.11.1) The maximum open circuit voltage shall be 210 Vdc.

3.7.11.2 Current limit. (See 4.7.11.2) The over-current protection circuitry shall trip at 420 mA minimum. The output shall shut down and discharge the output voltage through a resistance of 5000 ohms or less when the power feed current drops below 10 mA. After under current shut down, manual restart shall be required.

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3.7.12 Built-in Test Equipment (BITE) and Alarms. (See 4.7.12) The BITE shall monitor signals in and out of the unit and internal circuits. Go-no-go fault-location circuits shall isolate failures to a particular CCA of Lowest Replaceable Unit (LRU). Indicating devices shall be provided to ensure proper operation of the equipment and to monitor the dc voltages, go-no-go fault isolation, and other critical internal and external signals. The BITE shall operate without interrupting traffic or interfering with normal system operation. Removal of the BITE card shall not affect operation of the unit. Failure alarms shall be provided for channel failure, common power supply failure, and inter-unit signal malfunctions. Means shall be provided in the unit to check the operation of the alarms without interrupting the equipment operation.

3.7.12.1 BITE alarms. (See 4.7.12) Failures detected by the BITE circuitry cause relay contact closures accessible externally at a connector on the rear panel. The S and SA alarm outputs described below are available:

a. S alarm. The S alarm shall indicate any failure detected including loss of input power. This alarm shall also provide a visible indication on the front panel except when there is a loss of input power.

b. SA alarm. The SA alarm shall indicate that the malfunction is within the unit. If the unit power switch is set to off there shall be no SA alarm.

3.7.13 Operational test.

3.7.13.1 High Group error check. (See 4.7.13.1) There shall be no errors for a period of 10 seconds with the orderwires on.

3.7.13.2 Low group data error check. (See 4.7.13.2) There shall be no errors for a period of 10 seconds with the orderwires on.

3.7.13.3 Maintenance orderwire level. (See 4.7.13.3) The orderwire output level shall be between -10 and -13 dBm with a +21.3 dBm input level at 1.6 kHz.

3.7.13.4 DC voltage check. (See 4.7.13.4) The loop channel outputs, when enabled, shall provide power to the DSVT at a voltage between 46.9 Vdc and 56 Vdc.

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3.7.13.5 Sync/frame check. (See 4.7.13.5) The unit shall reframe when the power supply is interrupted and turned back on.

3.7.13.6 Power feed voltage check. (See 4.7.13.6) The power feed on the low group shall be 170 to 190 Vdc into 600 ohms.

3.7.13.7 Analog channel check. (See 4.7.13.7) With all level switches set to 0, the gain shall be 0 dBm ±1 dB of the room temperature value at a frequency of 820 Hz.

3.7.14 Environmental operational test. (See 4.7.14)

3.7.14.1 Digital channel/high group check. (See 4.7.14.1) There shall be no errors for a period of 10 seconds with the orderwires on.

3.7.14.2 Low group data error check. (See 4.7.14.2) There shall be no errors for a period of 10 seconds with the orderwires on.

3.7.14.3 Maintenance orderwire levels. (See 4.7.14.3) The orderwire output level shall be between-13 and -10 dBm with +21.3 dBm input level at 1.6 kHz.

3.7.14.4 DC voltage check. (See 4.7.14.4) The loop channel outputs,when enabled, shall provide power to the DSVT at a voltage between 46.9 Vdc and 56 Vdc.

3.7.14.5 Sync/frame check. (See 4.7.14.5) The unit shall reframe when the Group Signal is interrupted and turned back on.

3.7.14.6 Power feed voltage check. (See 4.7.14.6) The power feed on the low group shall be 170 to 190 Vdc into 600 ohms. The polarity shall be positive on both center conductors and negative on the shield.

3.7.14.7 Analog channel check. (See 4.7.14.7) With all level switches set to 0, the gain shall be within ±1 dB of the room temperature value at a frequency of 820 Hz.

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3.8 Service conditions. (See 4.8) The operational test of 3.7.14 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +51.7°C (+125°F) plus the heating effects of solar radiation at a rate of 360 BTU/ft^2/hr. The equipment shall withstand exposure (non-operating) to ambient air temperatures as high as +71.1°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient temperature as low as -45°C (-50°F). The equipment shall withstand exposure (non-operating) to ambient temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust (See 4.8.4) The equipment shall with-stand, in both operating and non-operating conditions, exposure to fine dust particles with wind speeds of 3542 ft/min (35 knots).

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

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3.8.7 Immersion. (See 4.8.7) Prior to and following vibration and shock testing, the equipment in its transit case shall be immersion-proof to a covering depth of 3 feet of water, for a period of not less than 2 hours.

3.8.8 Vibration and shock. (See 4.8.8) The equipment shall withstand vibration and shock induced during field transport by military vehicle and by rough handling. The equipment shall withstand vibration encountered during transportation and shocks encountered during servicing.

3.8.9 Rain. (See 4.8.9) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during periods of heavy precipitation with intermittent wind.

3.9 Electromagnetic interference (EMI). (See 4.9) The equipment shall meet the following emission and susceptibility requirements of MIL-STD-461, Notice 4:

CE01			
CE02	CS01	RE02	RS02
CE03	CS02	RE02.1	RS03
CE04	CS06		RS03.1
CE05			

NOTES:

a. RE02.1 shall be applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 shall be applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to dc, 60 Hz and 400 Hz power lines.

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3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedances shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The TD-1234 shall demonstrate a specified Mean-Time-Between-Failure (MTBF), θ_o of 2500 hours when in the normal mode of operation

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies and replaceable parts shall meet Requirement 7 of MIL-STD-454.

3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements of their respective Test Requirement Specifications (TRS) as referenced in the following subassembly drawings:

CCA	Drawing
Multi Vdc	SM-D-884567
DC Input	SM-D-884575
AC Input	SM-D-884579
52/180 Vdc	SM-D-884585
Analog Applique Unit	SM-D-959323
Group Modem	SM-E-986232
Loop Modem	SM-E-986235
Timing Generator	*SM-E-986239
Bite 2	*SM-E-986241
MUX/DEMUX	*SM-E-986247
Group Framing	*SM-E-986250

* Indicates drawing contains automatic test equipment (ATE) acceptance procedures and programs.

3.13 Operational test. (See 4.7.13) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.13 after all other inspections and tests are completed to determine that the unit is operating properly prior to packaging.

3.14 System safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

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3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive material. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires (electrical)

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

3.16 Nuclear survivability. (See 4.16) The equipment shall satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed measurements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Packaging inspection (see 4.14).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 18 first article units of the completed Multiplexer TD-1234(P)/G. The first article quantity shall be as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table I and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in process Group A inspection prior to burn-in.

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TABLE I. First article inspection.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
1. Inspections by Subsidiary Documents	N/A	4.4	Insp to be performed on all units.												
2. Burn-In	3.6	4.6	Insp to be performed on all units.												
3. Group A inspection	(See Table II)		Insp to be performed on all units.												
4. Group B inspection	(See Table III)		Insp to be performed on all units.												
5. Group C inspection															
Subgroup 1															
High temperature	3.8.1.1	4.8.1.1	1	*											
Low temperature	3.8.1.2	4.8.1.2	2	*											
Subgroup 2															
EMI	3.9	4.9		1											
Immersion	3.8.7	4.8.7			1										
Vibration/shock	3.8.8	4.8.8			2	*									
Immersion	3.8.7	4.8.7			3										
Subgroup 3															
Humidity (Note c)	3.8.2	4.8.2						1			*				
Rain	3.8.9	4.8.9						2			*				
Subgroup 4															
Altitude	3.8.3	4.8.3							1		*				
Dust (Note c)	3.8.4	4.8.4							2						
Salt fog (Note c)	3.8.5	4.8.5							3						
Fungus (Note c)	3.8.6	4.8.6								1					
Subgroup 5															
Diphase Group Output	3.7.1	4.7.1										1			
Output Rise Time	3.7.1.2	4.7.1.2										2			
Output Impedance	3.7.1.3	4.7.1.3										3			

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TABLE I. First article inspection (condt)

Inspection		Rqmt Para	Test Para	1	2	3	4	5	6	7	8	9	10	11	12-19	20+
Input Impedancee		3.7.2.1	4.7.2.1										4			
Station Clock Input		3.7.3	4.7.3										5			
Receiver Dynamic Range		3.7.3.1	4.7.3.1										6			
Input Load Impedance		3.7.3.2	4.7.3.2										7			
Timing Modes		3.7.3.3	4.7.3.3										8	*		
Orderwire Interface		3.7.4	4.7.4										9			
Input Impedance		3.7.4.1	4.7.4.1										10			
Output Impedance		3.7.4.2	4.7.4.2										11			
Receive Level		3.7.4.3	4.7.4.3										12			
Bandwidth	**	3.7.4.4	4.7.4.4										13			
Signal-to-Noise Ratio	**	3.7.4.5	4.7.4.5										14			
Transmit Level		3.7.4.6	4.7.4.6										15			
Output Waveshape		3.7.5.2	4.7.5.2										16			
Input and Output Impedance		3.7.5.3	4.7.5.3										17			
Idle Loop	**	3.7.5.5	4.7.5.5										18			
Analog Channel Input/ Output Impedance		3.7.7.1	4.7.7.1										19			
Longitudinal Balance		3.7.7.2	4.7.7.2										20			
Loop power feed (surge current)		3.7.6	4.7.6.1										21			
Frequency Response	***	3.7.7.3	4.7.7.3										22			
Envelope Delay Distortion	**	3.7.7.5	4.7.7.5										23			
Input-Output Linearity	***	3.7.7.6	4.7.7.6										24			
Peak Clipping	**	3.7.7.9	4.7.7.9										25			
Clock Rate	***	3.7.7.11.1	4.7.7.11.1										26			
Dynamic Range	***	3.7.7.11.2	4.7.7.11.2										27			
Output Level Control	***	3.7.7.11.3	4.7.7.11.3										28			
Input Level Control	***	3.7.7.11.4	4.7.7.11.4										29			
Format	**	3.7.8.1	4.7.8.1										30			
Frame Acquisition Time		3.7.8.2	4.7.8.2										31			
Frame Maintenance	**	3.7.8.3	4.7.8.3										32			
Detection of Out-of-Sync		3.7.8.4	4.7.8.4										33	*		
Idle Filler Pattern	**	3.7.8.5	4.7.8.5										34			
False Frame Lock	**	3.7.8.6	4.7.8.6										35			
Prime Power		3.7.9	4.7.9										36			
Current Protection	**	3.7.10	4.7.10										37			

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TABLE I. First article inspection (contd).

Inspection	Rqmt Para	Test Para	1 2 3 4 5 6 7 8	9	10	11	12-19	20+
Loopback Switch **	3.7.7.7	4.7.7.7			38			
Interchangeability	3.11	4.11			39			
Crosstalk Level **	3.7.7.8.2	4.7.7.8.2			40			
Current Limits	3.7.11.2	4.7.11.2			41			
Power Feed Output	3.7.11.1	4.7.11.1			42			
6. Group D Inspection								
Reliability	3.10	4.10					8 Units	

* Indicates use of an auxiliary unit

**One time engineering test to be covered by a separate engineering report not part of first article.

*** Test to be performed as part of TRS on PC board.

NOTES:

a. The first article inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The number in the unit columns in the table covered by Inspections 5 and 6 specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. All first article units shall be subjected to Group A and Group B inspections.

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4.3.3 First article inspection data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4. Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract of purchase order.

	Where Required
Printed wiring	3.2.2
Semiconductor devices and integrated circuits	3.2.3
Parts, materials, and processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5. through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specifications requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table II shall be inspected for conformance to the requirements specified in table II.

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4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table II. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table II.

b. Only for spares that are not installed into end equipment, the following shall be done: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspections of step a above shall be assembled into the end equipments, which shall then be subjected by the contrator to the burn-in test requirements of 4.6 and an end equipment 100% visual/mechanical and electrical inspection in accordance with table II, in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which have passed the 100% inspection and these shall be subjected to sampling utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to table III and to the special inspection levels of table 1 of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B ALQ shall be as shown in table III and the inspection level shall be S-4.

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TABLE II. Group A inspections.

Inspection	Rqmt Para	Insp Para	ALOs Major	ALOs Minor
Visual-Mechanical (Including spares)				
Case and Cover	3.15	4.13	2.5%	4.0%
Multi Vdc	3.15	4.13	2.5%	4.0%
52/180 Vdc	3.15	4.13	2.5%	4.0%
AC Input	3.15	4.13	2.5%	4.0%
DC Input	3.15	4.13	2.5%	4.0%
Timing Generator	3.15	4.13	2.5%	4.0%
BITE 2	3.15	4.13	2.5%	4.0%
Loop Modem	3.15	4.13	2.5%	4.0%
Mux/Demux	3.15	4.13	2.5%	4.0%
Group Framing	3.15	4.13	2.5%	4.0%
Analog Applique Unit	3.14	4.13	2.5%	4.0%
Group Modem	3.14	4.14	2.5%	4.0%
Electrical Performance (Including Spare Assemblies)				
Multi Vdc	3.12	4.12	2.5%	*
52/180 Vdc	3.12	4.12	2.5%	*
AC Input	3.12	4.12	2.5%	*
DC Input	3.12	4.12	2.5%	*
Timing Generator	3.12	4.12	2.5%	*
BITE 2	3.12	4.12	2.5%	*
Loop Modem	3.12	4.12	2.5%	*
Mux/Demux	3.12	4.12	2.5%	*
Group Framing	3.12	4.12	2.5%	*
Group Modem	3.12	4.12	2.5%	*
Analog Applique Unit	3.12	4.12	2.5%	*
Electrical Performance (Unit Level)				
Idle Channel Noise	3.7.7.8.1	4.7.7.8.1	1.0%	*
BITE and Alarms	3.7.12	4.7.12	1.0%	*
Operational test	3.7.13	4.7.13	1.0%	*

*All electrical defects are major.

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4.5.2.2 Order of inspection within Group B. Group B shall be performed in the order listed in table III.

TABLE III. Group B inspections.

Inspection	Reqt Para	Insp Para	AQL
Output Voltage	3.7.1.1	4.7.1.1	4%
Received Signal Level	3.7.2.2	4.7.2.2	4%
Output Voltage	3.7.5.1	4.7.5.1	4%
Input Voltage	3.7.5.4	4.7.5.4	4%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to immersion, vibration, shock and immersion testing with the unit subjected to testing in that order. The unit selected every eighth month shall be subjected to EMI testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity and rain testing.

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4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to Altitude, Dust and Salt Fog testing and one unit shall be subjected to Fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table IV.

4.5.3.2 Order of inspection within Group C inspection. Group C inspection shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

TABLE IV. Group C inspections.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High Temperature	3.8.1.1	4.8.1.1
Low Temperture	3.8.1.2	4.8.1.2
Subgroup 2		
Immersion	3.8.7	4.8.7
Vibration/Shock	3.8.8	4.8.8.
Immersion	3.8.7	4.8.7
EMI	3.9	4.9

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TABLE IV. Group C inspections. (Cont'd)

Inspection	Reqt Para	Insp Para
Subgroup 3		
Humidity	3.8.2	4.8.2 (Note 1)
Rain	3.8.9	4.8.9
Subgroup 4		
Altitude	3.8.3	4.8.3 (Note 1)
Dust	3.8.4	5.8.4 (Note 1)
Salt/Fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)
Subgroup 5		
Output Rise Time	3.7.1.2	4.7.1.2
Output Impedance	3.7.1.3	4.7.1.3
Input Impedance	3.7.2.1	4.7.2.1
Receiver Dynamic Range	3.7.3.1	4.7.3.1
Input Load Impedance	3.7.3.2	4.7.3.2
Timing Modes	3.7.3.3	4.7.3.3
Input Impedance	3.7.4.1	4.7.4.1
Output Impedance	3.7.4.2	4.7.4.2
Receive Level	3.7.4.3	4.7.4.3
Transmit Level	3.7.4.6	4.7.4.6
Output Waveshape	3.7.5.2	4.7.5.2
Input and Output Impedance	3.7.5.3	4.7.5.3
Analog Channel Input/ Output Impedance	3.7.7.1	4.7.7.1
Longitudinal Balance	3.7.7.2	4.7.7.2
Loop Power Feed	3.7.6	4.7.6
Frame Acquisition Time	3.7.8.2	4.7.8.2
Detection of Out-of-Sync	3.7.8.4	4.7.8.4
Prime Power	3.7.9	4.7.9
Interchangeability	3.11	4.11
Power Feed Output Voltage	3.7.11.1	4.7.11.1
Current Limits	3.7.11.2	4.7.11.2

NOTE: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.5.4 Reinspection of conforming Group C sample units. Unless orderwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection This inspection shall consist of the tests specified in table V and shall be performed on units from lots which have been subjected to and have met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE V. Group D inspection.

Inspection	Reqt Para	Insp Para
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Eight units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsequent Group D inspections, a maximum of eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.6 Electrical preconditioning (burn in). (See 3.6) During electrical preconditioning, the equipment shall be connected per figures 2 and 3. Test conditions of 4.10.3 shall apply except that duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred. The requirements of 3.6 shall be met. The procedure shall be as follows:

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a. Set Group Simulator as follows:

GROUP RATE 7	7
TIMING MODE	EXT
TRANSMIT FORMAT	MSF
CHANNEL RATE	32 kbps
TYPE	1
DATA SELECT	RANDOM

b. Set the UUT's as follows:

GROUP RATE	7
LOW GROUP	ON
TIMING	STATION
CHANNEL RATE	32 kbps
CHANNEL POWER FEEDS	ON
RMT POWER OUT	ON

c. Set the MD-1026 as follows:

DPHS A RATE	7
DPHS B SOW/MOW	MOW

d. Turn power on to all UUT's and test equipment. Run the units for 48 hours, using BITE to indicate failure. The last 24 hours of burn-in must be failure free. Provisions shall be made to latch all S and SA alarms.

e. After completion of burn-in, perform the operational test procedures outlined in 4.7.13.

4.7 Electrical. (See 3.7) Unles otherwise specified all tests shall be performed at 115V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in table VI or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table VII and figure 4) shall be provided by the contractor subject to approval by the Government. The Governement shall be responsible for supply of the Government Furnished Equipment (GFE) required for test. (See table VIII). Table IX identifies the external interface connections with the test figures. Table X annotates the interconnections on the MD-1026; table XI annotates the interconnections on the TD-1233.Unless otherwise noted all resistors in test figures shall be .5 watt.

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TABLE VI. Standard test equipment.

Description	Mfg/Model No.(or equivalent)
Oscilloscope	TEKTRONIX 7904
Ammeter, dc	Weston 1931 3, 15 &30 Amps
Voltmeter, digital	HP 3440A
Frequency counter	HP 5328A
Square-wave/pulse generator	TEKTRONIX PG-505
Voltmeter, ac	HP 3400A
Supply, dc, 0-40 volts	Harrison Lab 809A
Supply, ac, variable frequency	Elgar 1001
Audio Oscillator (Qnt 2)	HP 200CD
Attenuator 50-ohm (Qty 2)	HP 355C
Supply, dc, 0-10 volts	Trygon Hr 40-500
Milliammeter (5, 20, 100, 500 mA)	Weston 1931
Attenuator	
True rms Voltmeter	Ballantine 323
Wattmeter	Weston 432
Audio Oscillator	HP 200CD
Transmission Measuring Set	HP 3555B
Frequency Synthesizer	HP 3325A
Envelope Delay Test Set	HP 4942A

TABLE VII. Special test equipment.

Name	Drawing No.	Functional Description
Group Simulator	A3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It generates either a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination. A TTL error output signal is provided for counting errors external.
Timing Source		The Timing Source schematic is shown in figure 4.
Error Inserter	A3051665	The Error Inserter provides a means of injecting a .001 to .1 continuous BER with 20% random bursts. Balanced data and Timing input and output in interfaces are provided.

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TABLE VII. Standard test equipment. (continued)

Name	Drawing No.	Functional Description
DSVT Simulator	A3051668	The DSVT simulator provides a pseudo-random generator and correlator data at 2, 16 and 32 kbps with a balanced signal interface. A conditioned diphase interface is also provided at 16 and 32 kbps. An TTL error output signal is provided for counting errors external.

Table VIII. Government furnished equipment (GFE).

Equipment	Description
Alternate (for DSVT Simulator): Generator, Digital Data SG-1139/G	As an alternate for the DSVT Simulator, the Digital Data Generator provides pseudo-random data patterns of 72 to 4,608 kb/s balanced signals and 16/32 kb/s condtioned diphase signals. It provides correlation of digital data to perform bit-error measurements.
Modem, Digital Data MD-1026(P)/G	The Digital Data Modem provides electrical signal compatibility between balanced NRZ digital signals on its equipment side and conditioned diphase signals on its line side. This equipment must be configured with conditioned diphase CCAs in all four group slots.
Modem, Cable Driver MD-1024/G	The Cable Driver Modem provides high voltage cable power at 90 ma for repeatered cable transmission.
Cable Assembly CX-11230/G	Dual coax cable in 1/4 mile reel lengths.
Cable, Telephone WF-16/U	Four wire twist pair telephone cable.

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Table IX TD-1234 interface connections.

Connector	Function	PIN LO	PIN HI	PIN Ground	Signal Type
High Group	Data In		*(M)		Diphase
	Data Out		*(F)		Diphase
Low Group	Data In		*(M)		Diphase
	Data Out		*(F)		Diphase
Channel 1-8	Data In Ch 1	B	b	L,q	Diphase
	Data Out Ch 1	F	X		Diphase
	Data In Ch 2	C	c		Diphase
	Data Out Ch 2	G	Y		Diphase
	Data In Ch 3	D	J		Diphase
	Data Out Ch 3	H	Z		Diphase
	Data In Ch 4	K	a		Diphase
	Data Out Ch 4	E	d		Diphase
	Data In Ch 5	T	e		Diphase
	Data Out Ch 5	M	h		Diphase
	Data In Ch 6	f	i		Diphase
	Data Out Ch 6	N	u		Diphase
	Data In Ch 7	R	j		Diphase
	Data Out Ch 7	P	v		Diphase
	Data In Ch 8	A	S		Diphase
	Data Out Ch 8	W	g		Diphase
Station Lock and Alarm	Station Clock In	A	B	C	Square Wave
	Alarm S	D	E	H	Alm-Close Contact
	Alarm SA	F	G		Alm-Close Contact
Power	Low(-) High (+)	C	A	B	
Order Wire Low Group	AVOW In	A	B	C	
	AVOW Out	D	E		
	AVOW Control	F	G		Note 1
Order wire High Group	AVOW In	A	B	C	
	AVOW Out	D	E		
	Avow Control	F	G		Note 1

NOTE 1. These pins connected together by COU cable.

*(M) Male, (F) Female; Type determined by center conductor.

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TABLE X MD-1026 Interface connections.

Connector	Function	Lo	HI	GND	Signal Type
LINE SIG IN 1	Line Data In #1	-	-	-	
LINE SIG OUT 1	Line Data Out #1	-	-	-	
LINE SIG IN 2	Line Data In #2	-	-	-	
LINE SIG OUT 2	Line Data Out #2	-	-	-	
LINE SIG IN 3	Line Data In #3	-	-	-	
LINE SIG OUT 3	Line Data Out #3	-	-	-	
LINE SIG IN 4	Line Data In #4	-	-	-	
LINE SIG OUT 4	Line Data Out #4	-	-	-	
GROUP 1 GROUP 2 GROUP 3 GROUP 4	Group Data In	B	A	M	NRZ
	Group Tim in	D	C	N	SQ WV
	Group Data out	F	E	P	NRZ
	Group Tim out	H	G	R	SQ WV
	BLACK Sta Clk out	K	J	S	SQ WV
ANALOG VOICE ORDERWIRE	Gr 1 AVOW in	B	A		Analog
	out	D	C		Analog
	Gr 2 AVOW in	F	E		Analog
	out	H	G	T,U	Analog
	Gr 3 AVOW in	K	J		Analog
	out	M	L		Analog
	Gr 4 AVOW in	P	N		Analog
	out	S	R		Analog
POWER		A	C	B	

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Table XI TD-1233 Interface connections.

Connector	Function	LO	HI	GND	Signal Type
Group	Data In		*(M)		Diphase
	Data Out		*(F)		Diphase
Channel 1-4	Data In Ch 1	B	b	L,q	Diphase
	Data Out Ch 1	F	x		Diphase
	Data In Ch 2	C	c		Diphase
	Data Out Ch 2	G	y		Diphase
	Data In Ch 3	D	J		Diphase
	Data Out Ch 3	H	z		Diphase
	Data In Ch 4	K	a		Diphase
	Data Out Ch 4	E	d		Diphase
Station Clock and Alarm	Station Clock	A	B	C	Square Wave
	Alarm S	D	E	H	Alm=Close Contact
	Alarm SA	F	G	H	Alm=Close Contact
Power	Hi(+), LO(-)	C	A	B	
Orderwire	AVOW In	A	B	C	
	AVOW Out	D	E		
	AVOW Control	F	G		Note 1

Note 1: These pins connected together by COU cable.

*(M) Male, (F) Female; Type determined by center conductor.

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4.7.1 Diphase group output. (See 3.7.1)

4.7.1.1 Diphase group output voltage. (See 3.7.1.1)

a. Connect the equipment as shown in figure 5.

b. Set the UUT switches as follows:

Remote Power Switch	OFF
All Loop Power Feed switches	OFF
Channel Rate switch	32 kb/s
Low Group	ON
Timing switch	MASTER
Group Rate	7

c. Measure the High Group output peak-to-peak voltage.

d. Repeat b and c for the Low Group output.

e. Verify that the requirements of 3.7.1.1 are met.

4.7.1.2 Diphase group output rise and fall times. (See 3.7.1.2)

a. Connect the equipment as shown in figure 5.

b. Set the UUT switches as follows:

Loop Power Feed switches	OFF
Channel switch	32 kb/s
Timing switch	MASTER

c. Measure the group output rise and fall time (see figure 6).

d. Verify that the requirement of 3.7.1.2 is met.

4.7.1.3 Diphase group output impedance. (See 3.7.1.3)

a. Connect the UUT group output as shown in figure 7 with R1 =57.6 ohms.

b. Remove Timing Generator card, A10. (Removal of Timing Generator CCA causes S alarm and turns off output data).

c. Set the UUT switches as follows:

Loop Power Feed switches	OFF
Timing switches	MASTER

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d. Set the oscillator to 1 kHz and its output voltage to 2.00 volts as measured with the voltmeter connected to VI of the 57.6 ohm termination. Then measure V2.

e. Calculate the impedance using

$$Z = 57.6\frac{V2}{VL-V2} \text{ ohms.}$$

Record the value.

f. Repeat d and e with the oscillator set to 300 kHz.

g. Connect the oscillator to Group Data Out through a 100 ohm, resistor, set it for 300 Hz and set its output voltage to 2.00 volts.

h. Calculate the impedance using

$$Z = 100\frac{V2}{V1-V2} \text{ ohms.}$$

Record the value.

i. Repeat g and h with the oscillator set to 1 kHz.

j. Re-insert the Timing Generator CCA in UUT location A10.

k. Verify that the requirements of 3.7.1.3 are met.

4.7.2.1 Diphase group input impedance. (See 3.7.2.1)

a. Connect the UUT group input as shown in figure 7 with R1 = 57.6 ohms.

b. Set the oscillator to 1 kHz and its output voltage to 2.0 volts as measured with the voltmeter connected to V1. Then measure V2.

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c. Calculate the impedance using

$$Z = 57.6 \frac{V2}{VL-V2} \text{ ohms.}$$

Record the value.

d. Repeat b and c with the oscillator set to 300 Hz.

e. Connect the oscillator to Group Data Out through a 100 ohm resistor and set it for 300 Hz and its output voltage to 2.0 volts.

f. Calculate the impedance using

$$Z = 100 \frac{V2}{V1-V2} \text{ ohms.}$$

Record the value.

g. Repeat e and f with the oscillator set to 1 kHz.

h. Verify that the requirements of 3.7.2.1 are met.

4.7.2.2 Received Signal Level. (See 3.7.2.2)

a. Connect the equipment as shown in figure 8 and set the attenuator to 4 dB attenuation. Set the oscillator to 21.3 dBm at 1 kHz.

b. Set the UUT switches as follows:

Loop Power Feed switches	OFF
Channel Rate switch	32 kb/s
Group Rate	7
Timing switch	MASTER
Low Group	ON

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c. Set the Auxiliary TD-1234 for 32 kb/s Channel Rate, Loop Rate 5 Timing and all Power Feed switches OFF.

d. Measure the bit error rate on digital channels for 30 seconds.

e. Reduce all cable lengths to zero(minimum) (not counting adapters) and eliminate the attenuators.

f. Repeat d.

g. Verify that the requirements of 3.7.2.2 are met.

4.7.3 Station clock input. (See 3.7.3)

4.7.3.1 Receiver dynamic range . (See 3.7.3.1)

a. Connect the equipment as shown in figure 9.

b. Set the UUT for 32 kb/s Channel Rate, Master Timing, Rate 7, Low Group ON, and all Power Feed switches OFF and set the timing source to 576 kHz.

c. Adjust the attenuator to provide a $\pm$50 mV peak voltage to the Station Clock input.

d.Verify that the group output waveform is locked to the Station Clock input.

4.7.3.2 Input impedance. (See 3.7.3.2)

a. Connect the equipment as shown in figure 10.

b. Set the UUT switches as shown in 4.7.3.1.

c. Adjust the oscillator at a frequency of 1 kHz for an output level of 1 Volt RMS (V_1). Then measure the V_2 voltage.

d. Calculate the input impedance using the formula:

$$Z_{IN} = 100 \frac{V2}{V1} \text{ ohms}$$

e. Verify that the requirements of 3.7.3.2 are met.

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4.7.3.3 Timing modes. (See 3.7.3.3)

a. Loopback mode. Connect the equipment as shown in figure 11.

b. Set the UUT to 32 kb/s Channel, Rate, Loopback Timing, Loop Power Feed switches OFF and Power ON.

c. Set frequency synthesizer to 576.029 kHz.

d. Verify that Group Data In and Timing from I/O pin 23 of Timing Generator CCA A15 are in sync. Verify that there is no LED fault to CCA A15.

e. Set frequency synthesizer to 575.971 kHz and repeat step d.

f. Master mode. Switch the UUT Timing to Master (up position) on CCA A16. Loopback to the group output. Measure the frequency at pin 23 of the Timing Generator CCA (A15). Verify that the frequency is within the range of 575.977 kHz to 576.023.

g. Station clock mode. This test is performed as part of the receiver sensitivity test as described in 4.7.3.1.

4.7.4 Orderwire interface test. (See 3.7.4)

4.7.4.1 Input impedance. (See 3.7.4.1)

a. Connect the UUT as shown in figure 12 using the high group.

b. Set the oscillator to 2 Vrms at a 1 kHz rate.

c. Measure and record V1 and V2.

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d. Calculate the input impedance from:

$$Z_{in} = 600 \frac{V_2}{V_1}$$

e. Repeat b through d for the Low Group.

f. Verify that the requirements of 3.7.4.1 are met.

4.7.4.2 Output impedance. (See 3.7.4.2)

a. Connect the UUT as shown in figure 12 using the high group.

b. Set the oscillator to 2 Vrms at 1 kHz rate.

c. Measure and record V_1 and V_2.

d. Calculate the output impedance from:

$$Z_{in} = 600 \frac{V_2}{V_1}$$

e. Repeat b through d for the Low Group.

f. Verify that the requirements of 3.7.4.2 are met.

4.7.4.3 Receive Levels. (See 3.7.4.3) Done as part of 4.7.7.6.

4.7.4.4 Bandwidth. (See 3.7.4.3)

a. Connect the output of the oscillator to the UUT High group.

b. Orderwire as shown in figure 13. Loop back the High Group.

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c. Adjust the oscillator to 21.3 dBm at 1000 Hz.

d. Measure and record the signal level appearing at the orderwire output.

e. Repeat c through d with the oscillator set at 300, 400, 3000 and 3400 Hz.

f. Repeat a through e for the Low Group.

g. Verify that the requirements of 3.7.4.4 are met.

4.7.4.5 Signal-to-noise ratio. (See 3.7.4.5)

a. Connect the equipment as shown in figure 14.

b. Set the UUT as follows:

All Power Feed switches:	OFF
Channel Rate switch:	16 kb/s
Group Rate:	3
Timing switch:	MASTER
Low Group:	ON

c. Set the Group Simulator as follows:

Group Rate:	1
Timing Mode:	LOOP
Transmit Format:	MSF
Type:	2
Channel Rate:	16 kb/s
DATA Select:	RANDOM

d. Set the MD-1026 as follows:

DPHS A RATE.	1 (72 kb/s)
DPHS B SOW/MOW:	MOW

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e. Set the Noise measuring set for

600 Ohm termination

C- Message weighted

f. Measure the noise in dBm with the test oscillator OFF, then measure the signal with the oscillator set to -4 dBm at 1000 Hz.

g. The Signal-to-Noise ratio is dBm (oscillator ON) minus the dBm (oscillator OFF).

h. Repeat e through g with the Group Simulator set for Group Rate 3 and the UUT set for Channel Rate 32 kb/s and Group Rate 5.

i. Verify that the requirements of 3.7.4.5 are met.

4.7.4.6 Transmit level. (See 3.7.4.6)

a. Connect the equipment as shown in figure 15 and remove Timing Generator CCA A15.

b. Turn the unit on, set the oscillator for 572 mV at 1 kHz (58 ohms).

c. Measure the output in dBm (58 ohms).

d. Verify that the active orderwire requirements of 3.7.4.6 are met.

e. Repeat b through d with the High and Low Groups interchanged.

f. Turn the UUT OFF.

g. Connect the equipment as shown in figure 15(B).

h. Set the oscillator output to +21.3 dBm at 1 kHz.

i. Measure the output in dBm (58 ohms).

j. Repeat g through u for the Low Group.

k. Verify that the requirements of 3.7.4.6 are met.

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4.7.5 Digital channel characteristics. (See 3.7.5)

4.7.5.1 Output voltage. (See 3.7.5.1)

a. Connect the equipment as shown in figure 16.

b. Set the DSVT simulator to 32 kb/s rate all zeroes. Set the UUT to a 32 kb/s channel rate, Group Rate 5, Low Group OFF, and MASTER timing. All power feeds set to OFF.

c. Connect the DSVT simulator to the first digital channel input. Measure the peak-to-peak channel output voltage as shown in figure 6.

d. Verify that the requirements of 3.7.5.1 are met.

e. Repeat c and d for the remaining digital channels.

4.7.5.2 Output waveshape. (See 3.7.5.2)

a. The equipment connections are the same as in 4.7.5.1.

b. Set the DSVT simulator and UUT to 16 kb/s rate all zeroes. Set the UUT to 16 kb/s channel rate, Group Rate 3, Low Group OFF, and MASTER timing. All power feeds set to OFF.

c. Connect the DSVT simulator to the first digital channel input. Measure the channel output rise and fall times and droop. Refer to figure 6 for a definition of these characteristics.

d. Repeat c and d for the remaining digital channels.

e. Verify that the requirements of 3.7.5.2 are net.

4.7.5.3 Input and output impedance. (See 3.7.5.3)

a. Connect the equipment for the input test setup as shown in figure 17.

b. Set the UUT for 32 kb/s channel rates, MASTER Timing, Group Rate 5, Low Group OFF, and all Power Feed switches to OFF.

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c. With the input setup connected to the first digital channel, adjust the oscillator for 125 mV at a frequency of 8 kHz (V_1). Then measure the V_2 voltage.

d. Calculate the input impedance using

$$Z_{in} = 125 \frac{V_2}{V_1} \text{ ohms}$$

e. Repeat c and d with the oscillator frequency set to 16, 32 and 56 kHz.

f. Repeat c through e for each remaining digital channel.

g. Connect the output test setup to the first digital channel as shown in figure 16.

h. With the output setup connected to the first digital channel output, measure the output voltage V_1 with the switch (S1) open, then measure the V_2 voltage with the switch (si) closed.

i. Calculate the impedance using:

$$Z_{out} = 124 \frac{V_1}{V_2} - 1 \text{ ohms}$$

j. Repeat h and i for each remaining digital channel.

k. Verify that the requirements of 3.7.5.3 are met.

4.7.5.4 Input voltage test. (See 3.7.5.4)

a. Connect the equipment as shown in figure 18.

b. Set the DSVT simulator for 32 kb/s and adjust the Line Simulator for a 0.150V peak-to-peak signal with switch set to LOW and connected to first Digital Channel RCV terminals. Set UUT for 32 kb/s Channel Rate, Master Timing, Group Rate 5, Low Group OFF, and all Power Feeds OFF.

c. Monitor the DSVT Simulator error detector for a minimum of 30 seconds. Verify that there are no errors generated.

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d. Repeat b and c for each remaining Digital Channels.

e. Set the Line Simulator switch to HIGH and adjust the signal for a 4.0V peak-to-peak level at the first digital channel RCV terminals.

f. Monitor the DSVT Simulator error detector for a minimum of 30 seconds. Verify that there are no errors generated.

g. Repeat e and f for each remaining Digital Channel.

h. Verify that the requirement of 3.7.5.4 have been met.

4.7.5.5 Idle Loop. (See 3.7.5.5)

a. Connect equipment as shown in figure 20. Set the MD-1026 to Rate 5 and order wire to MOW.

b. Set UUT to 32 kb/s channels Group Rate 5, Low Group OFF, Master Timing and all Power Feeds OFF.

c. Verify that the Group channel bits are all digital ones at the MD-1026 Group data output. Refer to figure 28 for the Frame Format.

d. Disconnect the CX-11230 cable.

e. Verify that the channels 1 through 8 outputs are diphase ones (16 kHz square wave).

4.7.6 Loop Power Feed. (See 3.7.6)

a. Connect the Loop Power Feed Load Circuit to the first Digital Channel as shown in figure 21.

b. Set S1 to A and enable the first Digital Channel Power Feed switch on the UUT. Then measure and record the current (1).

c. Set S1 to B and measure the voltage (V), Verify that the voltage is less than 56 V.

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d. Set S1 to A and momentarily short either of the transmit lines to either receive line.

e. Verify that the current(I) returns to at least 68.8 mA.

f. Turn OFF the Power Feed switch on the UUT.

g. Connect the loop power surge test setup as shown in figure 22 to the first Digital Channel.

h. Turn ON the Power Feed switch and adjust the pulse generator pulse width to 45 ±1 ms. Verify that the loop power feed does not trip.

i. Repeat steps a through f for each remaining Digital Channel.

j. Verify that the requirements of 3.7.4 are met.

4.7.7 Analog channel characteristics. (See 3.7.7)

4.7.7.1 Input/output impedance and return loss. (See 3.7.7.1)

a. Connect the equipment as shown in figure 23. Do not connect the setup to the UUT.

b. Set the oscillator to 300 Hz and adjust the output for a V1 voltage of 0 dBm. Record the V2 (set) voltage in dBm (Z test = 00).

c. Connect the setup to the first Analog Channel input, reset the V1 voltage to 0 dBm and record the V2 (measured) voltage in dBm.

d. Repeat steps a through c for each remaining Analog Channel input.

e. Repeat steps a through d with oscillator frequency set to 3400 Hz.

f. Repeat steps a through e measuring each Analog Channel output.

g. Calculate the return loss, V2 (measured) minus V2 (set), in dB, and verify that it is not less than 26 dB.

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4.7.7.2 Longitudinal balance. (See 3.7.7.2)

a. Set up the bridge as shown in figure 24.

b. Check the bridge balance at 300 and 3400 Hz without connecting the UUT. It must be greater than 60 dB.

c. Connect the bridge to the first Analog Channel input.

d. Set V1 to 1 Vrms at 300 Hz. Then measure the V2 voltage.

e. Repeat d with the frequency set to 1000 Hz and 3400 Hz.

f. Repeat d and e with the bridge connected to the first Analog Channel output.

g. Repeat c through f for each of the remaining Analog Channels.

h. Calculate the balance using:

$$LB = 20 \log \frac{V1}{V2}.$$

i. Verify that the LB is not less than 40 dB for each measurement.

4.7.7.3 Frequency response. (See 3.7.7.3) Frequency response requirement is demonstrated as part of the individual CCA TRS testing performed on 100% of the analog CCAs.

4.7.7.4 Analog channel input and output test tone. (See 3.7.7.4) The test tone level and frequency shall be demonstrated as part of the individual CCA TRS tests performed on 100% at the analog CCAs.

4.7.7.5 Envelope delay distortion. (See 3.7.7.5)

a. Connect the equipment as shown in figure 25.

b. Set the Analog Channel and Output/Input Level switches to 0dB and the Channel Loopback switches to LOOPBACK. Set the channel rate to 32 kbps, Group Rate 5, Timing switch to MASTER, and Low Group OFF.

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c. Measure the envelope delay distortion of the input and output filters at 800 Hz and at 3000 Hz.

d. Repeat a and b for the remaining Analog Channels.

e. Verify that the delay is less than 100 us for each measurement.

4.7.7.6 Input-output linearity. (See 3.7.7.6) The linearity requirement is demonstrated as part of the individual CCA TRS test performed on 100% of the CCAs.

4.7.7.7 Loopback switch. (See 3.7.7.7) Operation of loopback switches is demonstrated as part of 4.7.7.5 (for loop mode) and 4.7.7.8.1 (for normal mode).

4.7.7.8 Noise. (See 3.7.7.8)

4.7.7.8.1 Idle channel noise. (See (3.7.7.8.1)

a. Connect no equipment as shown in figure 26 for idle channel noise.

b. Set the Analog Channel Input and Output Level switches to 0 dB and the Channel Loopback switches to NORMAL. Set the Channel Rate to 32 kb/s, Group Rate 5, Timing switch to MASTER, and Low Group OFF.

c. Set the noise measuring set to 600 ohms terminated and C-message weighting.

d. Measure the channel 1 output noise.

e. Repeat a through d for each remaining Analog Channel.

f. Verify that the noise measured does not exceed 33 dBrnC.

4.7.7.8.2 Crosstalk. (See 3.7.7.8.2) For this test, the UUT must contain 4 Analog Channels.

a. Connect the equipment as shown in figure 26.

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b. Set the UUT input level switches to -10 dB, the Output Level Switches to 0 dB, Channel Loopback switches to NORMAL, Channel Rate to 32 kHz, Group Rate 5, Station Clock Timing, and Low Group OFF.

c. With the oscillator connected to Channel 1, adjust the oscillator at 820 Hz to produce an output voltage V1 of +6 dBm. (This gives a channel input of 0 dBm at 820 Hz).

d. Set the transmission measuring set at 600 ohms termination, C-message weighted. Measure the voltage at Channel 2 Out.

e. Repeat Step c and d for Channel outputs 3 through 8.

f. Verify that the voltages do not exceed 34 dBrnC.

g. Connect the equipment as shown in figure 27. Set the Auxiliary TD-1234 to Loop Timing and the UUT to MASTER Timing.

h. With the oscillator connected to Analog Channel 1, adjust the oscillator at 820 Hz to produce an output voltage V_1 of +6 dBm. (This gives a channel input of 0 dBm at 820 Hz).

i. Set the transmission measuring set to 600 ohms termination, C-message weighted. Measure the voltage at Channel 1 OUT. Verify that the voltage does not exceed 34 dBrnC.

4.7.7.9 Peak clipping. (See 3.7.7.9)

a. Connect the equipment as shown in figure 28.

b. Set the Analog Channel Input Level switch to -10dB, and the Output Level switch to +10 dB. Set the Channel Loopback switch to NORMAL the Channel Rate switch to 32 kHz, MASTER Timing, and Group Rate 5.

c. With the oscillator connected to the first Analog Channel input, set the oscillator output at 200 Hz for +10 dBm, (this gives a Channel input of +4 dBm).

d. Measure total distortion.

e. Repeat a through d for each Analog Channel.

f. Verify that the total distortion is less than 10% for each channel.

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4.7.7.10 Full load tone. (See 3.7.7.10) The full load tone is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.11 Analog channel signal processing. (See 3.7.7.11)

4.7.7.11.1 Clocking rate. (See 3.7.7.11.1) Clocking rate is demonstrated as part of 4.7.7.11.2.

4.7.7.11.2 Dynamic range. (See 3.7.7.11.2) The dynamic range is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.11.3 Output level control. (See 3.7.7.11.3) The output level control is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.7.11.4 Input level control. (See 3.7.7.11.4) The input level control is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.8 Framing. (See 3.7.8)

4.7.8.1 Format. (See 3.7.8.1) The unit output group patterns and overhead channels will be verified by examination of the group output data displayed on the oscilloscope. Channel positions will be verified by sequentially applying a diphase all zero pattern. Then, observing that the corresponding bit position in the group signal responds to the zeros, the channels will be located. The telemetry subchannel positions will be verified by applying PRG to the first overhead subchananel (the framing subchannel counts as zero). Set all digital channel Power Feed to OFF. For this test, configure the UUT with 8 digital channels.

a. Connect the equipment as shown in figure 29, configuration A.

(1) Set the UUT and DSVT Simulator as shown in table XII for Test 1.

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Table XII. Format test setup.

UUT SETTINGS						GROUP SIMULATOR						DSVT SIMULATOR	
TEST	CHANNEL RATE	GROUP RATE	LOW GROUP	POWER FEEDS	TIMING	GROUP RATE	TIM-ING	FORMAT	CHANNEL RATE	DATA	TYPE	RATE	DATA
1	32 kb/s	5	OFF	OFF	MASTER			OFF				32 kb/s	Zeroes
2	16 kb/s	3	OFF	OFF	MASTER			OFF				16 kb/s	Zeroes
3	32 kb/s	4	OFF	OFF	MASTER			OFF				32 kb/s	Zeroes
4	16 kb/s	2	OFF	OFF	MASTER			OFF				16 kb/s	Zeroes
5	32 kb/s	5	ON	OFF	MASTER				OFF			32 kb/s	Zeroes
6	16 kb/s	3	ON	OFF	MASTER				OFF			16 kb/s	Zeroes
7	32 kb/s	4	ON	OFF	MASTER				OFF			32 kb/s	Zeroes
8	32 kb/s	7	ON	OFF	MASTER				OFF			32 kb/s	Zeroes
9	16 kb/s	5	ON	OFF	MASTER				OFF			16 kb/s	Zeroes
10	32 kb/s	6	ON	OFF	MASTER				OFF			32 kb/s	Zeroes
11	16 kb/s	4	ON	OFF	MASTER				OFF			16 kb/s	Zeroes
12	32 kb/s	7	ON	OFF	LOOP	7	INT	Sub Chnl	32 kb/s	RAN-DOM	TYPE 1	N/A	
13	32 kb/s	5	ON	OFF	LOOP	3	INT	Sub Chnl	32 kb/s	RAN-DOM	TYPE 2	N/A	
14	32 kb/s	7	ON	OFF	LOOP	5	INT	Sub Chnl	32 kb/s	RAN-DOM	TYPE 1	N/A	

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Table XIII. TD-1234 High Group and format.

TEST	UUT CHNLS	LOW GROUP NO OF CHNLS	LOW GROUP INPUT GROUP RATE	HIGH GROUP NO OF CHNLS	HIGH GROUP OUTPUT GROUP RATE	OUTPUT FORMAT
						A= UUT CHANNEL INPUT SP= SPARE (SET TO DIGITAL ONE) B= CHANNEL INPUT LOW GROUP O= OVERHEAD
1-2	8	-	-	9	288	0 A1 A2 A3 A4 A5 A6 A7 A8 0
3-4	7	-	-	8	256	0 A1 A2 A3 A4 A5 A6 A7 0
5-6	4	4	144	9	288	0 A1 B1 A2 B2 A3 B3 A4 B4 0
7-	3	4	144	8	256	0 B1 A1 B2 A2 B3 A3 B4 0
8-9	8	8	288	18	576	0 SP A1 B1 A2 B2 A3 B3 A4 B4 A5 B5 A6 B6 A7 B7 A8 B8 0
10-11	8	7	256	16	512	0 A1 B1 A2 B2 A3 B3 A4 B4 A5 B5 A6 B6 A7 B7 A8 0

NOTE: Group Rates shown for a 32kb/s channel rate. Rates are half for 16kb/s channel rate.

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(2) Observe the data displayed on the oscilloscope and locate the Framing Bit. Successively connect the DSVT Simulator to Channels 1 through 8.

(3) Verify that the channel locations are as shown in Table XIII Test 1-2.

(4) Repeat steps (1) through (3) for Test 2 as shown in Table XII.

(5) Set the UUT, Group Simulator, and DSVT Simulator as shown in table XII for Test 3.

(6) Observe the data displayed on the oscilloscope and locate the framing channel. Successively connect the DSVT Simulator to Channels 1 through 7.

(7) Verify that the channel locations are as shown in table XIII Test 3-4.

(8) Repeat (5) through (7) for Test 4 as shown in table XII.

b. Connect the equipment as shown in figure 29, configuration B with a TD-1233 as the auxiliary unit.

(1) Set the UUT, Group Simulator and DSVT Simulator as shown in table XII, Test 5. Set the TD-1233 for looped timing and channel rate to 32 kbps.

(2) Observe the data displayed on the oscilloscope and locate the framing channel. Successively connect the DSVT Simulator to UUT channels 1 through 4, and the TD-1233 channels 1 through 4.

(3) Verify that the channel locations are as shown in table XIII Test 5-6.

(4) Repeat steps (1) through (3) for Test 6 as shown in table XII with TD-1233 channel rate set to 32 kbps.

(5) Set the UUT, and DSVT Simulator as shown in table XII for test 7. Set TD-1233 channel rate to 32 kbps.

(6) Observe the data displayed on the oscilloscope and locate the framing channel. Successively connect the DSVT Simulator to UUT channel 1 through 3, and the TD-1233 channels 1 through 4.

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(7) Verify that the channel locations are as shown in table XIII, Test 7.

c. Connect the equipment as shown in figure 29 configuration B with a TD-1234 as the auxiliary unit.

(1) Set the UUT, and DSVT Group Simulator as shown in table XII for Test 8. Set the auxiliary TD-1234 for looped timing and channel rate 32 kbps Low Group Group Rate 5.

(2) Observe the data displayed on the oscilloscope and locate the framing channel. Successively connect the DSVT Simulator to channels 1 through 8, and the auxiliary TD-1234 channels 1 through 8.

(3) Verify that the channel locations are as shown in table XIII, Test 8 through 9.

(4) Repeat (1) through (3) for Test 9 as shown in table XII with auxiliary TD-1234 channel rate set to 16 kbps and group rate to 3.

(5) Set the UUT, and DSVT Simulator as shown in table XII for Test 10. Set the auxiliary TD-1234 channel rate to 32 and group rate to 4.

(6) Observe the data displayed on the oscilloscope and locate the framing channel. Successively connect the DSVT Simulator to UUT channels 1 through 8, and the auxiliary TD-1234 channels 1 through 7.

(7) Verify that the channel locations are as shown in table XIII, test 10-11.

(8) Repeat steps 5 through 7 for Test 11 as shown in table XII. Set the auxiliary TD-1234 channel rate to 16 kbps and group rate 2.

d. Connect the equipment as shown in figure 30, configuration c. Set the Group Simulator to provide random data activity in subchannel 5. Loopback the Low Group. Set the UUT and Group Simulator as shown in table XII, Test 12.

(1) Verify that there are no errors for 30 seconds.

(2) Verify the location of subchannel 5 as shown in figure 31.

(3) Repeat (1) through (2) for subchannels 6 and 7.

e. Connect the equipment as shown in figure 30, configuration D Set the UUT and Group Simulator as shown in table XII test 13.

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(1) Supply random data to subchannel 5. Verify that there are no errors for 30 seconds. Verify a TD-1234 overhead format.

(2) Set the UUT Group Simulator as shown in table XII Test 14, supply random data to subchannels 5,6 and 7 one at a time and verify no errors for 30 seconds.

(3) Verify that the overhead formats are as shown in figure 31.

4.7.8.2 Frame acquisition time. (See 3.7.8.2)

a. Connect the equipment as shown in figure 32. Configure both units for 32 kb/s Channel Rate. Set the UUT to MASTER Timing and the Auxiliary Unit to LOOPED Timing. Group 7, Low Group ON. Set the MD-1026 to rate 7, MOW.

b. Set up the error inserter for 0.1% BER continuous operation.

c. Squelch the data at the error inserter to cause a loss of frame; release squelch to restore data.

d. Measure the time from the resumption of received group data on error inserter to the resumption of data at I/O pin 48 on the group framing card, CCA A14. Use the universal counter, take 50 measurements and verify that no more than 5 of the 50 measurements exceed 50 ms at 32 kb/s or 100 ms at 16 kb/s.

e. Set the error inserter for 20% BER random bursts interspersed with 0.1% BER ramdom error intervals.

f. Repeat step d, verify that no more than 5 of 50 measurements exceed 100 ms at 32 kb/s or 150 ms at 16 kb/s.

g. Set the MD-1026 to MOW and both units for 16 kbps channel rate.

h. Repeat steps b through f.

4.7.8.3 Frame maintenance. (See 3.7.8.3)

a. The equipment setup is the same as shown in figure 32. Configure both units for 32 kb/s channel rate Group Rate 7. Set the UUT to MASTER Timing, the Auxiliary Unit for LOOPED Timing, and the MD-1026 to Rate 7, MOW.

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b. Connect the event counter to TP6 on the group framing cards (CCA A14) and adjust it to trigger on the positive edge of the signal. Squelch the data, this will cause the UUT to reframe. The counter should advance by one. Reset the counter.

c. Set the error inserter for 20% BER, continuous error rate.

d. Run the unit for a period of 12 hours. Verify that no declarations of Out-of-Sync are generated, as indicated by a count of zero on the counter.

e. Verify that the requirements of 3.7.8.3 are met.

4.7.8.4 Detection of Out-of-Sync. (See 3.7.8.4)

a. The equipment setup is the same as shown in figure 32.

b. Set the error inserter to no errors.

c. Use the data squelch signal (scope sync) at the error inserter to start the time interval measurement and use the signal at TP3 on the group framing card to stop the counter.

d. Squelch the data, and measure the time from loss of data to out-of-frame indication.

e. Verify that no more than 5 of 50 measurements exceed 150 ms.

4.7.8.5 Idle filler pattern. (See 3.7.8.5) The Idle Filler Pattern requirements are done as part of the Idle Loop Tests, 4.7.5.5.

4.7.8.6 False frame lock. (See 3.7.8.6)

a. Connect the equipment as shown in figure 33.

b. Set the DSVT simulator to 32 kbps, random data. Set the UUT to 32 kb/s Channel Rate, Group Rate 5, Low Group ON and Master Timing. Set the TD-1233 to 32 kb/s Channel Rate and Loop Timing.

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c. Monitor the DSVT Simulator, verify the presence of activity with no errors.

d. Turn the TD-1233 on and off 50 times, verify channel activity and no errors each time the TD-1233 is on.

e. Repeat steps c and d for Channels 2 through 4.

4.7.9 Prime power interface. (See 3.7.9)

a. Connect the equipment as shown in figures 34 and 35.

b. Set the DSVT simulator to 32 kbps, random data. Set the UUT to a 32 kbps channel rate, Group rate 7, Low Group OFF, MASTER Timing, and all power feeds OFF.

c. Connect the Digital Channels to the Loop Power Feed Load Box, such that the DSVT input and output is connected to channel 1 through 8.

d. Turn on the Digital Channel Power Feed switches.

e. For each combination of input voltage and frequency given in table XIV measure the input power and monitor the Digital Channel Errors over a 30 second period.

Table XIV. Prime power interface input voltage and frequency combinations

Input Voltage	Frequency
115 Vac	60 Hz
103.5 Vac	60 Hz
126.5 Vac	60 Hz
103.5 Vac	47.5 Hz
103.5 Vac	420 Hz
126.5 Vac	47.5 Hz
126.5 Vac	420 Hz
22 Vdc	
28 Vdc	
32 Vdc	

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4.7.10 Current protection. (See 3.7.10)

a. Connect the equipment as shown in figure 36 (A).

b. With the UUT ac power turned OFF connect the MD-1024 Modem, cable Driver Line signal out to the High Group data input of the UUT.

c. Turn on the MD-1024 Power Feed for 1 minute.

d. Verify that there is no physical damage to the UUT.

e. Repeat c through e for the High Group data output.

f. Connect the equipment as shown in figure 36(B). Set the DSVT simulator to 32 kbps, random data. Set the UUT to a 32 kbps channel rate, Group Rate 7, Low Group ON, MASTER Timing, and all power feeds OFF.

g. Verify that there are no errors for 30 seconds on either DSVT Simulators.

h. Repeat b through g with the MD-1024 connected to the Low Group of the UUT as shown in figure 36 (A).

i. Verify that the requirements of 3.7.10 are met.

4.7.11 Power feed. (See 3.7.11)

4.7.11.1 Power feed output over-voltage. (See 3.7.11.1)

a. Connect the equipment as shown in figure 37.

b. Set the UUT Power Feed ON.

c. Measure the Power Feed voltage and current through the 600 ohm load.

d. Verify that the voltage is within the limits of 3.7.11.1.

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4.7.11.2 Current limits. (See 3.7.11.2)

a. Connect the UUT Low Group Data Output as shown in figure 37.

b. Set the Load Select Switch S1 to position 2. Turn on UUT. Turn on the remote power out switch, verify that the Low Group power feed voltage is between 170 and 190 Vdc.

c. Set Load Select Switch S1 to position 1. Verify that the RMT Power Out indicator extinguishes high and low group power feed voltage turns off.

d. Set Load Select switch S1 to position 2 and reset UUT remote power out switch. Verify that the low group power feed voltage is on.

e. Set Load Select Switch to position 3. The RMT power out indicator should remain on. Verify that the power feed current is between 10 and 20 mA.

f. Set Load Select switch S1 to position 4. Verify that the RMT power out indicator extinguishes and the low group power feed voltage turns off.

g. Repeat b through f for low group data input signal.

h. Verify that the requirements of 3.7.11.2 are met.

4.7.12 Built-in Test Equipment (BITE) and Alarms. (See 3.7.12)

a. Connect the equipment as shown in figure 38.

b. Set the UUT Power Feed switches OFF, the Channel Rate switches to 32 kb/s, MASTER Timing, Group Rate 7, and Low Group ON. Do not connect the Analog Channel inputs. Set the Group Simulator to Rate 5, Channel Rate 32, Type 1, MSF RANDOM. Set the Group Modem to Rate 5, and orderwire to MOW.

c. Connect the unit to 115 Vac Power and turn the unit on. Verify that the following BITE conditions are met.

S must stay open.
SA must stay open.
Summary Fault Lamp must be OFF.
Power Supply Out-of-Tolerence Lamp must be OFF
All CCA LEDs must be OFF.

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d. Press the SUMMARY FAULT switch on the UUT. Verify that the BITE requirements listed below are met.

S must stay closed.
SA must stay closed.
Summary Fault Lamp illuminates.
Power Out of Tolerance is ON.
BITE indicators on each of the CCAs illuminate in a sequential order during self-test sequence.
All indicators are OFF at the completion of the self-test sequence.

e. Turn OFF the power at the source. Verify that the following BITE requirements are met.

S must stay closed.
SA must stay closed.

f. Turn off the UUT power switch. Verify that the following BITE requirements are met.

S must stay closed.
SA must stay open.

4.7.13 Operational check. (See 3.7.13)

a. The equipment setup is the same as in 4.7.12.

b. Set all Analog Channel Loop Back switches to normal and level switches to 0. Set all Digital Loop Power Feed switches ON,

c. Set the UUT for 32 kb/s Channel Rate, Group Rate 7, Low Group ON, and MASTER Timing and loopback the High Group signal.

d. Set the Group Simulator to Group Rate 5, Type 1 MSF, RANDOM and 32 kb/s channel rate. Set the Group Modem to Rate 5 and orderwire to NOW.

e. Set the Orderwire Oscillator output to +21.3 dBm at a frequency of 1600 Hz.

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4.7.13.1 Digital data error check. (See 3.7.13.1)

a. Monitor the DSVT Error Detector for a 30 second period.

b. Verify that there are no errors.

4.7.13.2 Low Group data error check. (See 3.7.13.2) Monitor the Group Simulator error detector for a 10 second period. Verify that there are no errors.

4.7.13.3 Maintenance OW output check. (See 3.7.13.3) Measure the OW OUT with the transmission measuring set. Verify that the requirements of 3.7.13.2 are met.

4.7.13.4 DC voltage check. (See 3.7.13.4)

a. Measure the DC voltage on each Digital Line at the Power Load Box.

b. Repeat a using 28 Vdc power.

c. Verify that the requirements of 3.7.13.4 are met.

4.7.13.5 Sync frame check. (See 3.7.13.5) Monitor the DSVT Error Detector while turning the unit off and on. When the power is turned on there shall be no errors or alarms indicating that the unit has reframed.

4.7.13.6 Power feed voltage check. (See 3.7.13.6)

a. Measure the power feed on the low group. Verify it is within the required specification in 3.7.13.6.

b. Repeat a using 28 Vdc power.

c. Verify that the requirements of 3.7.13.6 are met.

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4.7.13.7 Analog channel check. (See 3.7.13.7)

a. Connect the Analog Channel input to the first Analog Channel through 600 ohms. Set the oscillator at 820 Hz for an output of -11 dBm.

b. Connect the Analog Channel output to the voltmeter terminated in 600 ohms.

c. Measure the first channel in dBm. Verify that the requirements of 3.7.13.5 are met.

d. Repeat a through c for each remaining Analog Channel.

4.7.14 Environmental operational test. (See 3.7.14)

a. Connect the equipment as shown in figure 39.

b. Set the UUT Analog Channel Loop Back switches to Normal and Level switches to 0. Set the UUT to MASTER Timing and Auxiliary Unit to Looped Timing.

c. Set the UUT and Auxiliary Unit for 32 kb/s Channel Rate , Group Rate 7, Low Group ON.

d. Set the Group Simulator for Group Rate 5, Type 1, MSF, Random and 32 kb/s channel rate.

e. Connect the DSVT Simulator to the Looped Digital Channels on the Auxiliary Unit.

f. Set the audio oscillator output to -21.3 dBm at a frequency of 1600 Hz.

g. For all tests except temperature, connect the UUT and the Auxiliary Unit to 115 Vac power. For temperature, connect the UUT to 28 Vdc power.

4.7.14.1 Digital channel high group check. (See 3.7.14.1) Monitor the Digital Channel Error Detector for a 30 second period. Verify that there are no errors.

4.7.14.2 Low group data error check. (See 3.7.14.2) Monitor the Group Simulator Error Detector for a 10 second period. Verify that there are no errors.

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4.7.14.3 Maintenance OW output check. (See 3.7.14.3) When the oscillator set to 21.3 dBm at 1600 Hz, measure the OW OUT with the transmission measuring set. Verify that the requirements of 4.7.14.3 are met.

4.7.14.4 DC voltage check. (See 3.7.14.4) Measure the DC voltage on each digital line at the Power Feed Load Box. Verify that the requirements of 3.7.14.4 are met.

4.7.14.5 Sync frame check. (Seee 3.7.14.5) Monitor the UUT Summary Alarm while turning off the Auxiliary Unit. The UUT shall generate a Summary Fault Alarm. Turn on the power to the Auxiliary Unit. The UUT Summary Alarm shall extinguish, indicating that the unit has reframed. Verify that the requirements of 3.7.13.5 are met.

4.7.14.6 Power feed voltage check. (See 3.7.14.6) Measure the power feed on the low group. Verify it is within the required specification in 3.7.14.6.

4.7.14.7 Analog channel check. (Seee 3.7.14.7) For this test the analog channel gain in both directions, from Auxiliary RMC to UUT to Auxiliary Unit will be checked.

a. Connect the oscillator at 820 Hz for termination in 600 ohms to the first analog channel on the UUT.

b. Measure the first analog out channel on the Auxiliary RMC with the voltmeter terminated in 600 ohms.

c. Repeat a and b in the other direction in Auxiliary Unit Chl and out UUT CHl.

d. Repeat a through c for each remaining analog channel.

e Verify that the requirements of 3.7.14.7 are met.

4.8. Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

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4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2, Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +51.7°C (+125°F). Maintain internal chamber temperature for six hours at +51.7°C (+125°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +51.7°C (+125°F) within a time period of one hour. Repeat the above steps two additional times to make a total of three 12-hour cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +71.1°C (+160°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

4.8.1.2 Low temperature. (See 3.7.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C (-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation (paragraph II-3.2 of Method 502.2). Chamber temperature in step 1 of Procedure II shall be adjusted to -45°C (-50°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III. After completion of Procedure III, the test item shall be subjected to Procedure II. In step 4, cycle 5 of table 507.2-I shall be used.

4.8.3 Altitude The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. The air velocity in step 2 shall be 3542 $\pm$ 250 feet per minute. In step 5, reduce the air velocity to 300 $\pm$ 200 feet per minute and adjust the air temperature to + 63°C(+145°F). The equipment shall be reoriented periodically to equally expose all sides of the equipment to the dust stream. Equipment shall be operated before step 1, during step 8 and during step 11. There shall be no

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degradation in performance as a result of the dust test, and there shall be no binding or blocking of moving parts, non-operation of contacts or relays, clogging or dust accumulation which could form the nucleus for the collection of water vapor as a result of dust penetration of the test item.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C (+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall also be no clogging or binding of moving mechanical parts.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery-condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs, etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Immersion. (See 3.8.7) The equipment shall be subjected to the test of MIL-STD-810D, Method 512.2, Procedure I.

4.8.8 Vibration and shock. (See 3.8.8)

4.8.8.1 Vibration. (See 3.8.8) The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3 for Category 1 -Basic Transportation. The equipment in its transit case, if any, as restrained and/or tied down in its combat transport configuration shall be subjected to the random vibration environment for mission/field transportation. The vibration environment shall be simulated using the test schedules given in figures 514.3-4, 514.3-5, 514.3-6, and figures 514.3-10 through 514.3-22. Test duration for figures 514.3-4, 514.3-5, and 514.3-6 shall be 48 minutes per axis, for a test time of 144 minutes. Test

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duration for figures 514.3-10 through 514.3-22 shall be as given in the test schedules, for a test time of 3 hours. Total test time shall be 324 minutes. The test item shall not be operated during this test. The equipment in its transit case, if any, shall also be subjected to the test of MIL-STD-810D, Method 514.3 for transportation vibration, Category 3, Loose Cargo Transport. Prior to and following the tests for both Category 1 and Category 3, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance. The equipment, including transit case, shall not sustain any mechanical damage.

4.8.8.2 Transit drop. (See 3.8.8) The equipment in its transit case, if any, shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure IV. Prior to and following this test, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance. The equipment, including transit case, shall not sustain any mechanical damage.

4.8.8.3 Bench handling. (See 3.8.8) The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI. Prior to and following this test, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance and shall not sustain any mechanical damage.

4.8.9 Rain. (See 3.8.9) The equipment shall be subjected to the test of MIL-STD-810D, Method 506.2, Procedure I. In step 1, the test item shall be exposed to a simulated rain at a rate of 2 $\pm$ 0.5 inches per hour for 10 minutes. The rate of rainfall shall then be raised to 5 $\pm$ 1 inches per hour and held at this rate for 5 minutes. The rate shall then be reduced to 2$\pm$0.5 inches per hour for the next 15 minutes. Step 2 shall be changed as follows: starting 5 minutes after the initiation of the rain, the wind source shall be turned on and adjusted to produce a horizontal wind velocity of 40 miles per hour (3,500 feet per minute). The test item shall be operated in step 3.

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4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government-approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article testing reliability test. First article testing shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. The Remote Loop Group Multiplexer TD-1234()/TTC shall be operated back-to-back.

b. Duty cycle. Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24 hour cycle.

C. Chamber temperature shall be 50°C to 55%C (122°F to 131°F).

d. Vibration shall be 2.2 g ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

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4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.13.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12 assembly drawings.

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4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (System safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System safety engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table XV, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

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TABLE XV. Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Visual
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.15 Serial numbers. (See 3.5.1) The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA.

4.16 Nuclear tests. (See 3.16) Tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

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5. Packaging.

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2A, tables I through VII coded as follows:

5.1.1 Level A. 10-1-00-00-NS-X-ED

5.1.2 Level B. Same as level A.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES.

6.1 Intended use. The TD-1234 is a field exposed unit which combines up to eight (8) 16/32 kb/s loop circuits with a group from a TD-1233(P)/TTC Multiplexer and another TD-1234 into an output rate from 128 kb/s to 576 kb/s. It also provides transmission of maintenance orderwire and power feed for loop end instruments (DSVT, DNVT) and for a TD-1233(P)/TTC. At the shelter end, the unit interfaces with the MD-1026(P)/G, Modem, Digital Data, via CX-11230 cable.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article units required.

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e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirements.

i. MIL-Q-9858.

j. Length of cables (if any required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

l. GFE.

(1) MD-1026(P)/G Modem, Digital Data, with all diphase modules or require contractor to develop a test which precludes this GFE requirement.

(2) TD-1233(P)/TTC Multiplexer, with all loop modem channels.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. Initial shipments of end items will be contingent upon successful completion of all first article tests.

6.5 Location of operational inspection. It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contract's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table VIII.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject term (key word) listing.

Analog Applique
Continuous Variable Slope Delta
CVSD
Digital Group Multiplex
Diphase
Framing
Group Modem
Loop Modem
Multiplexer
Multiplexer-Combiner
Multiplexer-Combiner, TD-1234(P)/TTC
MUX/DEMUX
Orderwire
Remote Multiplexer-Combiner (RMC)

CUSTODIAN
ARMY-CR

Preparing Activity
ARMY-CR
Project No. 5820-A830

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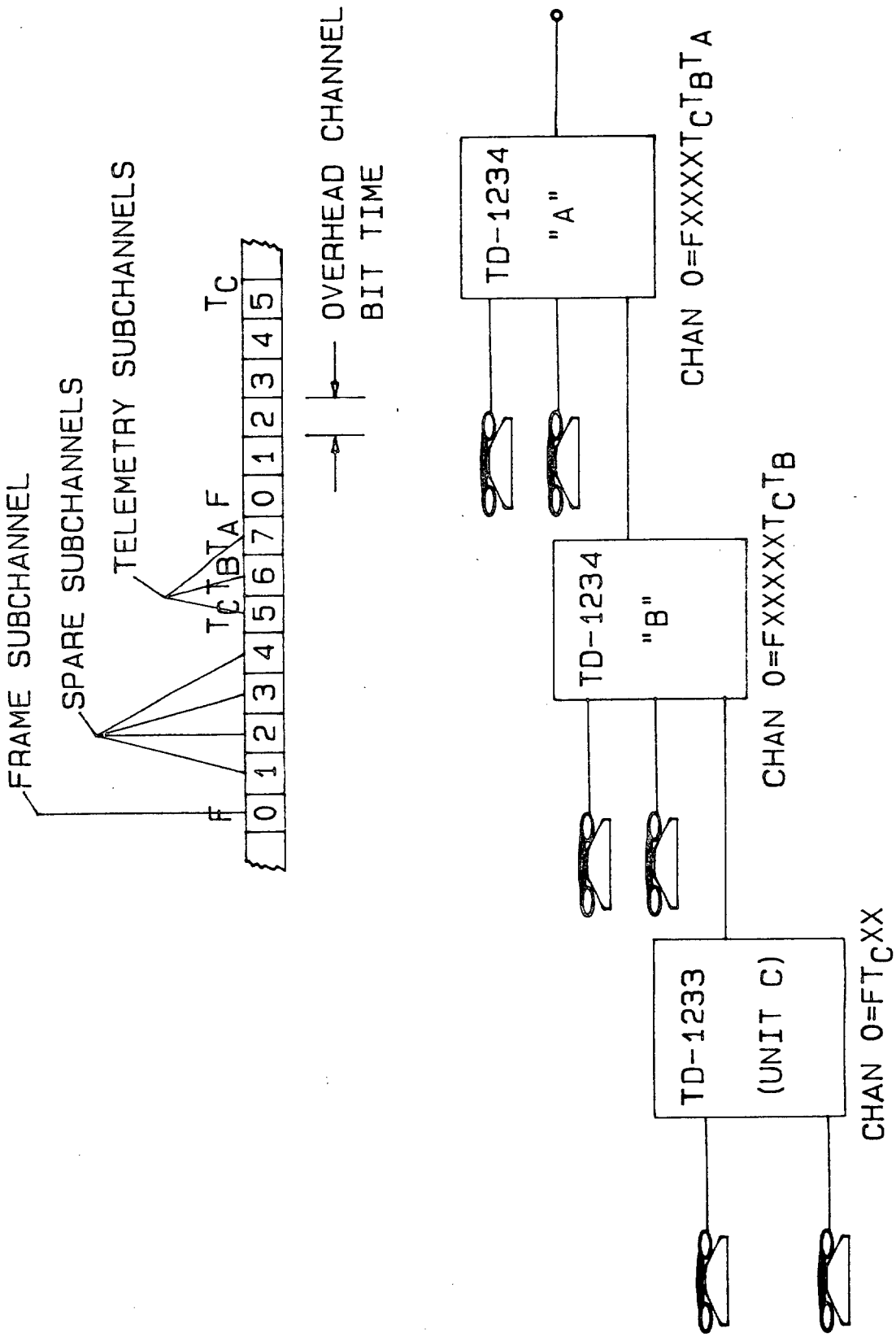


FIGURE 1. TD-1234 overhead channel formats.

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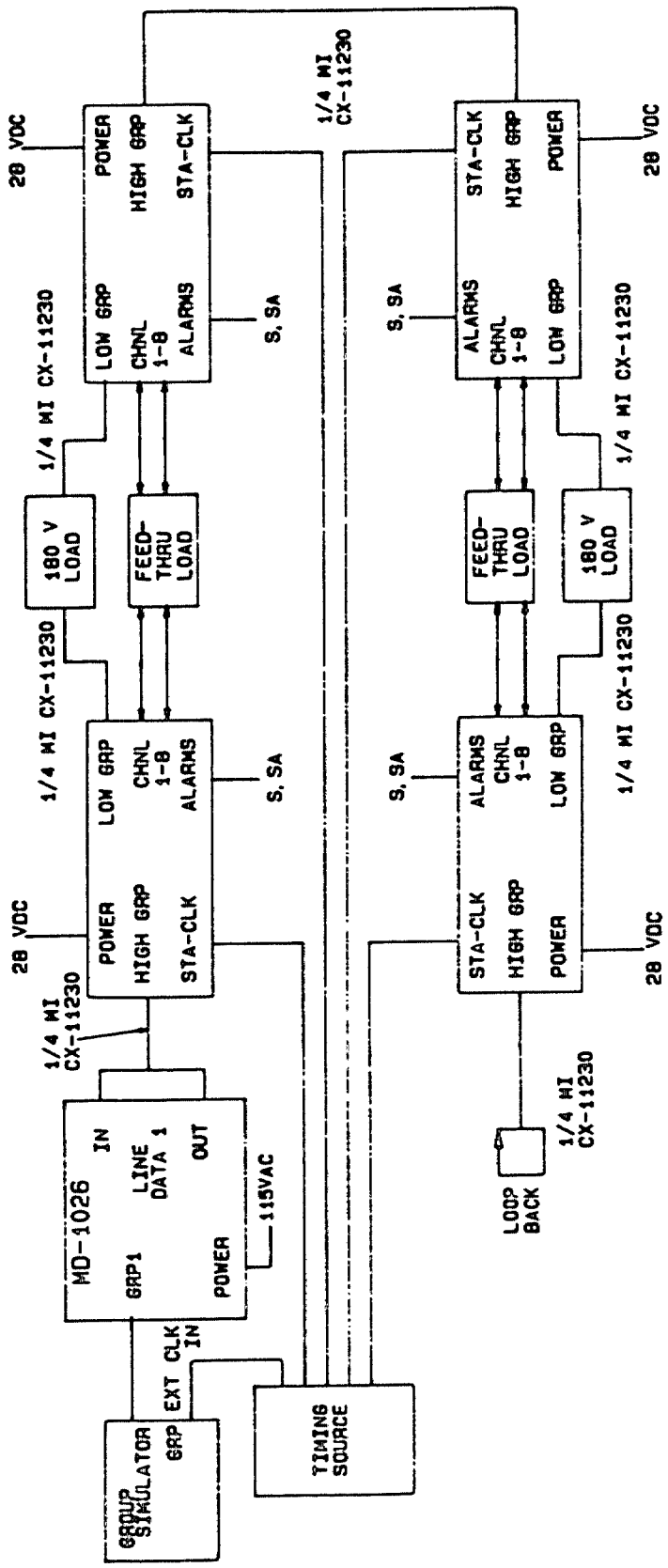


FIGURE 2. Burn-in.

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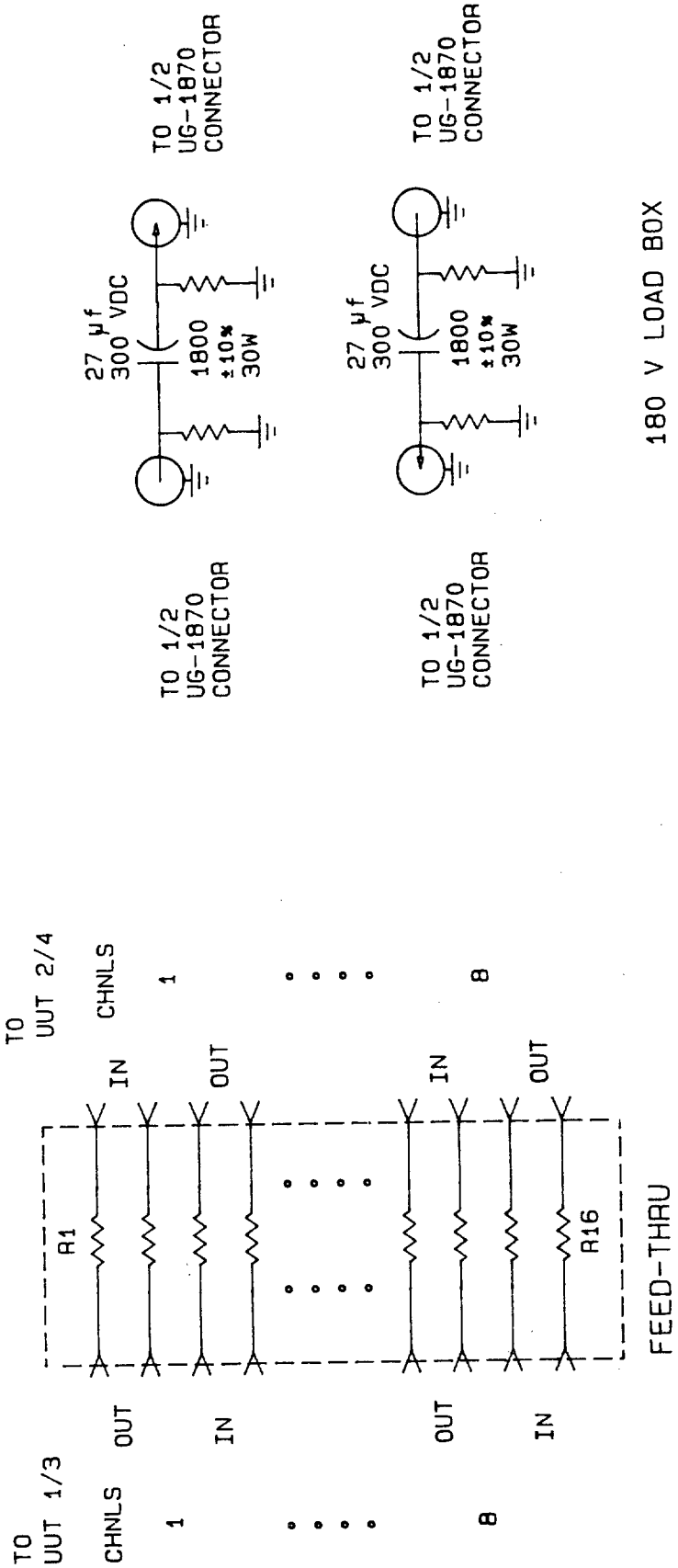
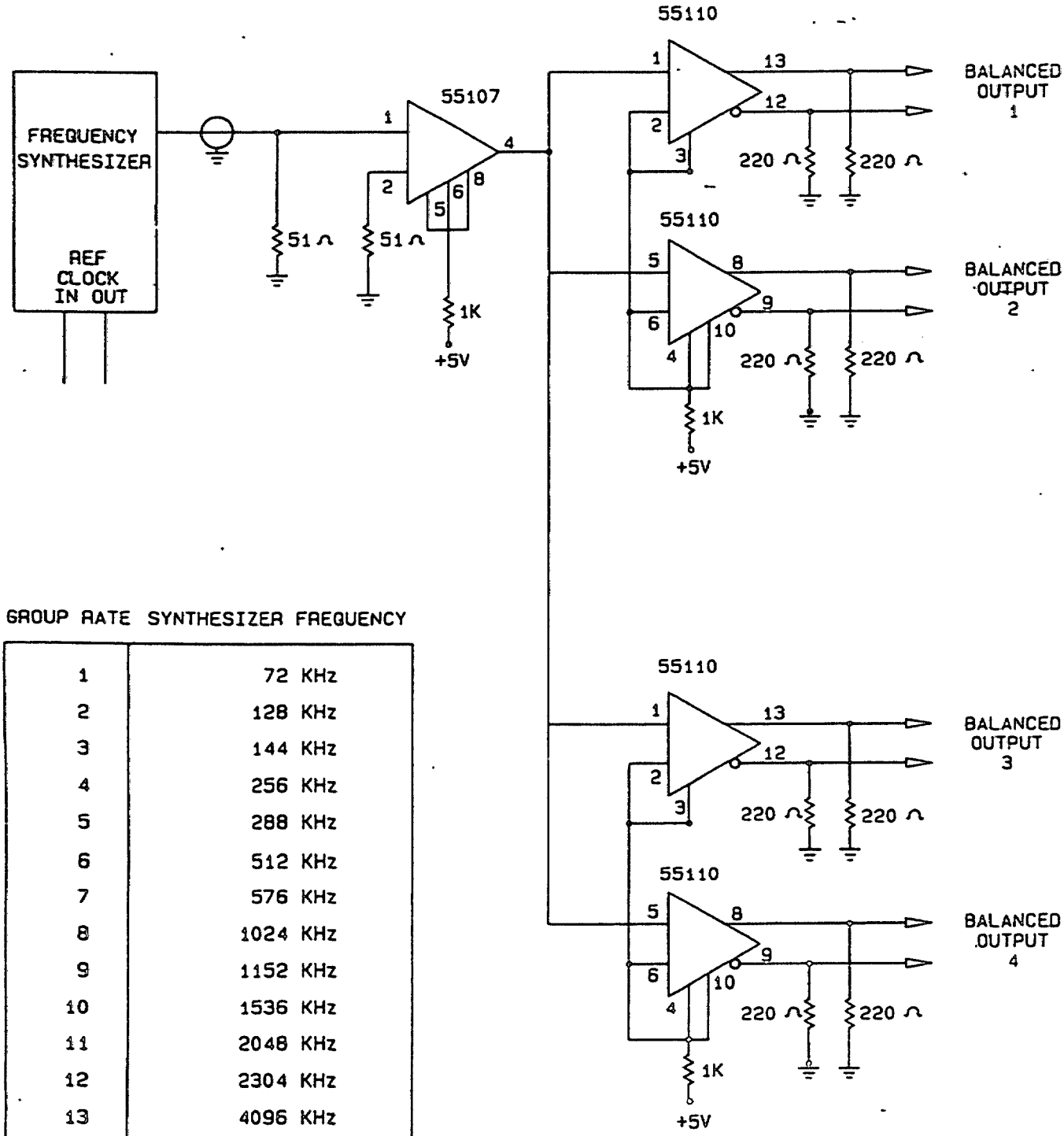


FIGURE 3. Test boxes.

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GROUP RATE SYNTHESIZER FREQUENCY

1	72 KHz
2	128 KHz
3	144 KHz
4	256 KHz
5	288 KHz
6	512 KHz
7	576 KHz
8	1024 KHz
9	1152 KHz
10	1536 KHz
11	2048 KHz
12	2304 KHz
13	4096 KHz
14	4608 KHz
15	4915.2 KHz

FIGURE 4. Timing source.

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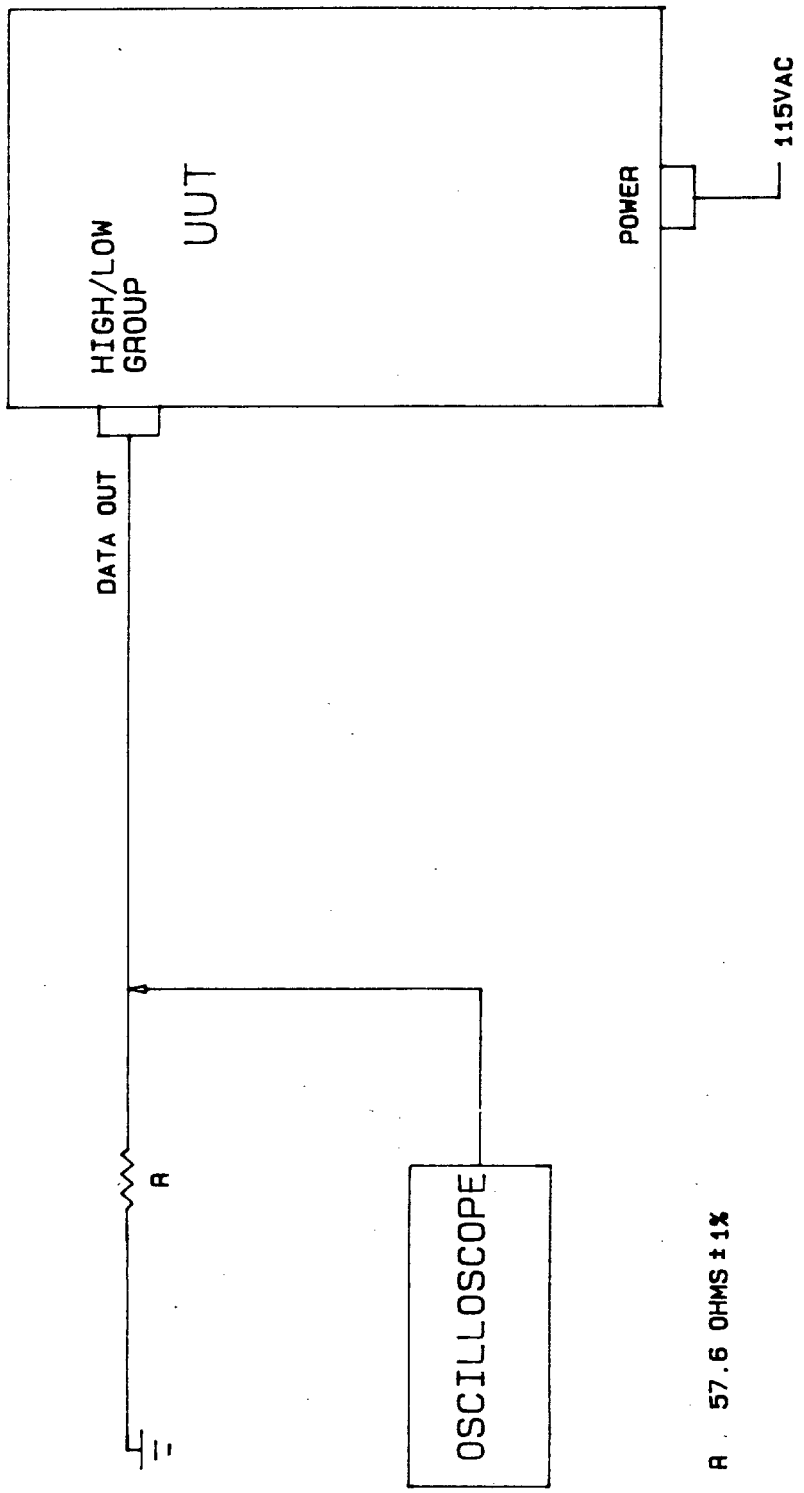


FIGURE 5. Diphase group output voltage and rise time.

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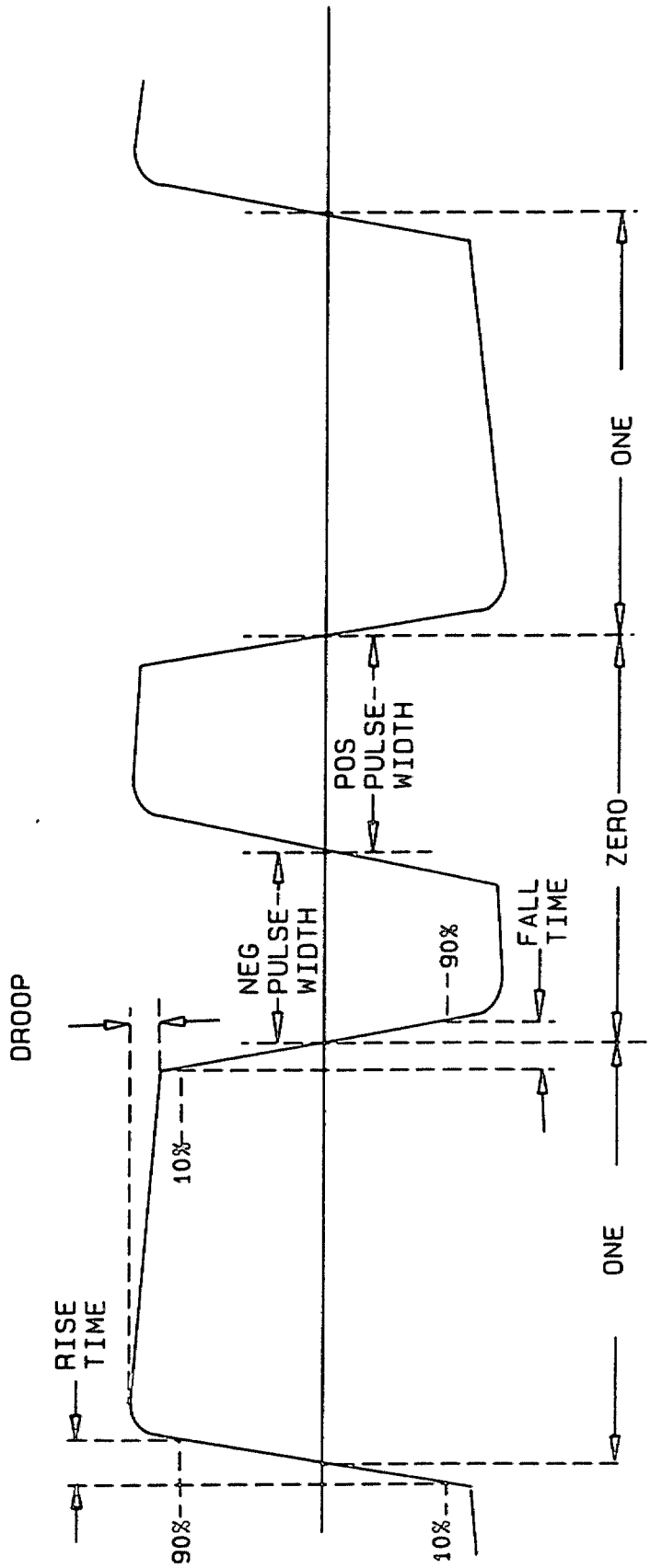
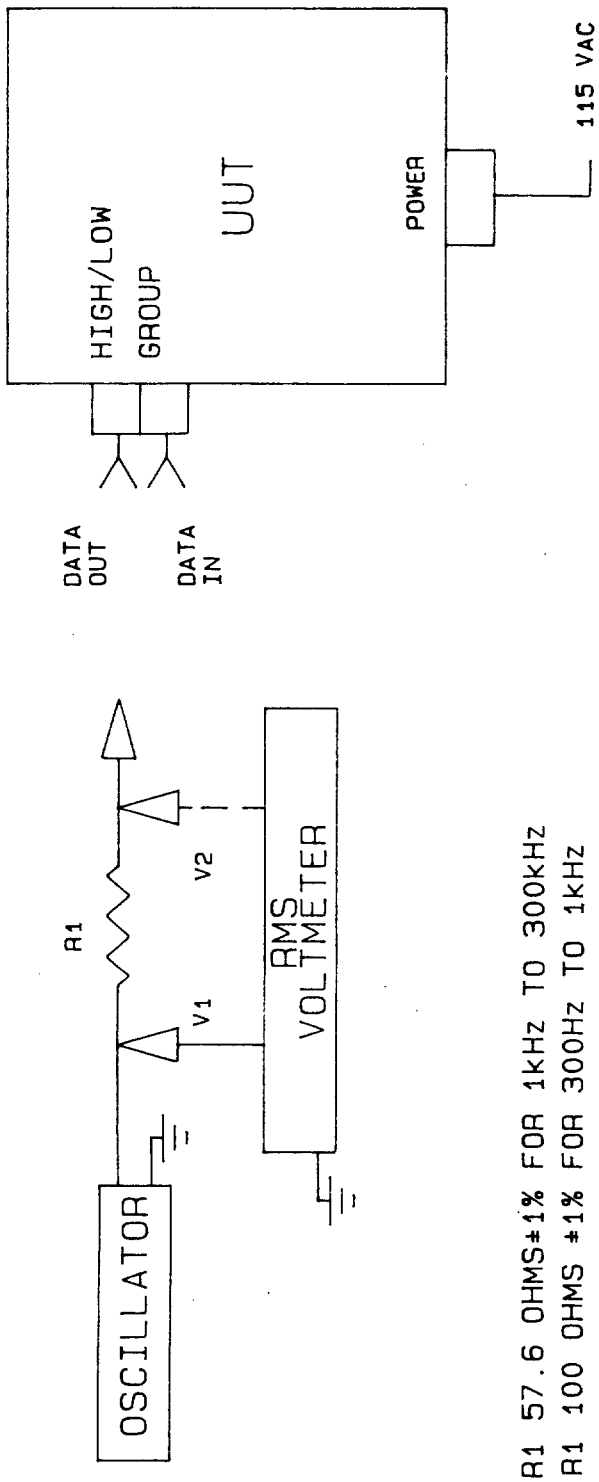


FIGURE 6. Diphase waveform (digital zero).

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R1 57.6 OHMS±1% FOR 1kHz TO 300kHz
R1 100 OHMS ±1% FOR 300Hz TO 1kHz

FIGURE 7. Diphase group output/input impedance.

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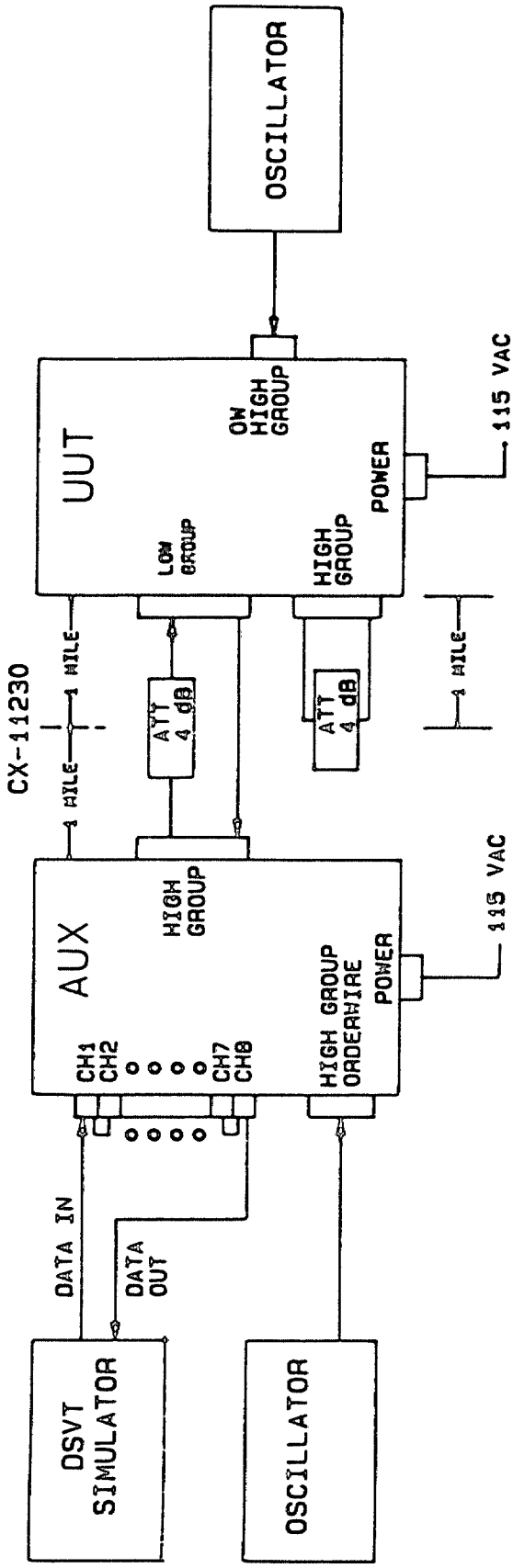


FIGURE 8. Received signal level.

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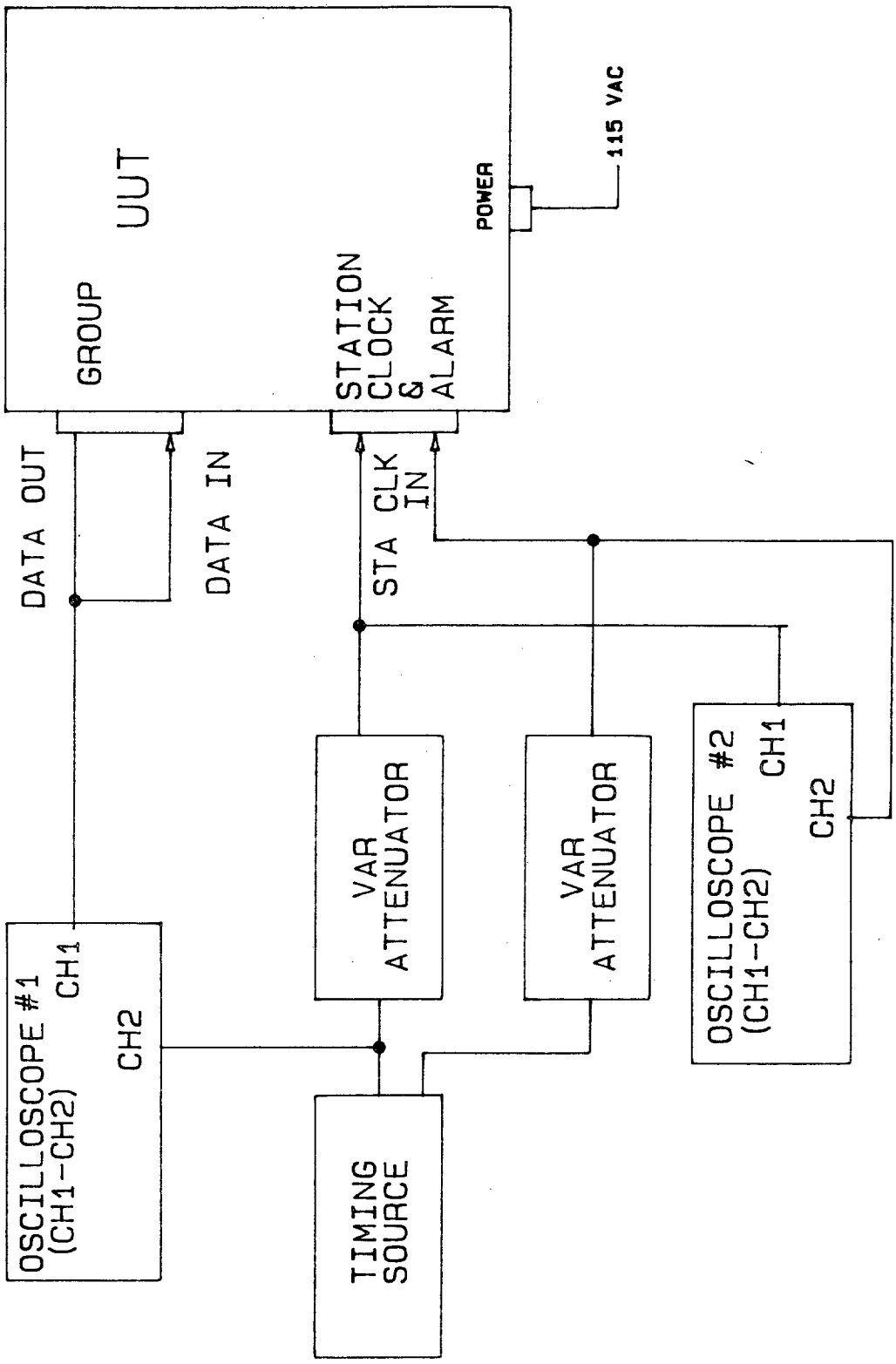


FIGURE 9. Receiver sensitivity and timing mode.

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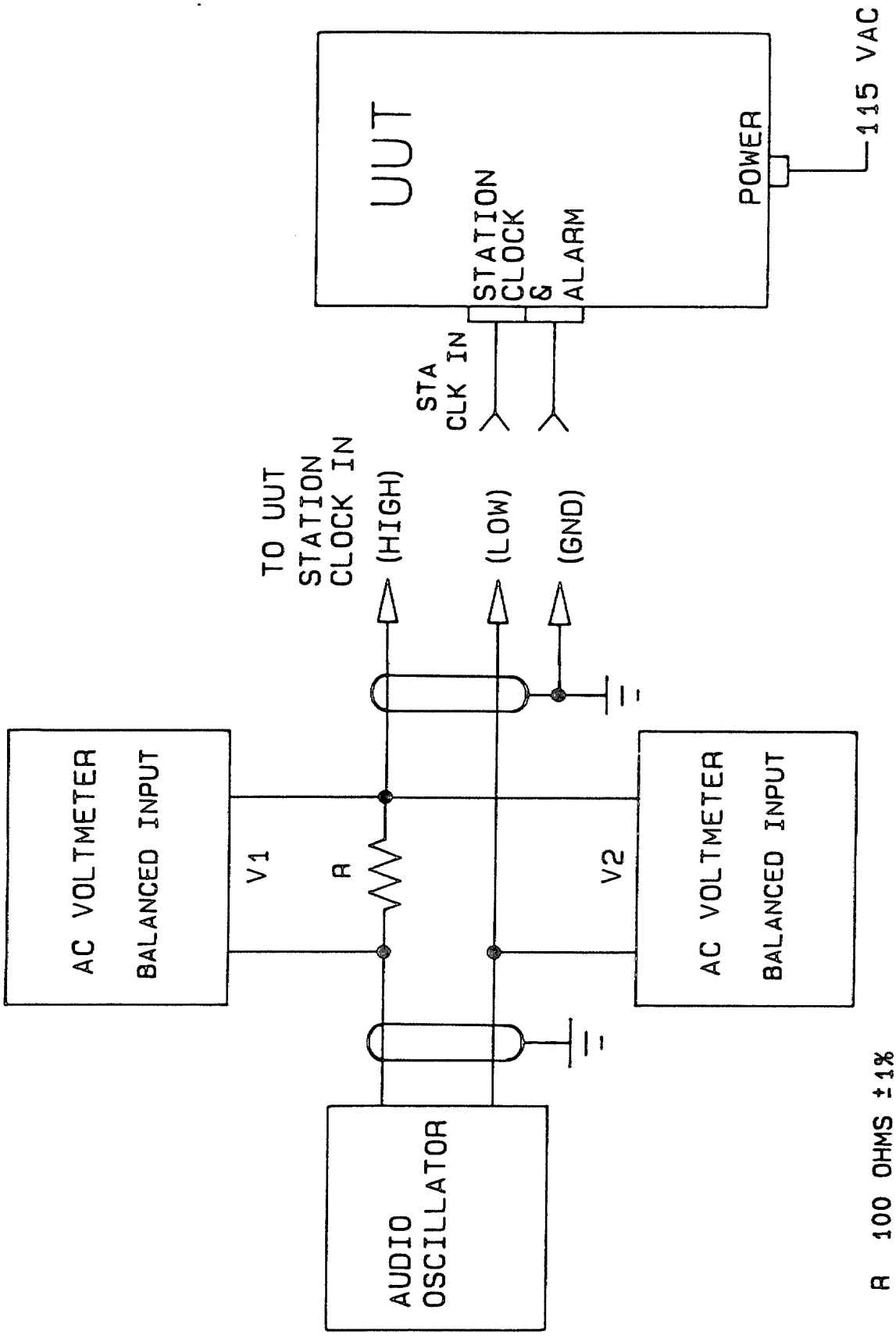


FIGURE 10. Input load impedance.

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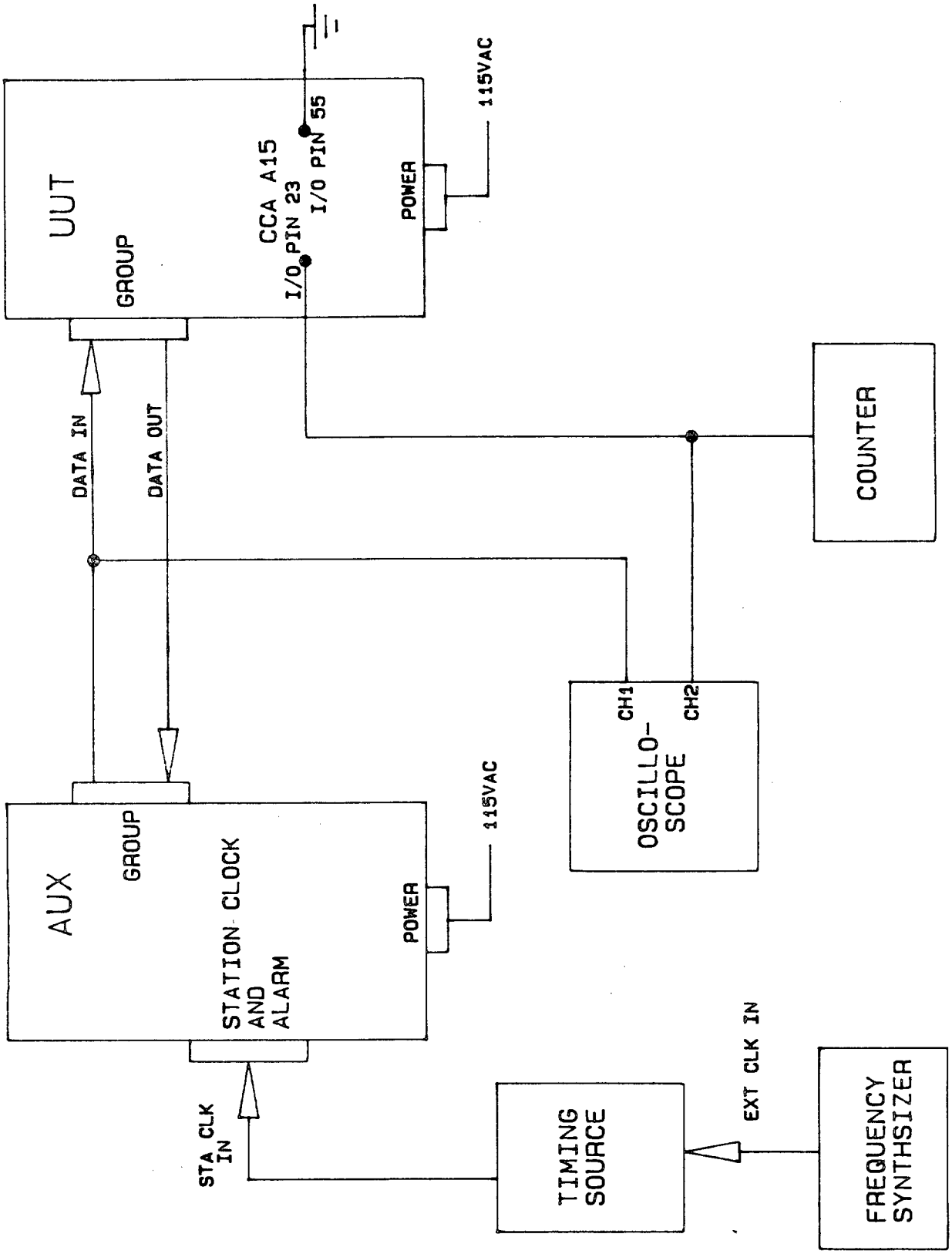


FIGURE 11. Timing modes.

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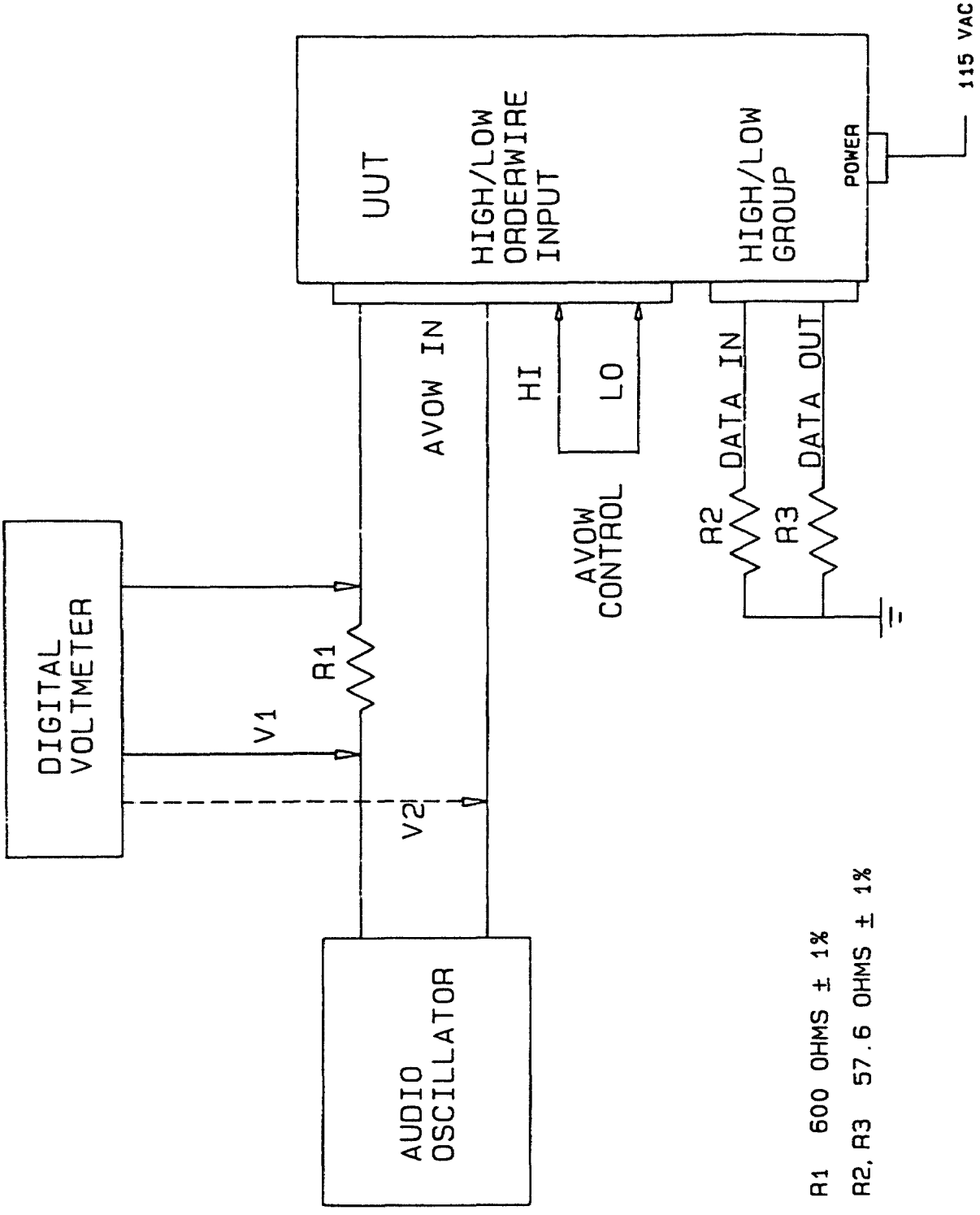
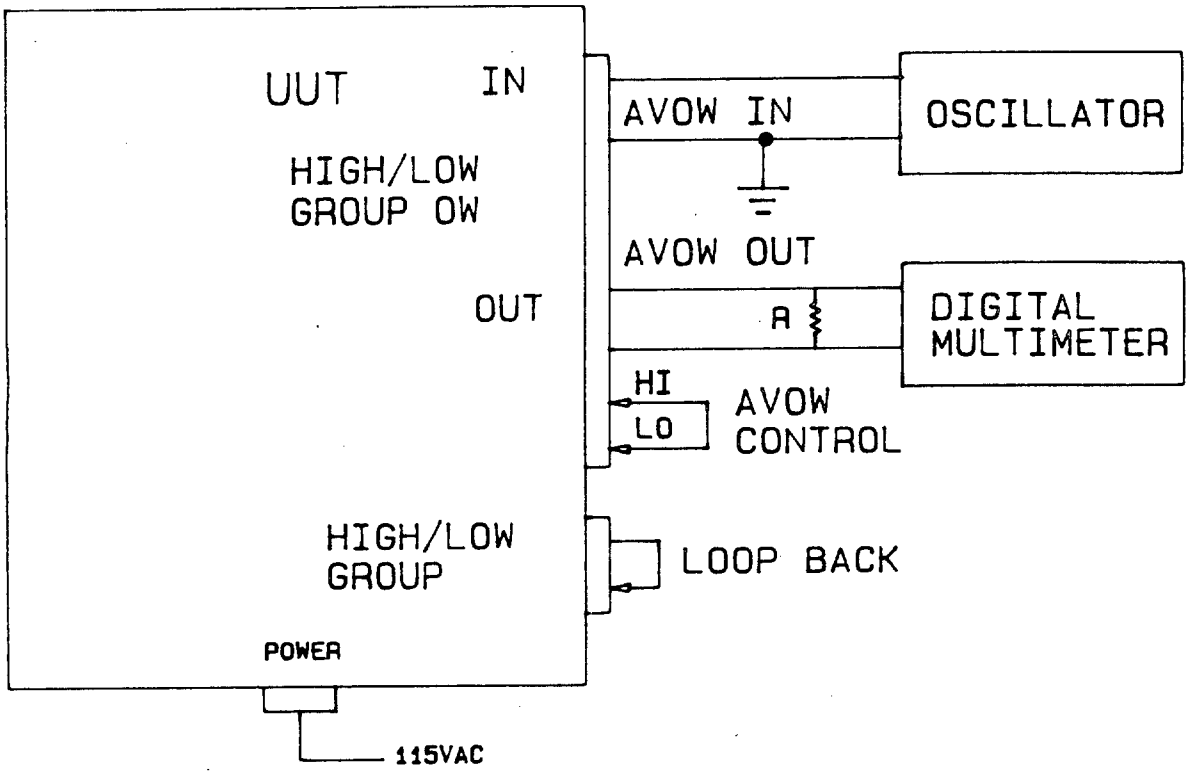


FIGURE 12. Orderwire input/output impedance.

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R 600 OHMS ±1%

FIGURE 13. Bandwidth.

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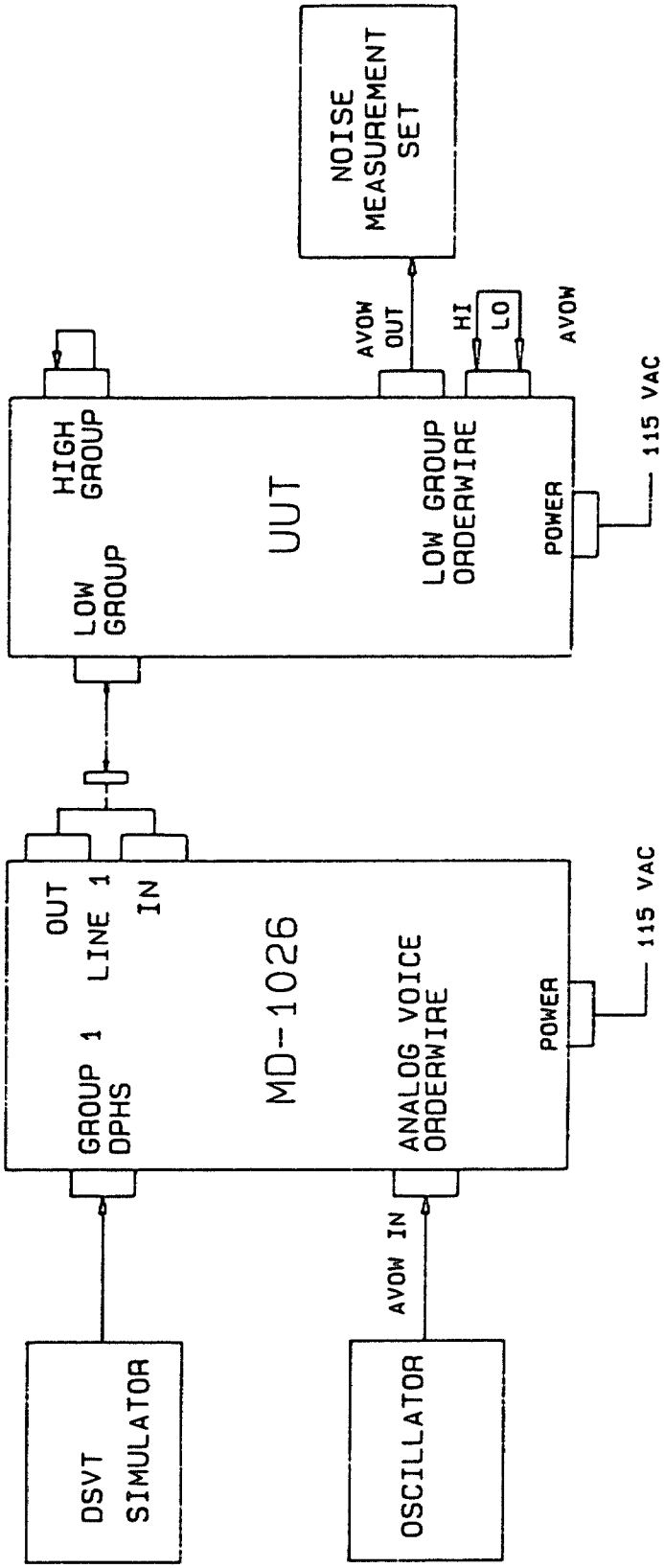


FIGURE 14. Signal-to-noise ratio.

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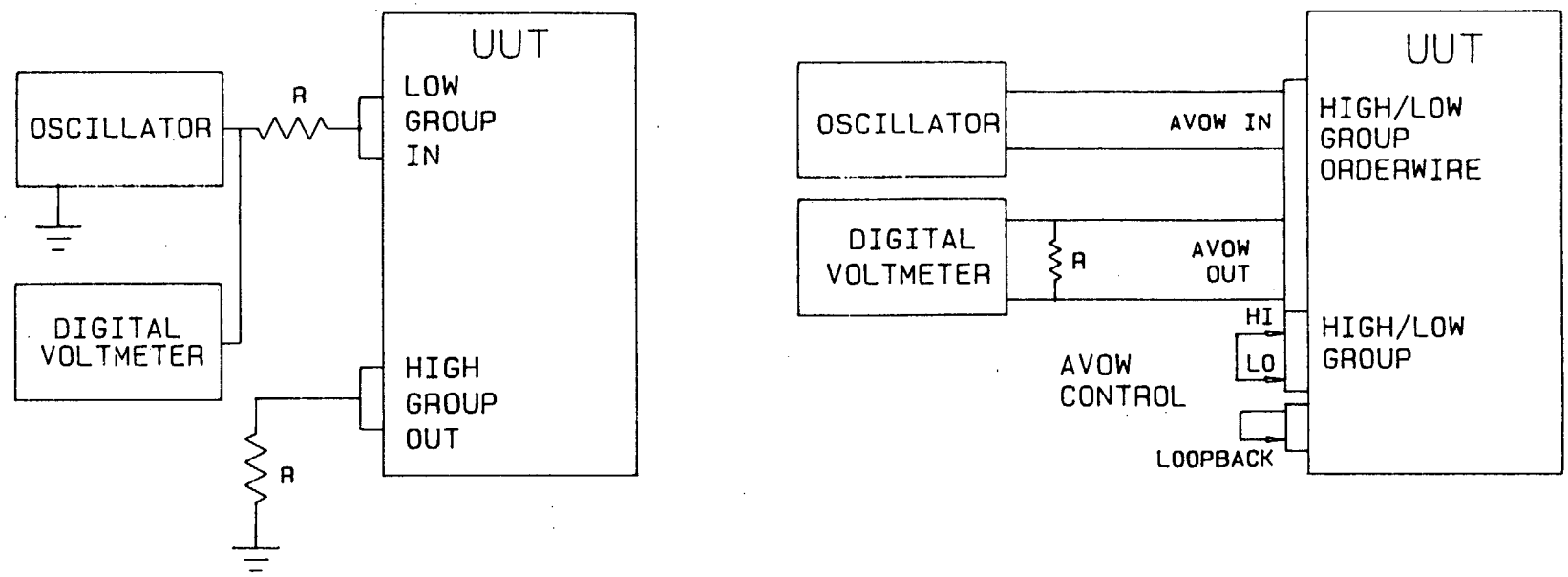
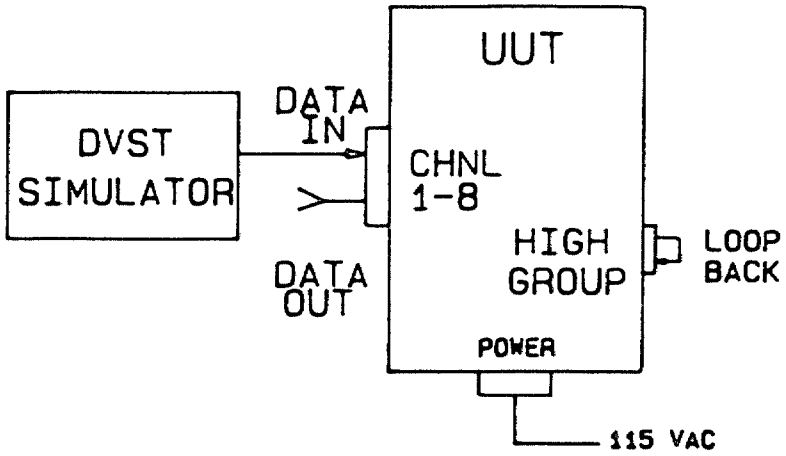
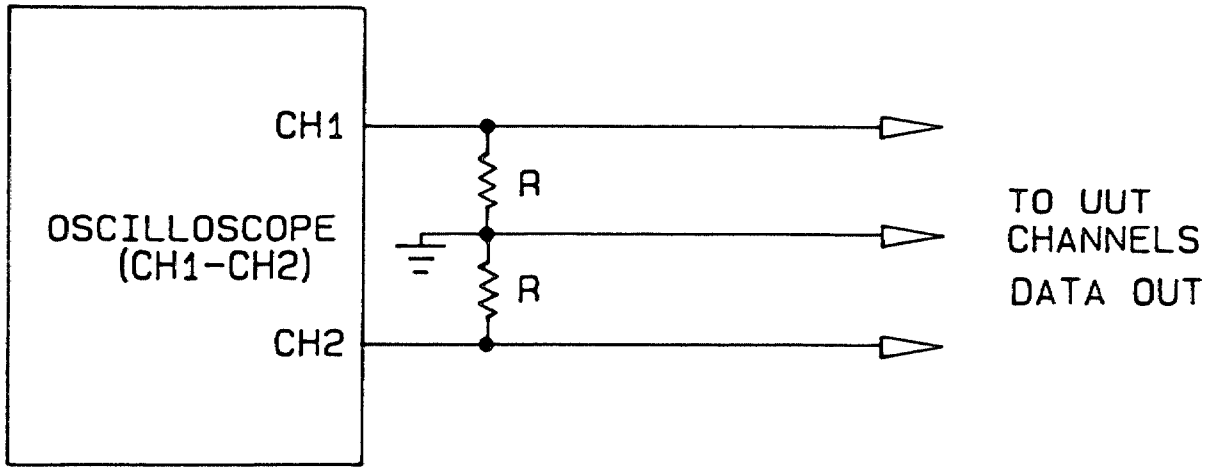


FIGURE 15. Transmit level.

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R 62.5 OHMS ± 1%

FIGURE 16. Output voltage/wave shape.

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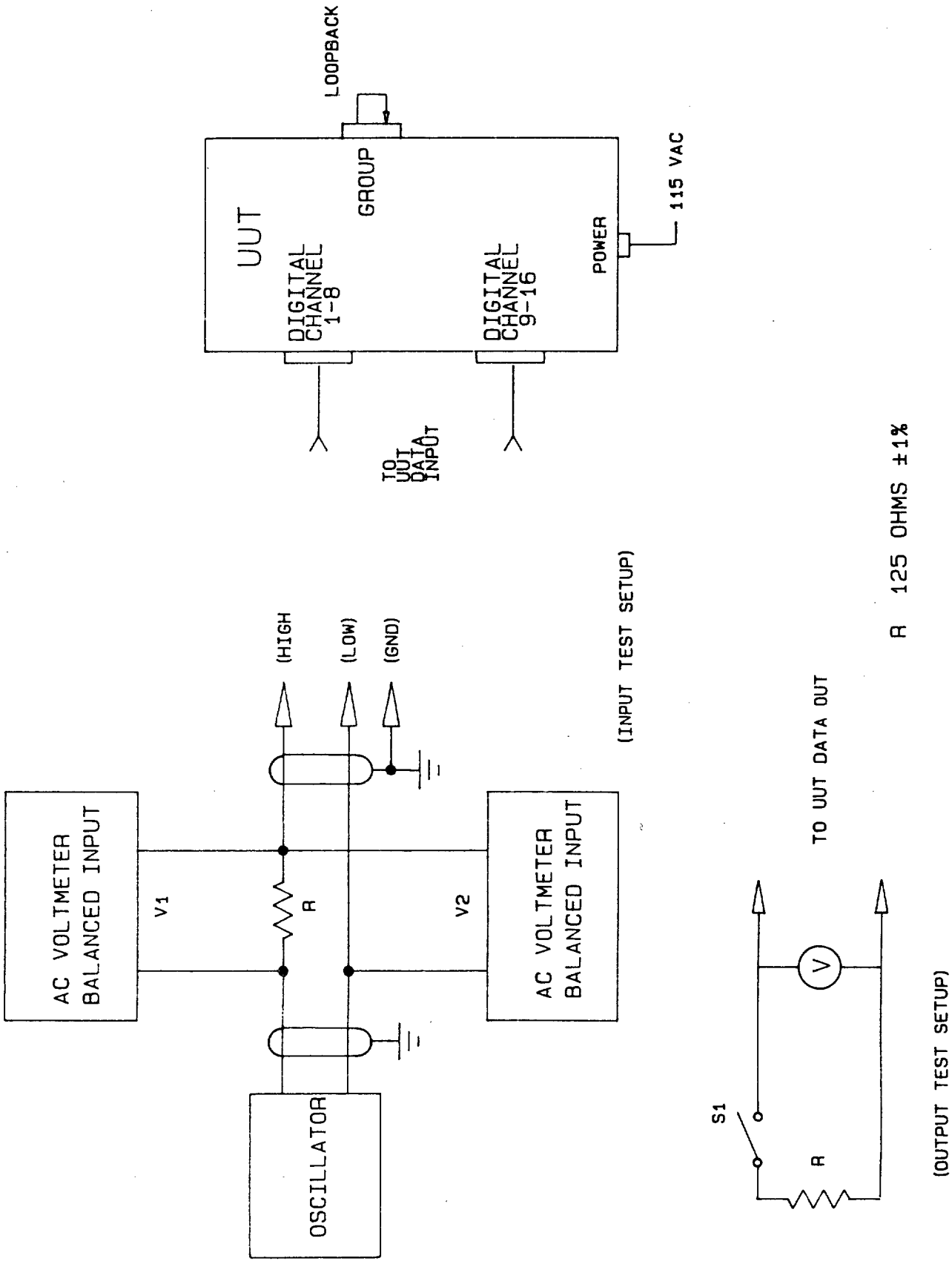


FIGURE 17. Input and output impedance.

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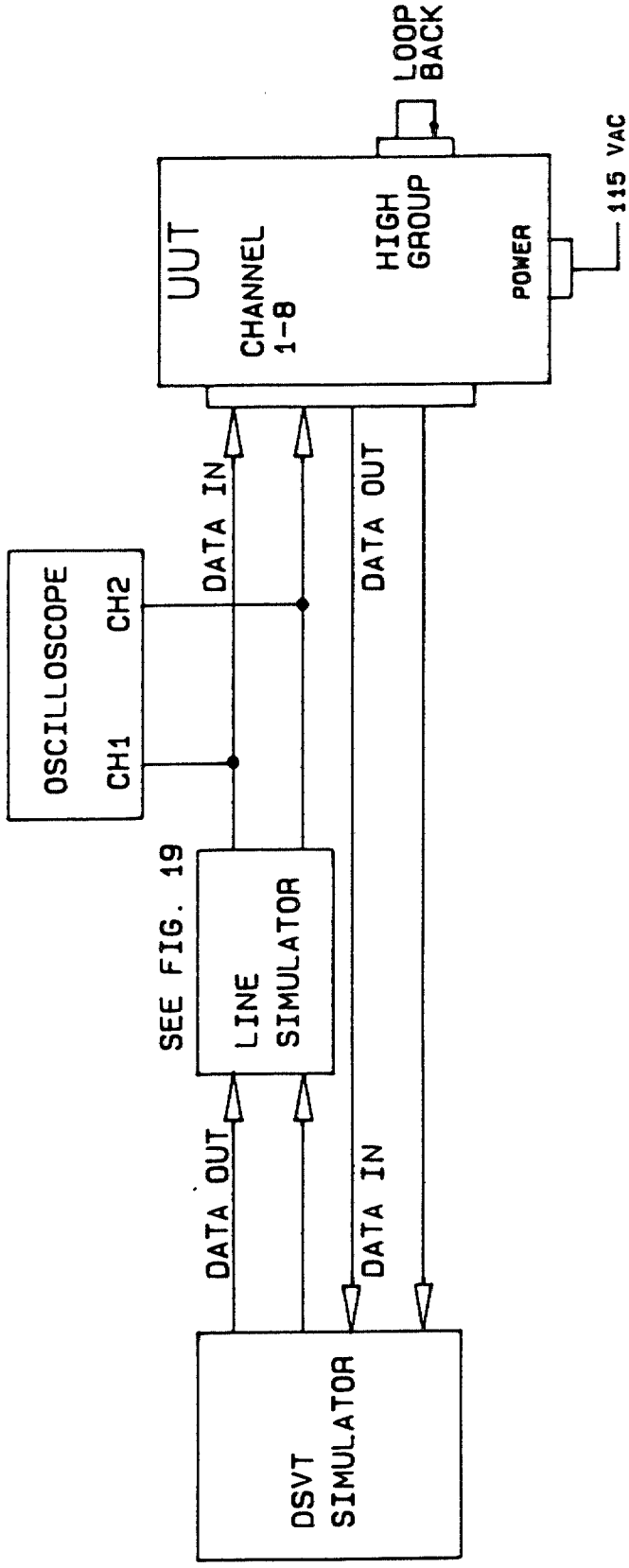


FIGURE 18. Input voltage.

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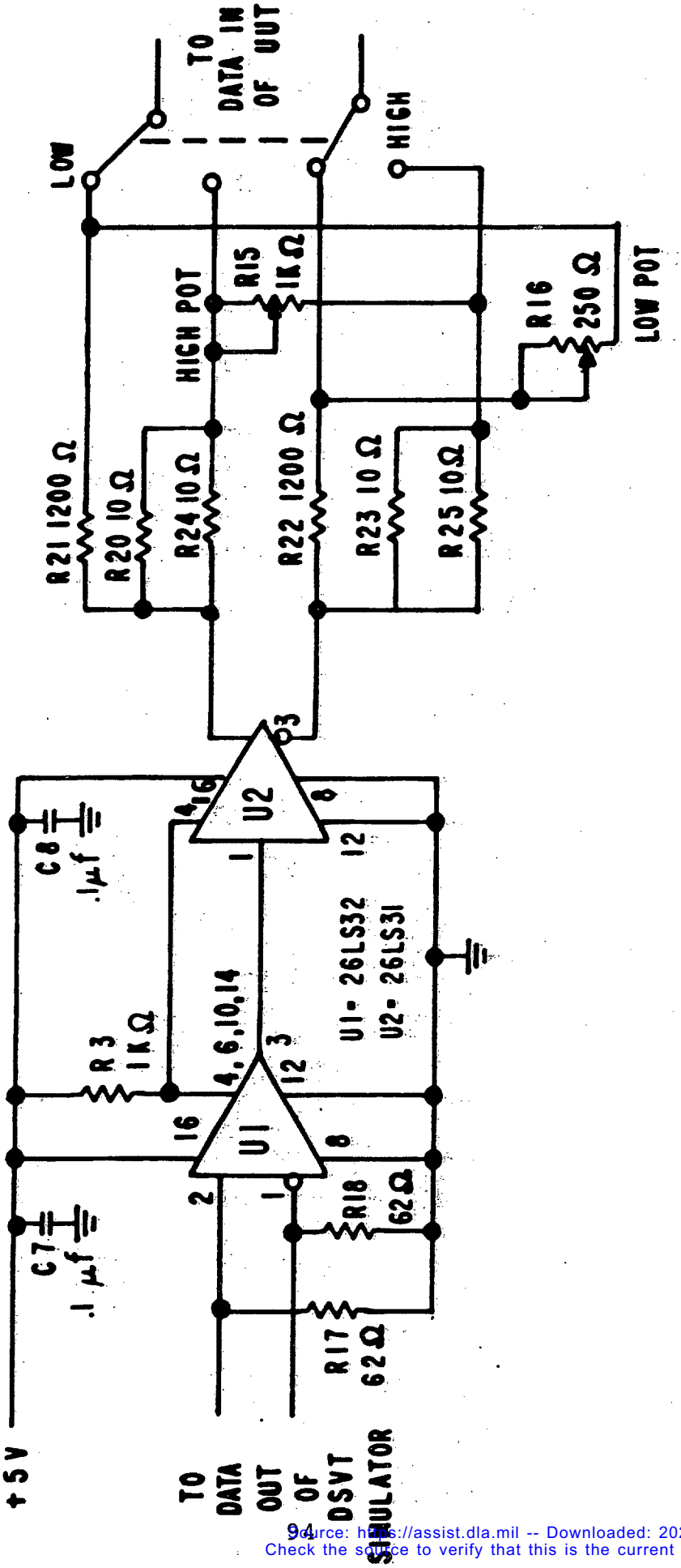


FIGURE 19. Line simulator detail.

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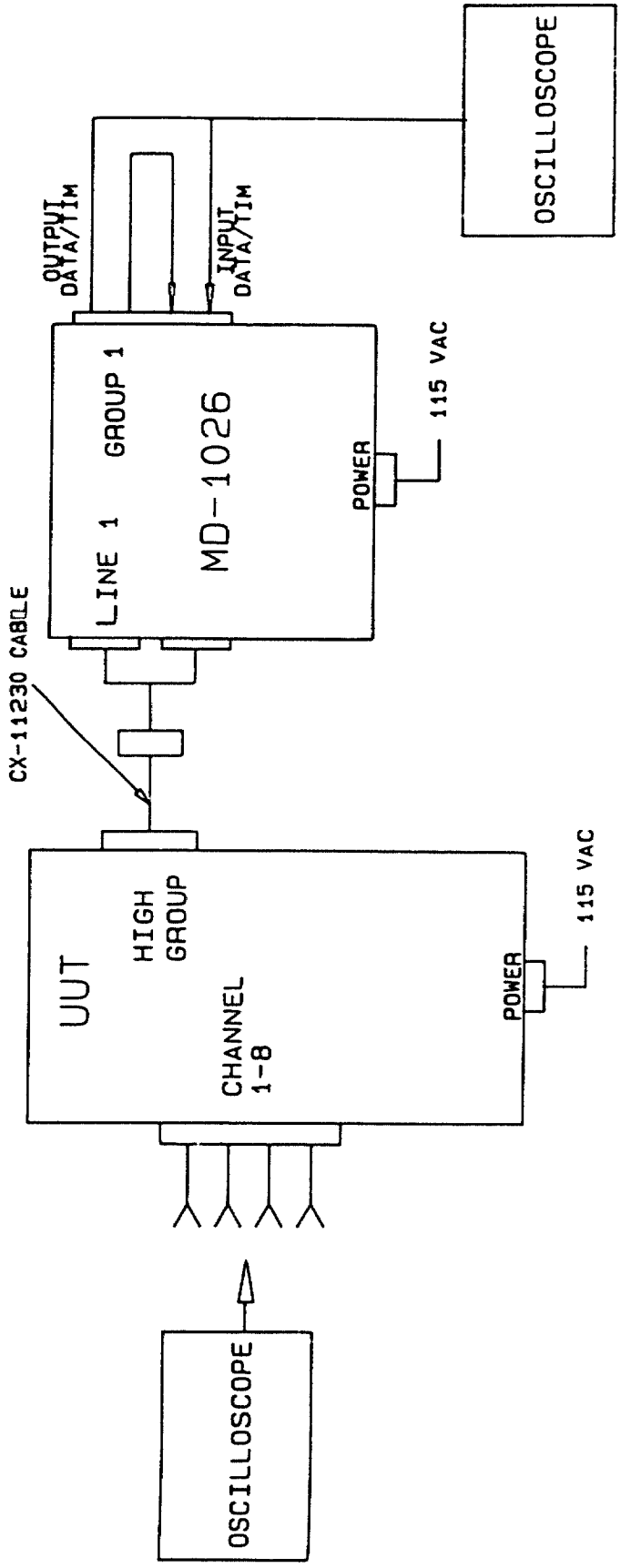


FIGURE 20. Idle loop/idle filler.

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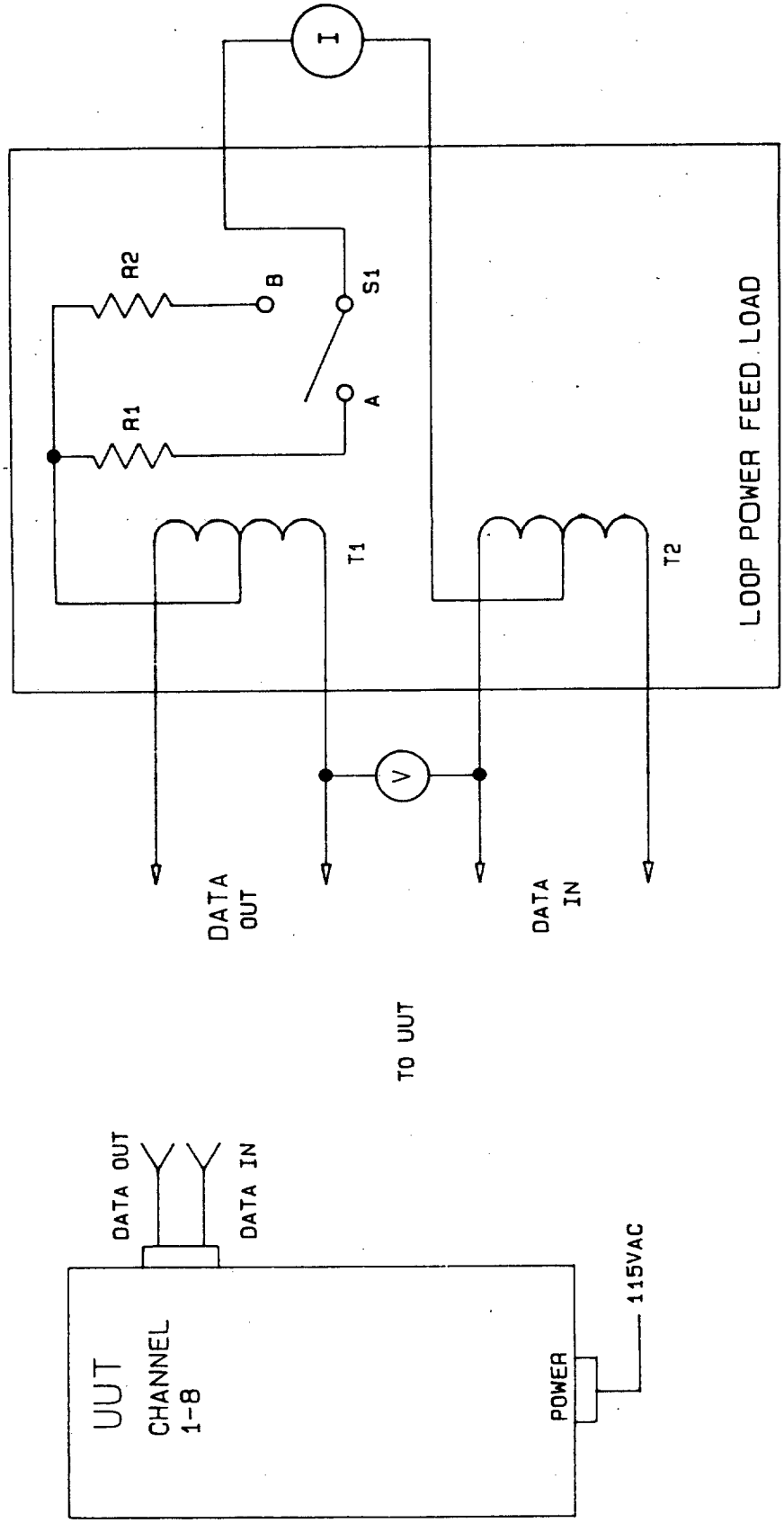
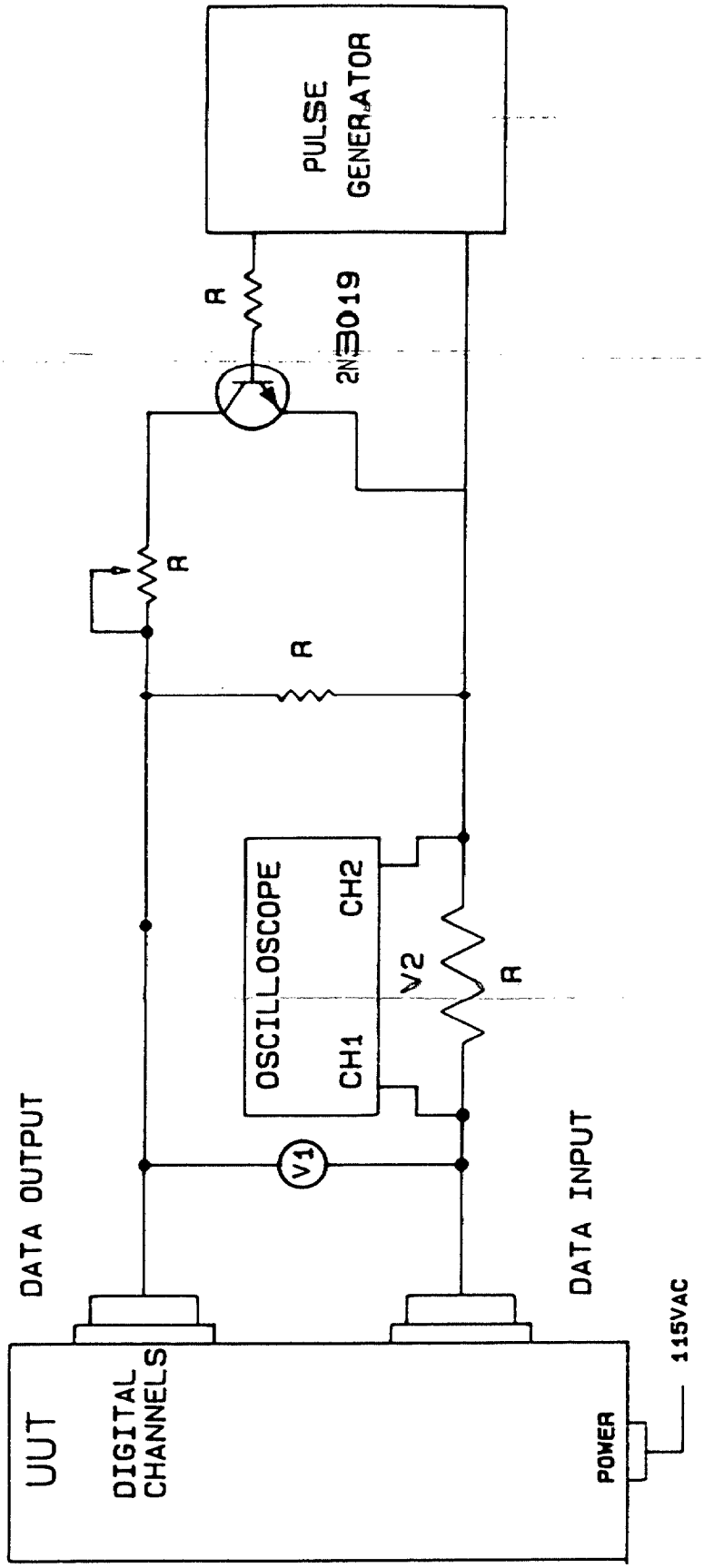


FIGURE 21. Loop power feed load.

NOTE: FOR T1 AND T2 SEE SM-A-985920
R1 11.2K ±1% 0.5W
R2 680 OHMS ±1% 5W

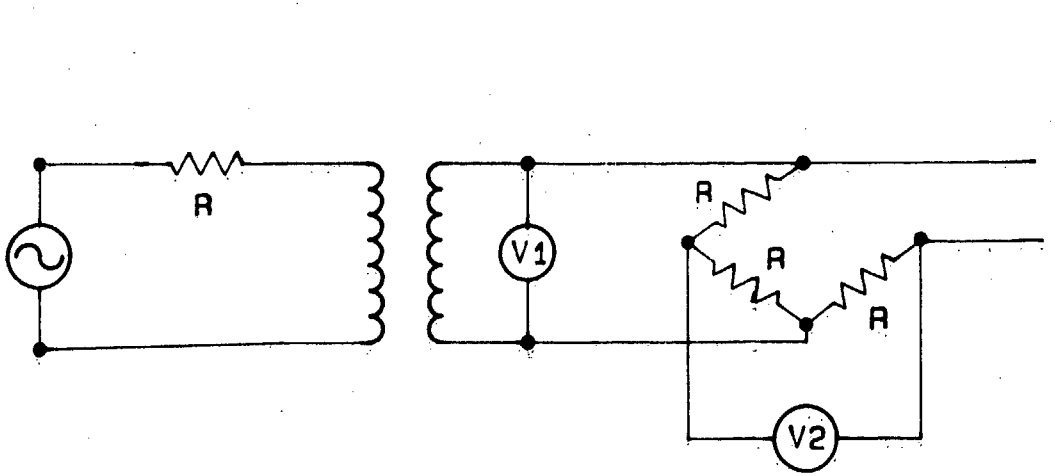
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
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R1 10 OHMS ± 1%
R2 680 OHMS ± 1% 5W
R3 5K OHMS ± 1% 2W
R4 750 OHMS ± 1%

FIGURE 22. Loop power surge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
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R 600 OHMS ± 1%

SET V1=0dBm

FOR TRANSFORMER SEE SM-A-984723

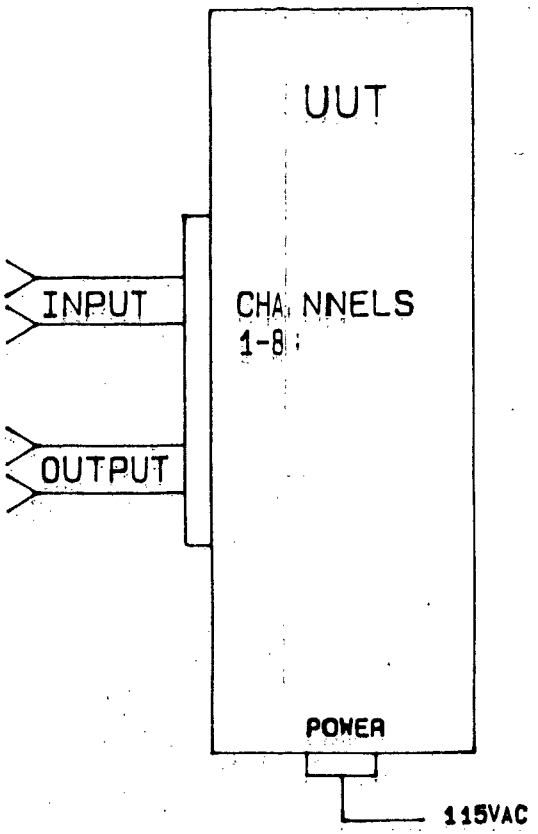
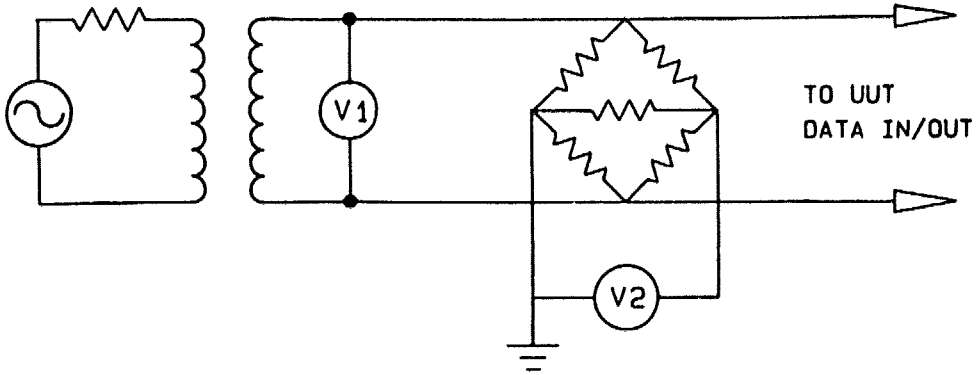
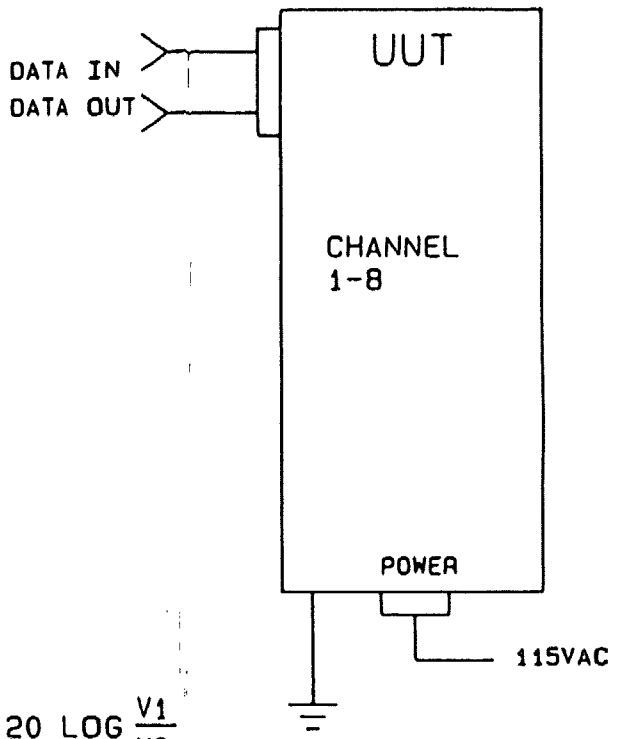


FIGURE 23. Impedance (return loss).

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$$\text{LONGITUDINAL BALANCE} = 20 \log \frac{V1}{V2}$$

ALL RESISTORS 600 OHMS ± 1%
BRIDGE MUST BE TRIMMED SO THAT
BALANCE IS GREATER THAN 60 db
WITH NO OUTPUT CONNECTED.
UUT CASE GROUND MUST BE CONNECTED TO TEST CIRCUIT GROUND.
FOR TRANSFORMER SEE SM-A-984723

FIGURE 24. Longitudial balance.

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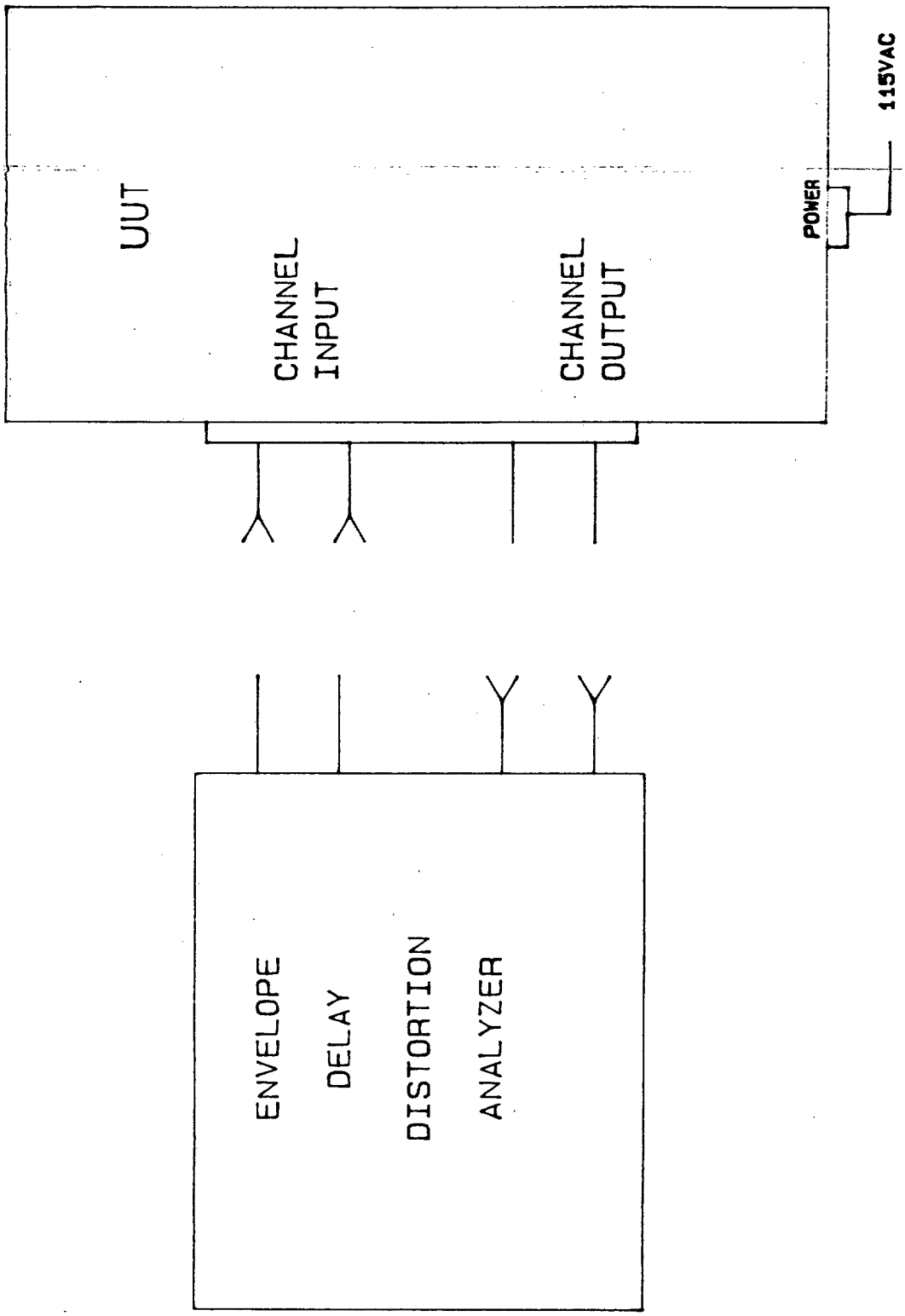


FIGURE 25. Envelope delay distortion.

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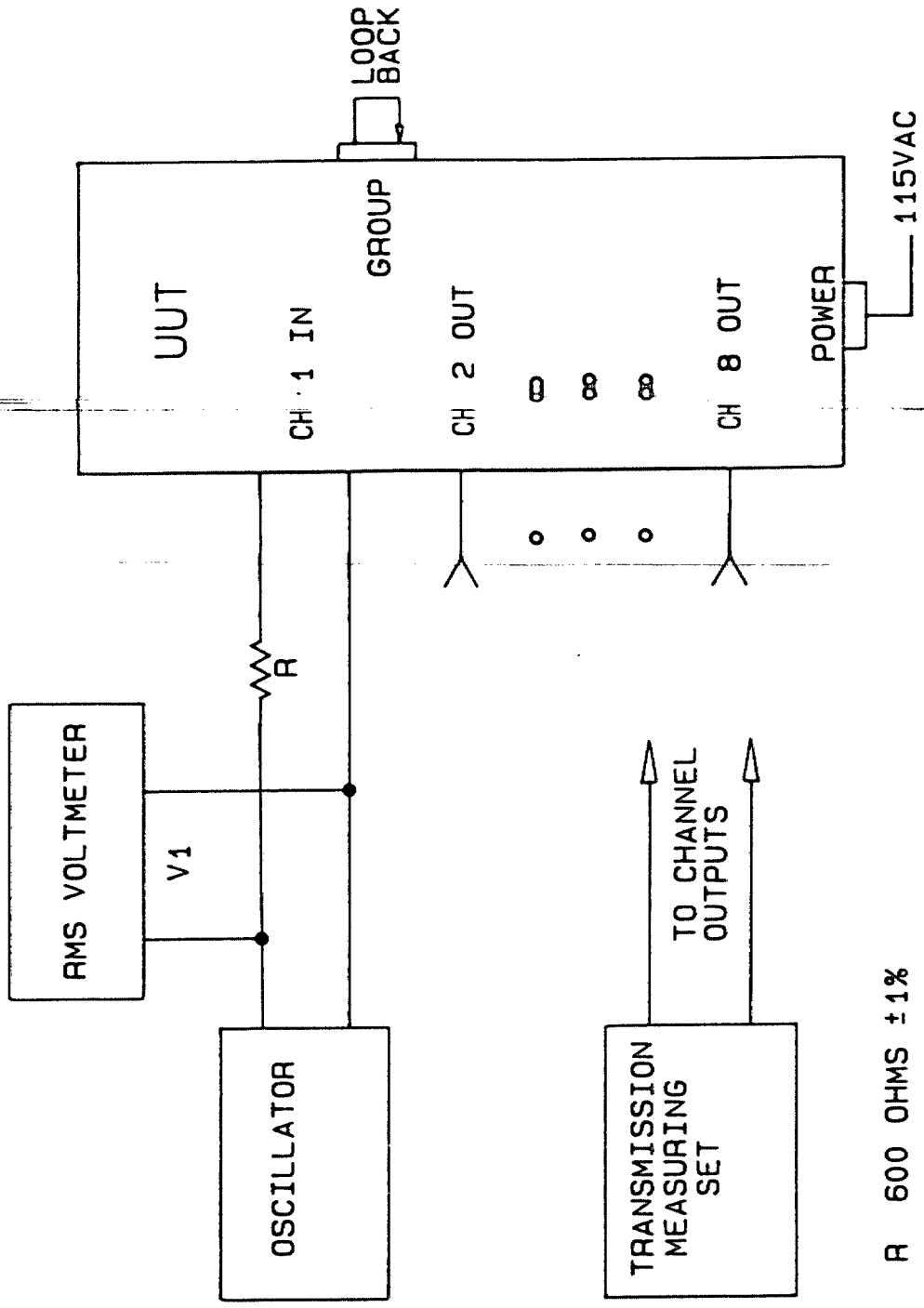
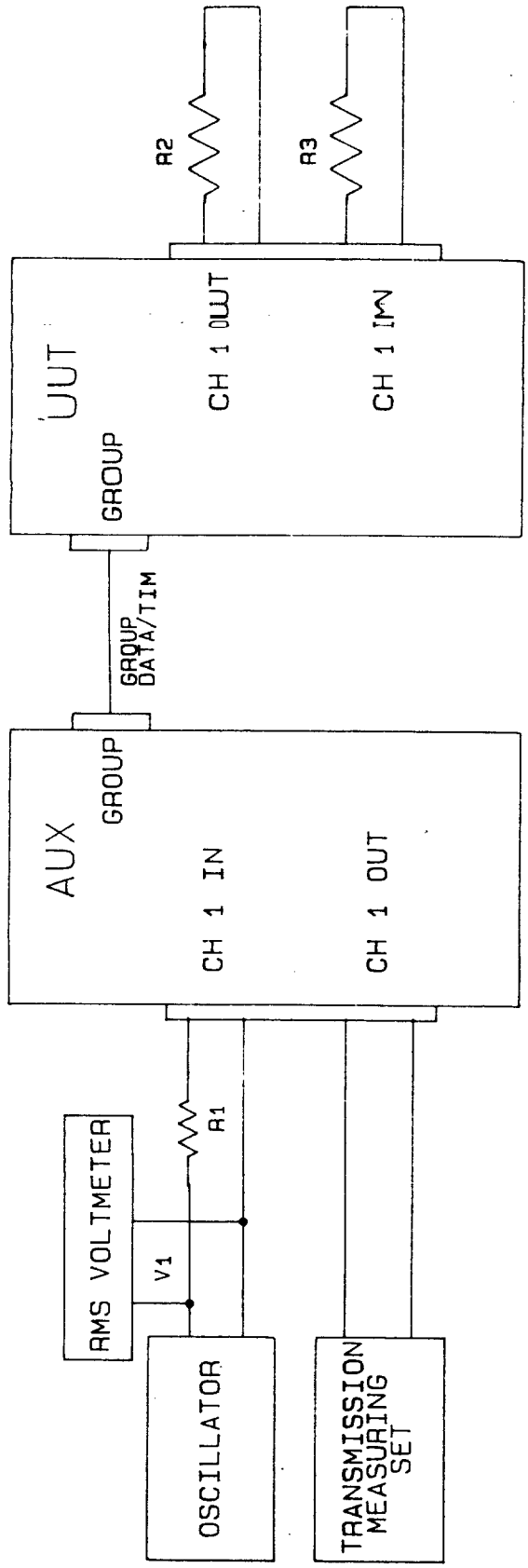


FIGURE 26. Idle channel noise; crosstalk.

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R 600 OHMS ±1%

FIGURE 27. Crosstalk, second configuration.

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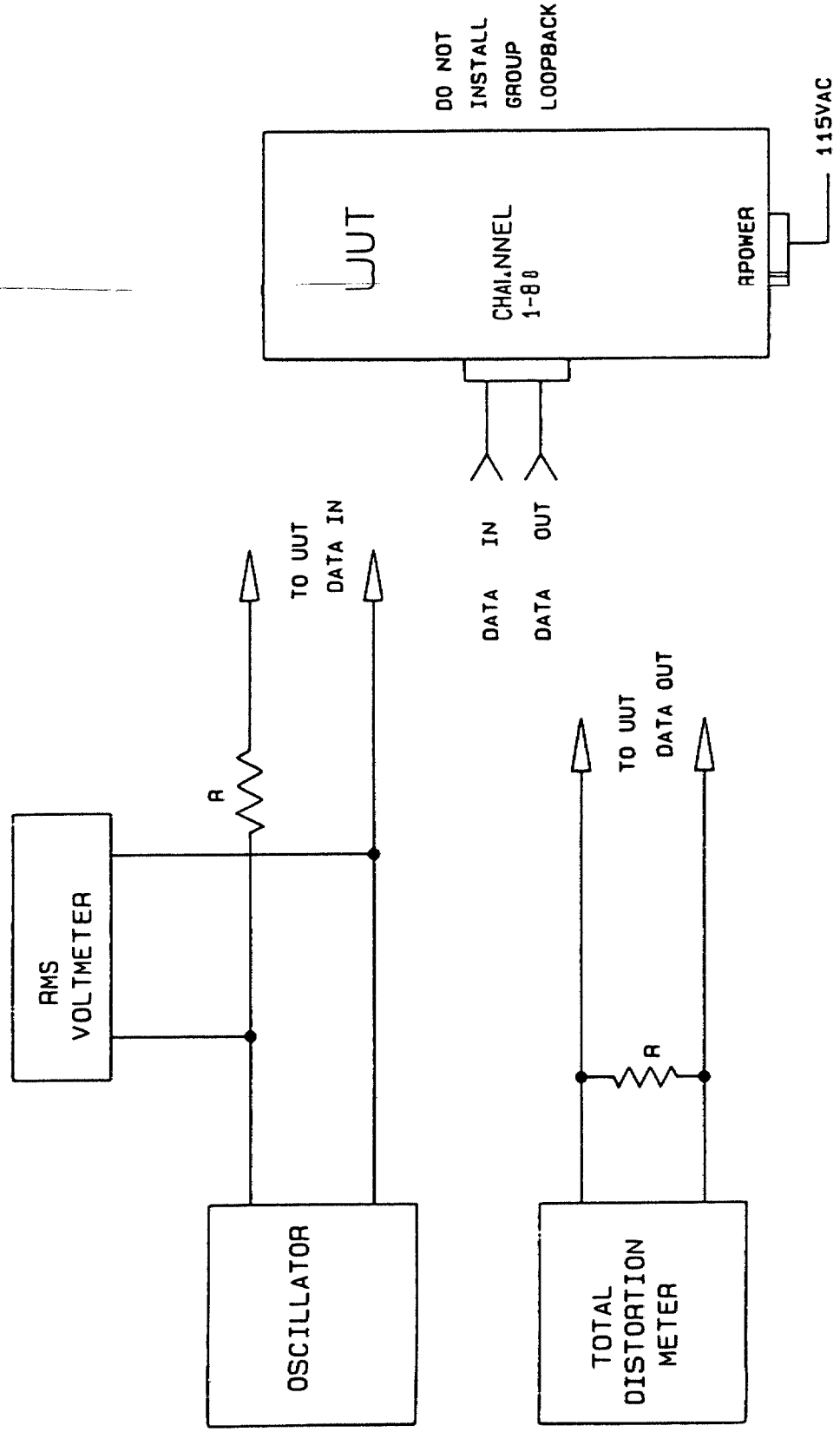
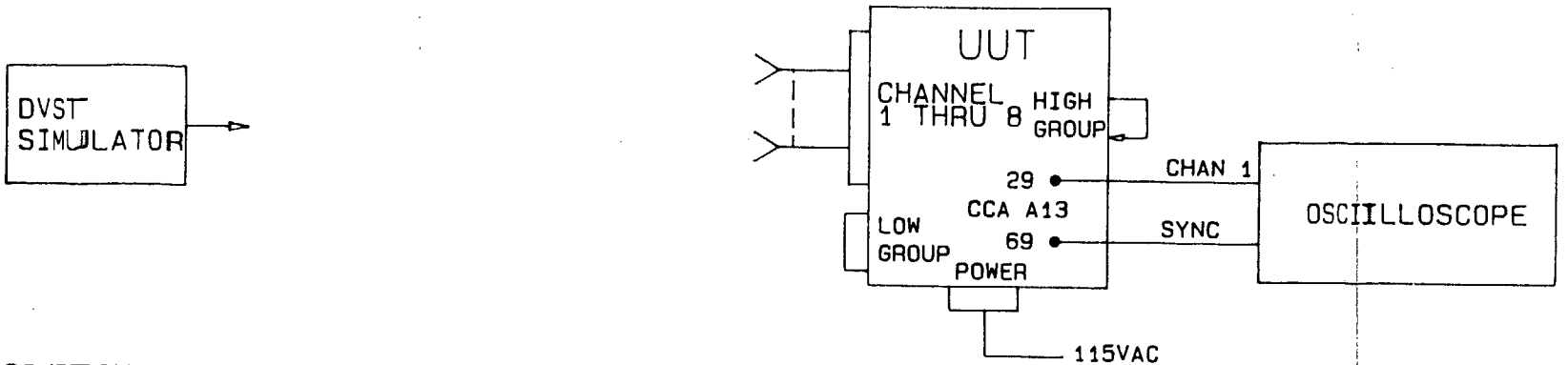


FIGURE 28. Peak clipping.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
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CONFIGURATION A.



CONFIGURATION B.

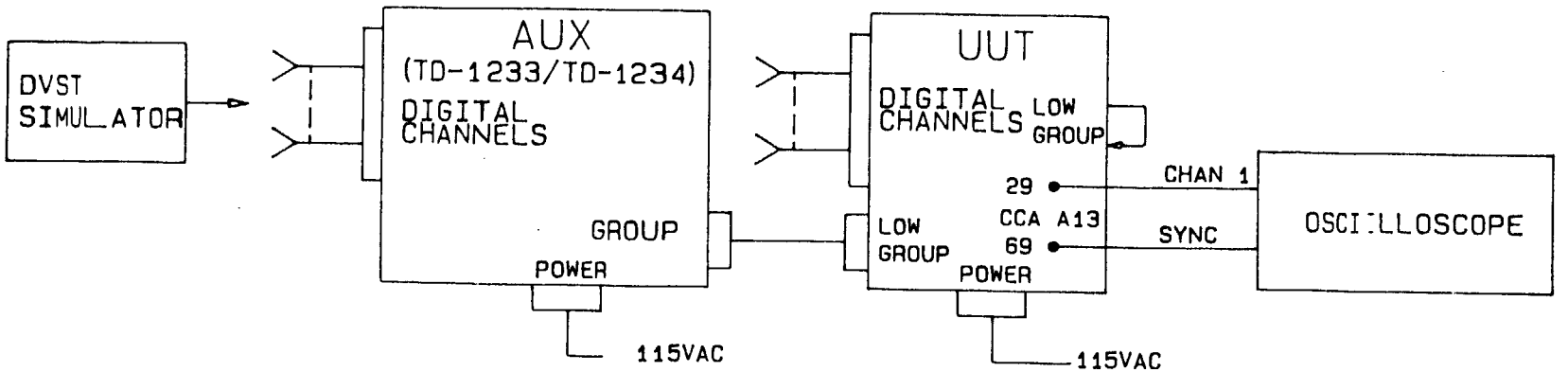


FIGURE 29. Format, configurations A and B.

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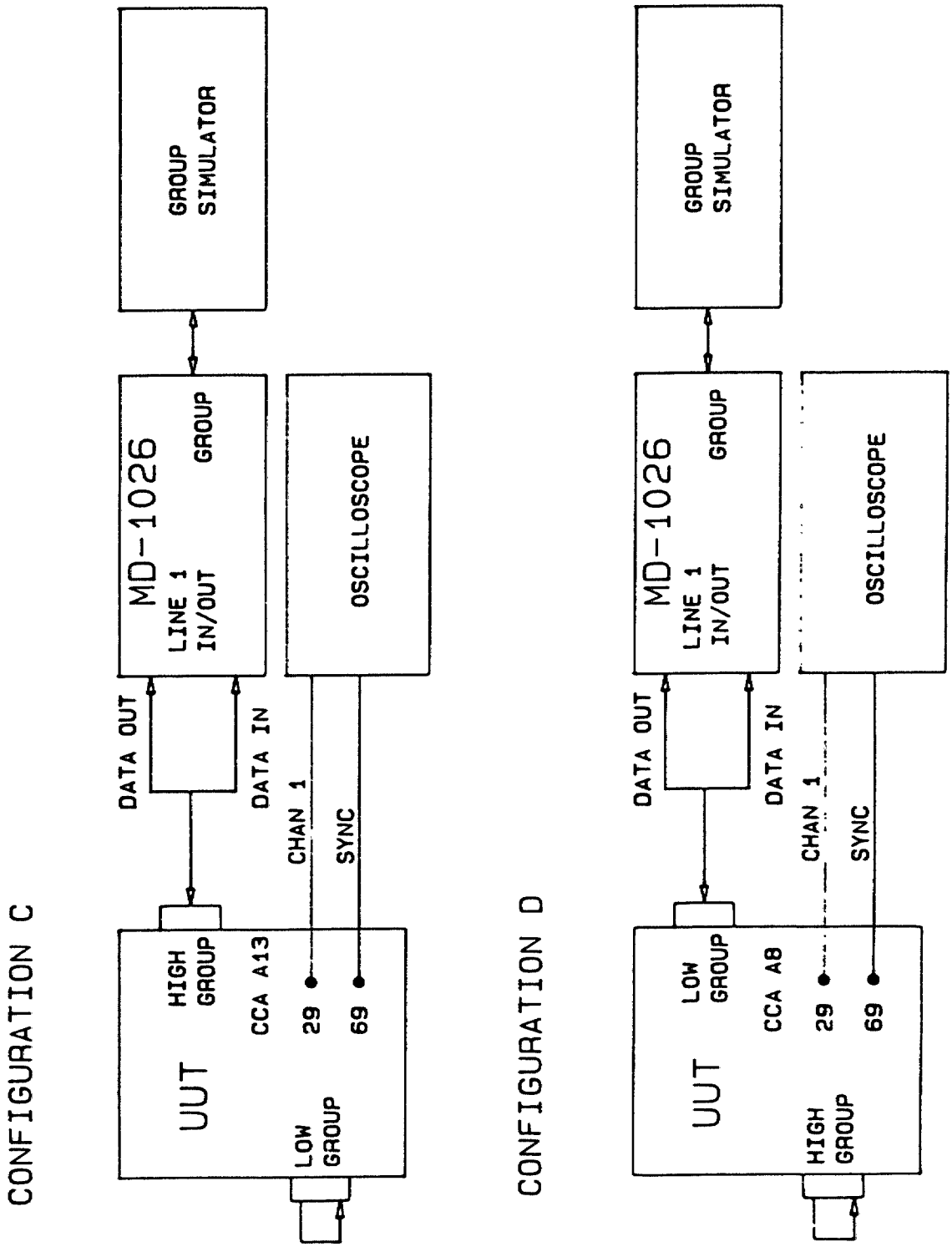


FIGURE 30. Format, configurations C and D.

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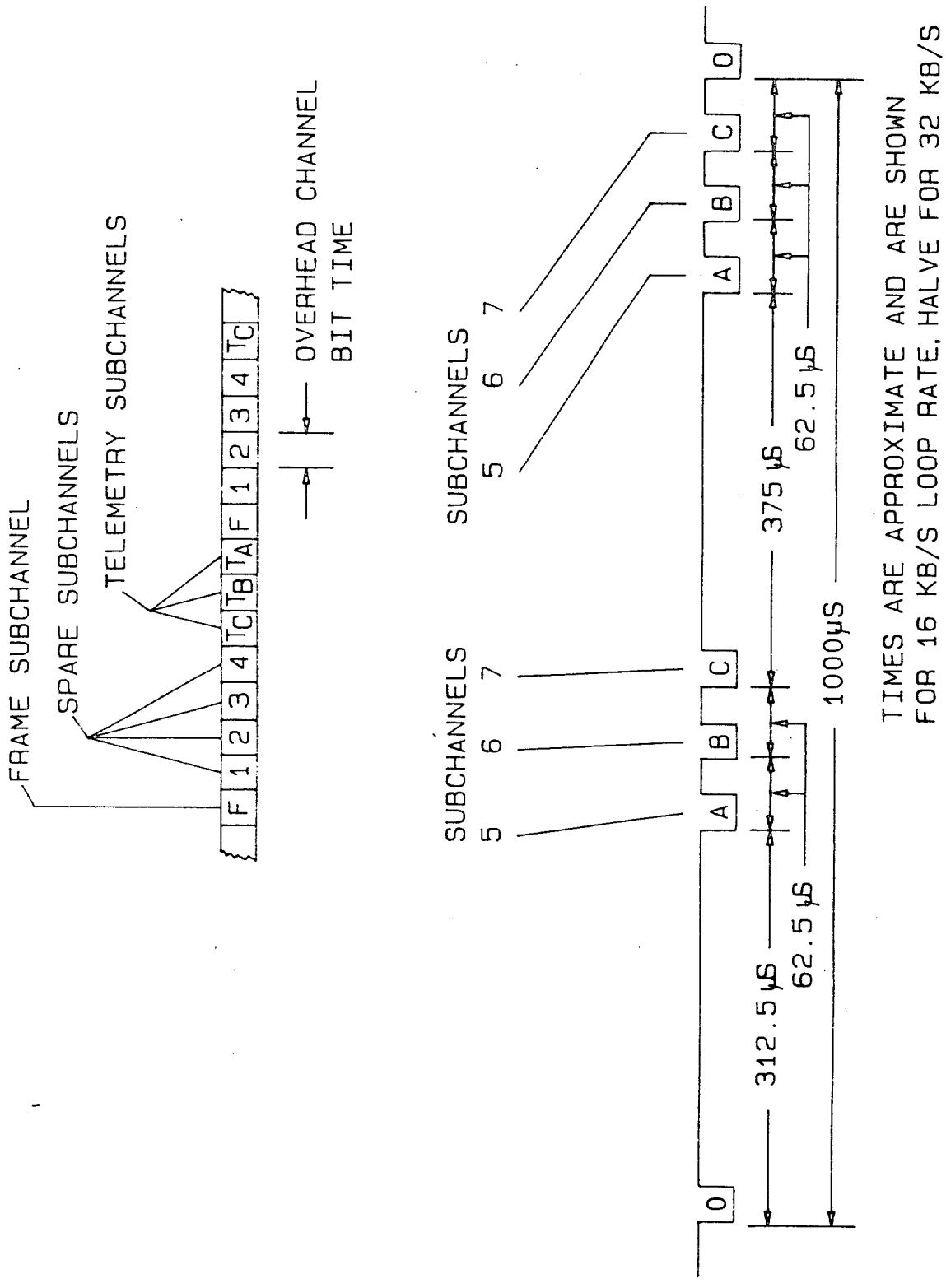


FIGURE 31. TD-1234 overhead channel formats.

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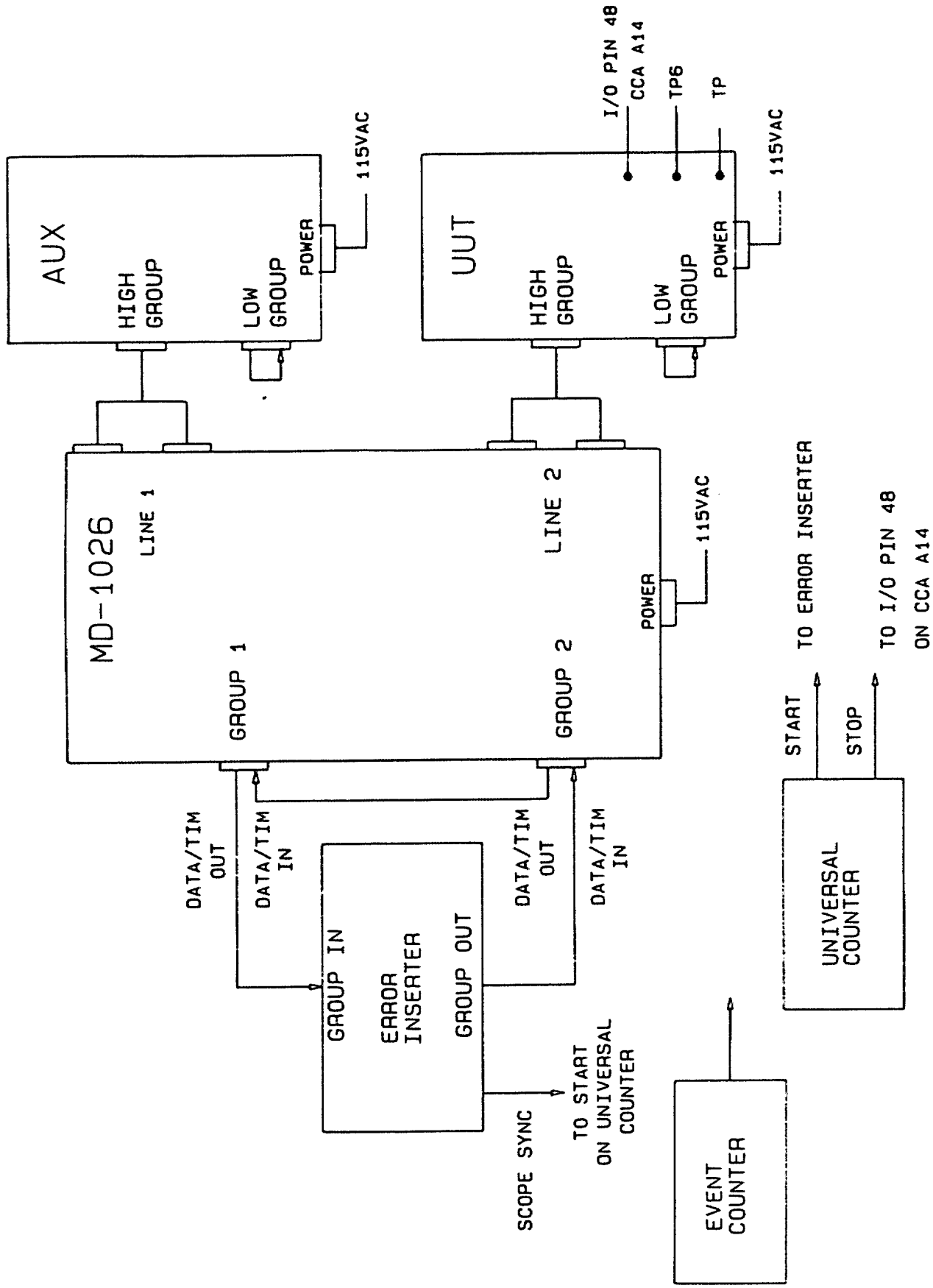


FIGURE 32. Frame acquisition; maintenance; detection of out-of-sync.

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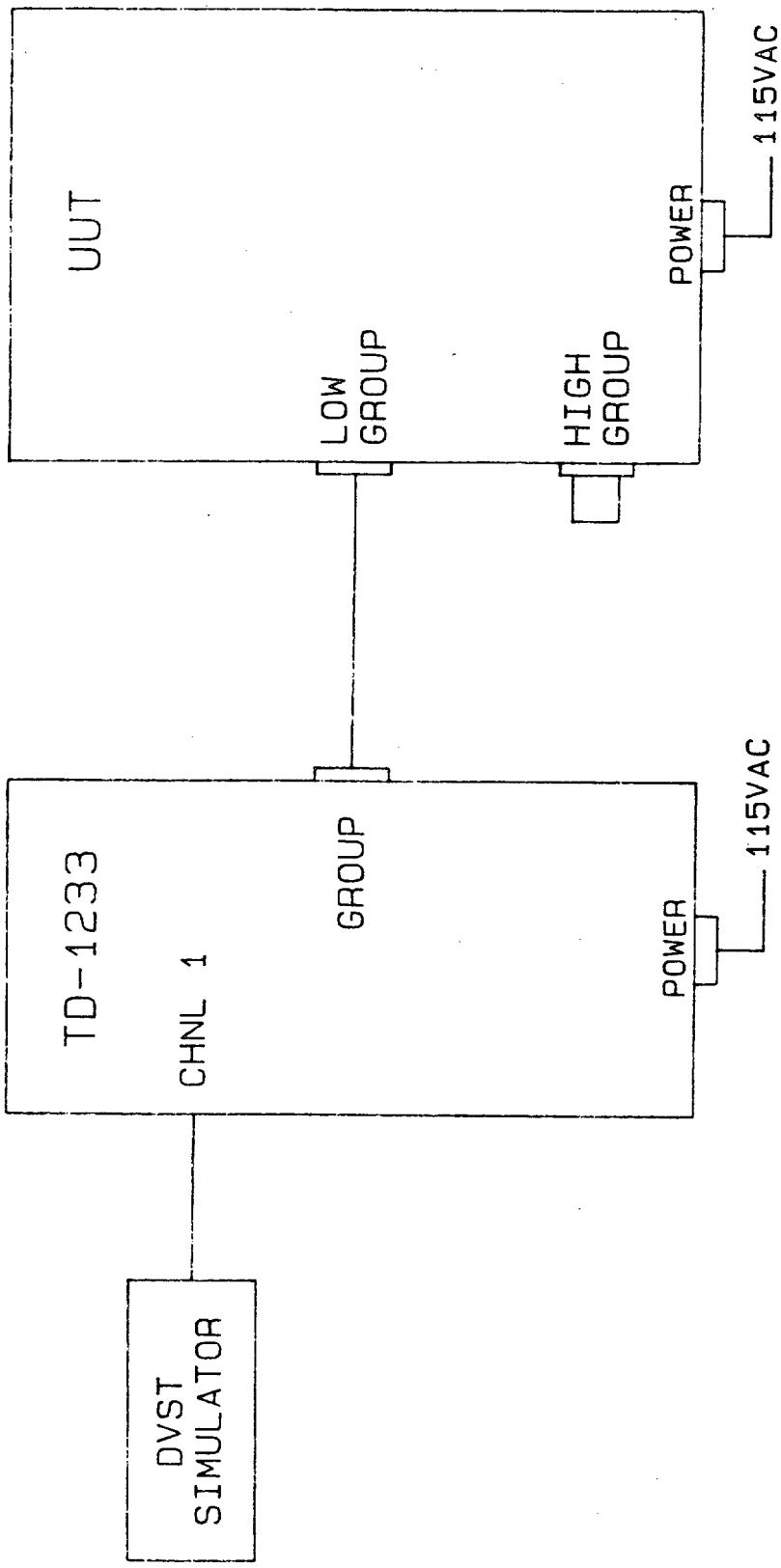


FIGURE 33. False frame lock.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.

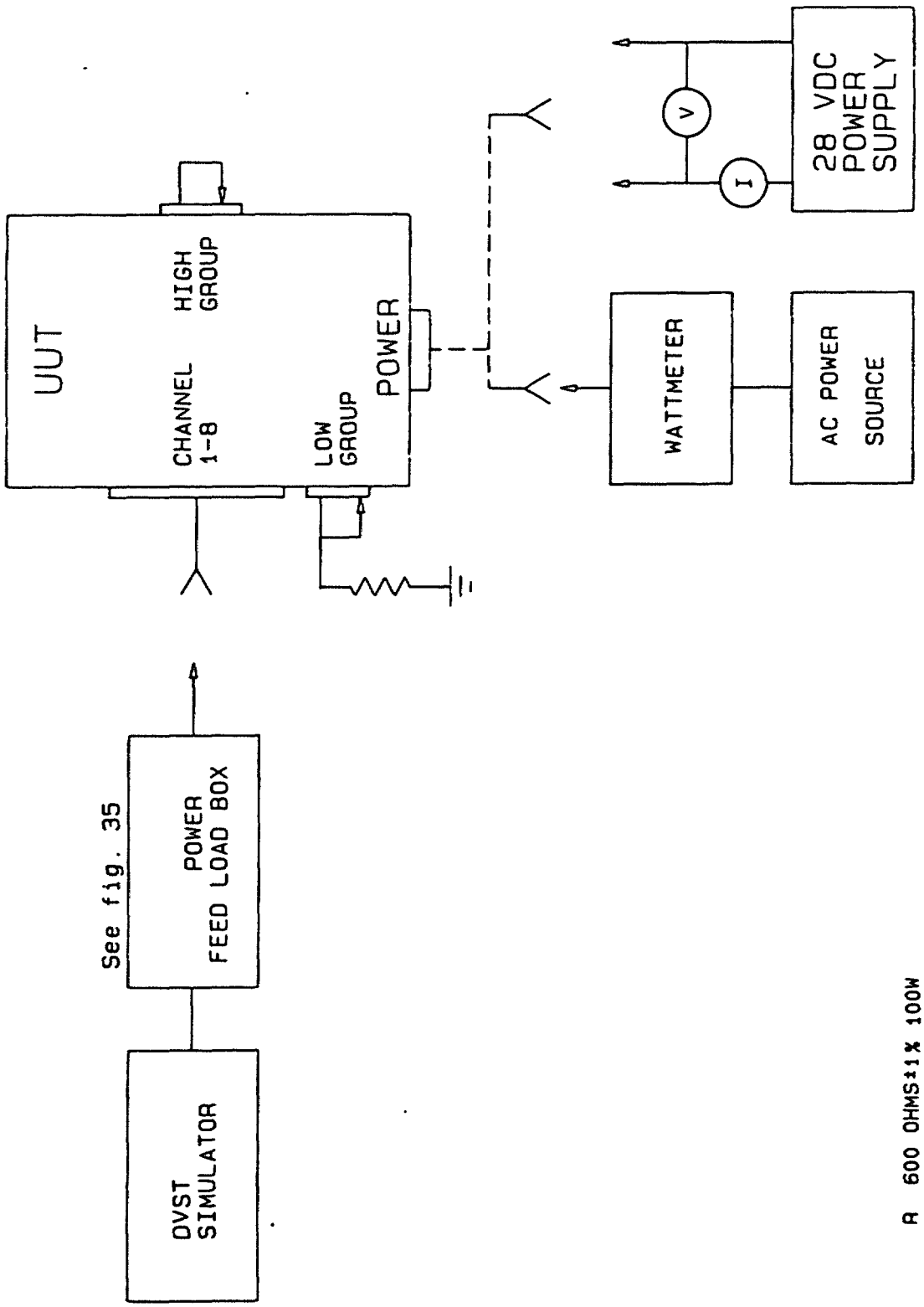
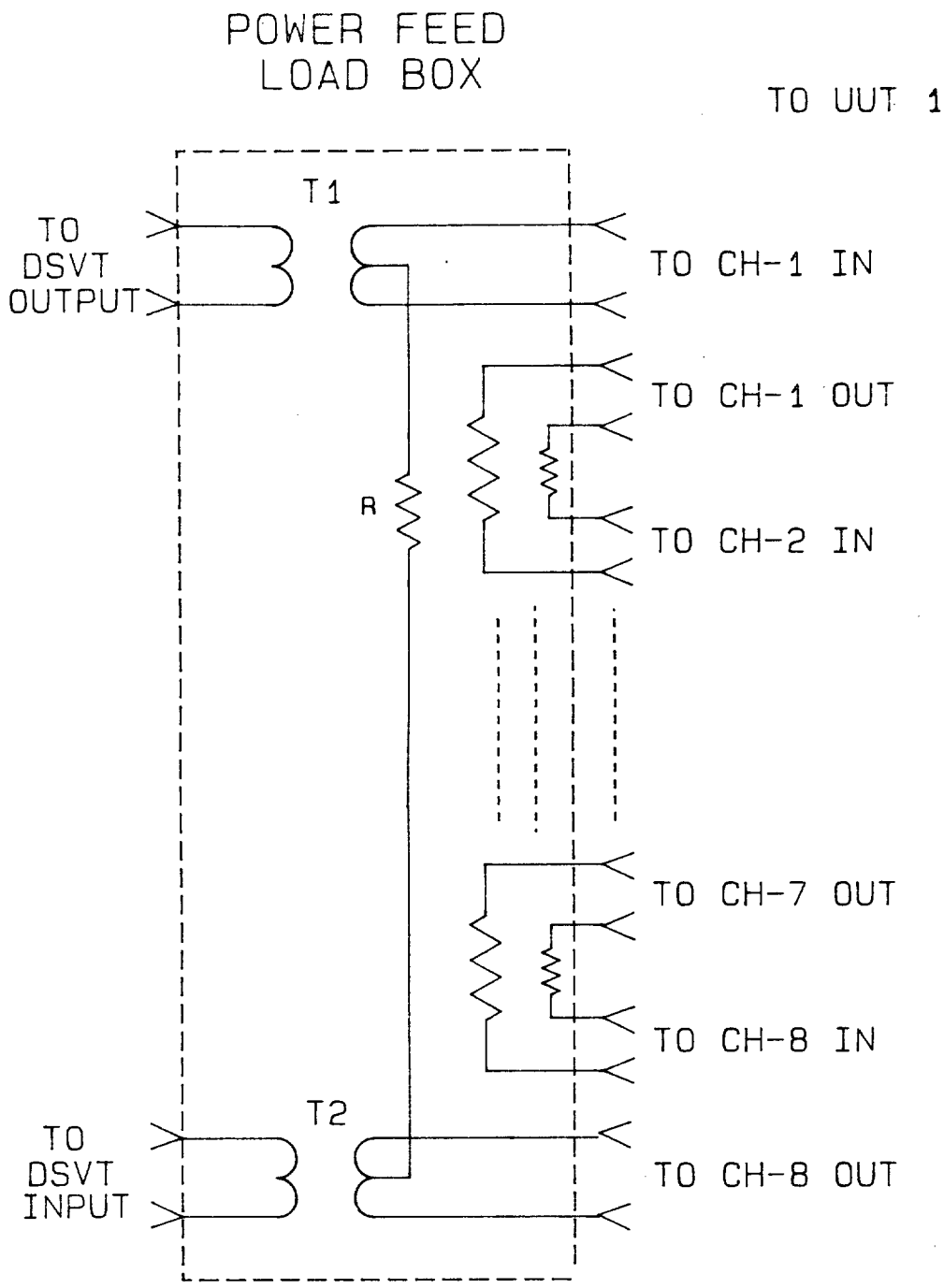


FIGURE 34. Prime power.

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ALL RESISTORS 1.36 K ± 1%, 2 W UNLESS OTHERWISE SHOWN.
FOR T1 AND T2 SEE SM-A-985920

R 680 OHMS ± 1%2W

FIGURE 35. Power feed load box.

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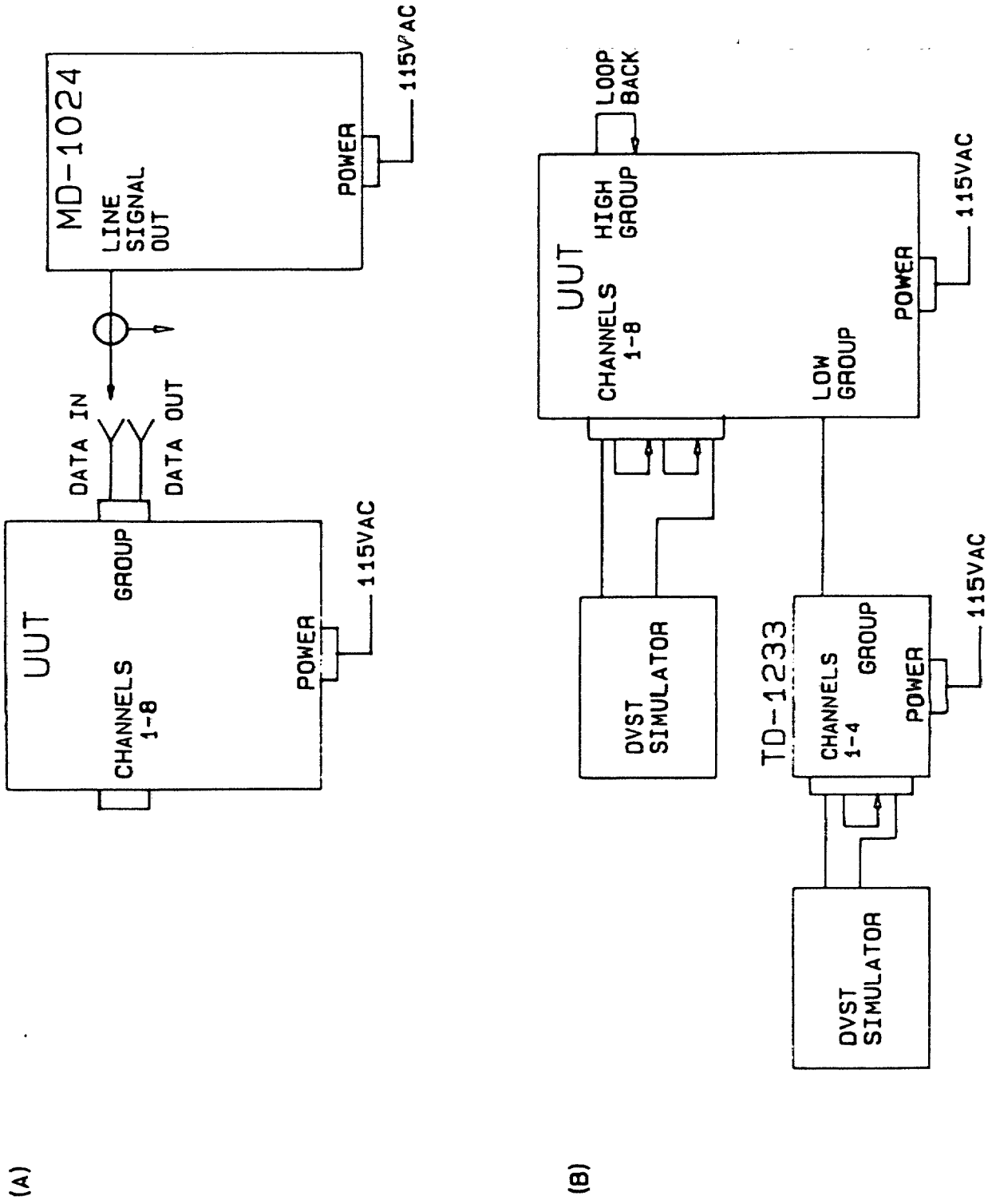
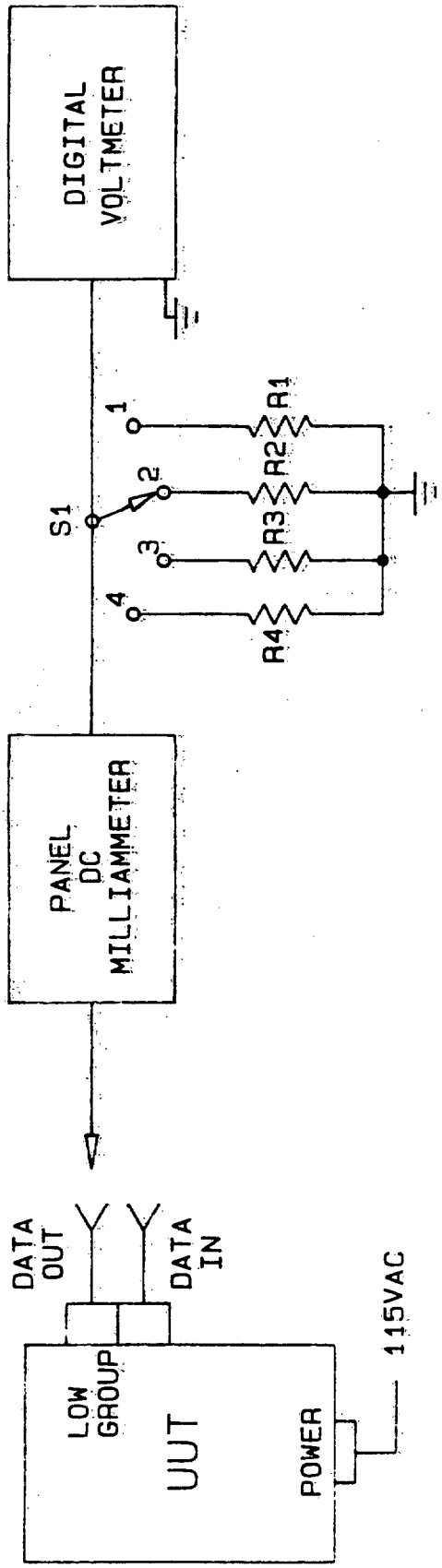


FIGURE 36. Current protection.

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SWITCH POSITION	
1	R1 402 OHMS ±5%, .95 W
2	R2 600 OHMS ±5%, .70 W
3	R3 12.5K OHMS ±5%, .5 W
4	R4 33K OHMS ±5%, .1 W

SWITCH MUST BE "MAKE BEFORE BREAK"

FIGURE 37. Power feed.

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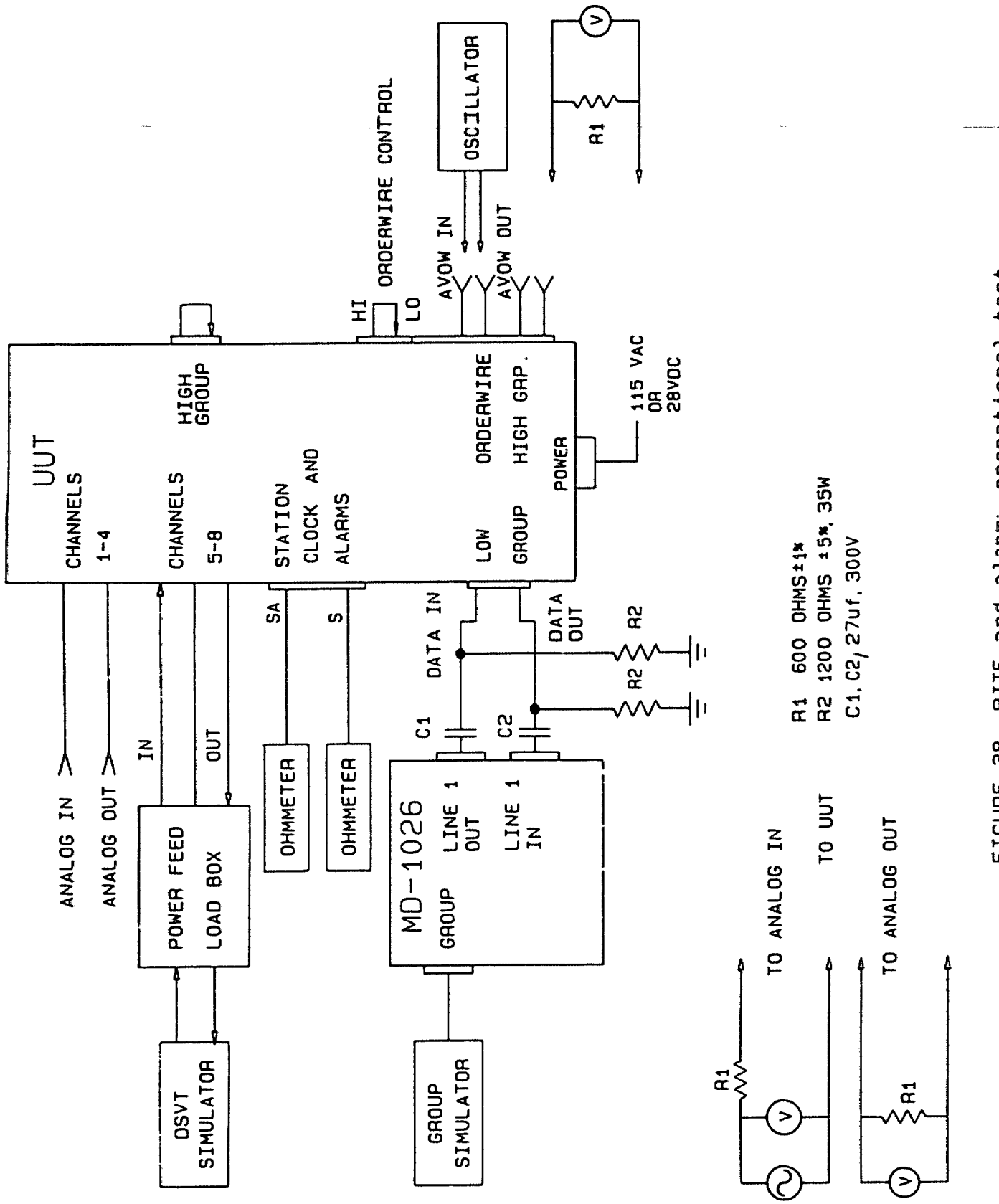
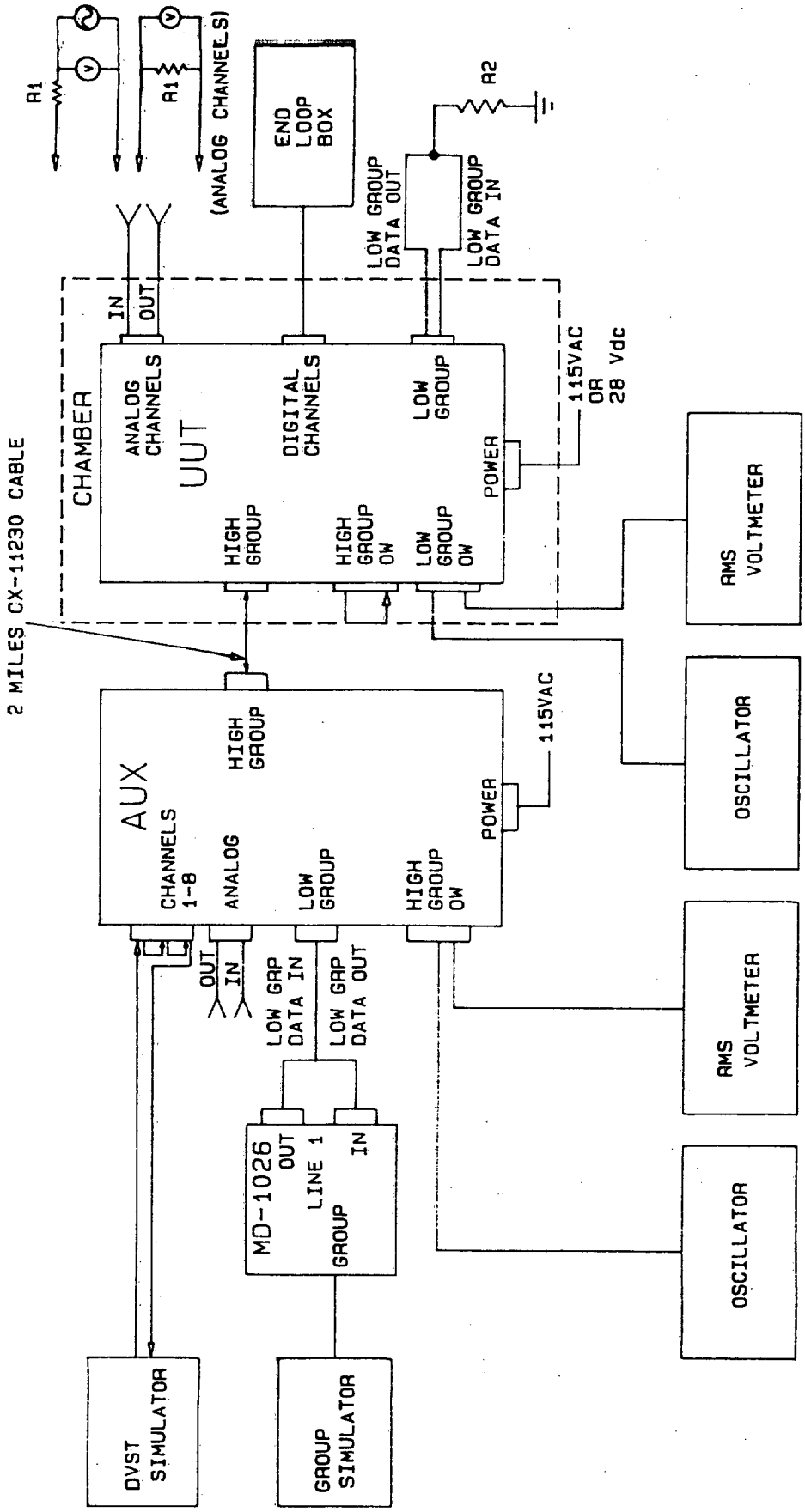


FIGURE 38. BITE and alarm; operational test.

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R1 600 OHMS±1% 70W
R2 600 OHMS±1%

FIGURE 39. Environmental operational test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-49239A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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